

claude-windows / 45ea3a91-654d-4972-9cd3-65f35db54caf

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\45ea3a91-654d-4972-9cd3-65f35db54caf\subagents\workflows\wf_2a64bc88-2a8\agent-a118ecf62ef5d4b46.jsonl`  
- SHA-256: `9d11b55bdd042702e04c1debcbcc53398a6db18baae1bdeccecc8ba770f577f2`  
- Source modified: `2026-06-10T06:09:07+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_2a64bc88-2a8`  
- session_id: `45ea3a91-654d-4972-9cd3-65f35db54caf`
```

Transcript

1. user (2026-06-10T05:59:28.046Z)

CONTEXT ? two sibling repos on a Windows machine:

- INDEXER: C:/Users/avidu/Projects/badminton-highlight-indexer ? local AI badminton(->racket-sports) highlight indexer + rally detector. FastAPI + SQLite + ffmpeg + GPU CV via WSL; Gemini = paid cloud AI. docs/ tree is rich. Current branch docs/next-steps-multisport (master = main branch).

- COLLECTOR: C:/Users/avidu/Projects/sports-data-collector ? golden-label acquisition/curation/ingestion CLI (system-of-record candidate for training corpus).

Project state (June 2026): scaffolding complete; owned-model roadmap agreed (docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md) but NO platform/owned-model implementation started; next committed platform slice = async job model. Licensing guardrail: MIT/Apache-2.0/BSD only for code+data+weights; paid LLMs are inference-only, never training-data sources. ENVIRONMENT RULES: this dev environment has NO GPU/torch/cv2 ? do NOT attempt to run the video pipeline or import torch/cv2 modules. Test suites are offline/fully-mocked and SAFE to run. You are READ-ONLY: never edit/create/delete repo files; running tests and git read commands is allowed.

QUALITY BAR: cite file paths (and line numbers where possible) for every finding. Report what IS in the code/docs, not what you assume. Be skeptical and concrete. Your final structured output is consumed programmatically by an orchestrator ? return raw data, not prose for a human.

TASK: STRATEGY-DOC CONSISTENCY sweep of the indexer. Read FULLY, in this order: CLAUDE.md (repo root), docs/README.md, docs/NEXT_STEPS.md (new, on current branch), docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md, docs/ROADMAP.md, docs/DEFERRED_HARDENING_PLAN.md, docs/BACKLOG.md, docs/COMMERCIALIZATION.md, docs/PLATFORM_ARCHITECTURE.md, docs/PMF_MARKET_RESEARCH.md, docs/archives/decisions/DECISIONS.md, docs/archives/checkpoints/2026-06-strategy-foundation/README.md, docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md (at least its correction banner + conclusions).

FIND, as findings: (a) CONTRADICTIONS between docs ? e.g. licensing policy wording (CLAUDE.md says 'open Apache-2.0 only' for training teachers/bases; later ratified policy is 'MIT/Apache-2.0/BSD' ? where do docs disagree?); what is 'the next slice' (async job model vs Phase-0 M0.x vs repo rename ? do docs agree on sequencing and on who decides?); v1 scope definitions; Gemma-4 stance (AMBER/bench-only vs candidate); WASB checkpoint provenance stance across docs; conversational-QA punt-vs-build framing. (b) CLAIMS CONTRADICTED BY REPO REALITY ? verify with Grep/Glob: does training.label_candidates still carry Gemini-derived labels the plan says must be purged (check backend + DB-related code/docs for what actually exists ? note output/ DB itself is gitignored so check code paths + docs claims)? does config.json default_segmenter match what docs claim? do doc-referenced files/flags exist (e.g. output/human_lovo_manifest.json is gitignored ? do docs point at artifacts a fresh clone won't have)? does the repo-rename action item appear anywhere actionable? (c) ROADMAP GAPS / missed gotchas ? sequencing conflicts (async-job slice vs Phase-0 both 'next'), undefined ship-gate F1@tIoU target, cost-to-Gen-2 unquantified, per-sport detector blocker vs the new finding that WASB transfers to table tennis (docs/NEXT_STEPS.md) ? is that finding consistently reflected or does it contradict older 'binding blocker' framing?

Each finding must QUOTE both sides (doc A says X at path:line-ish; doc B / code says Y).

Severity: high = a reader following docs would act wrongly; medium = confusing drift; low = cosmetic.

2. user (2026-06-10T05:59:33.721Z)

```
1      # CLAUDE.md ? project memory / agent handoff
2
3      Local, AI-powered badminton (? racket-sports) highlight indexer + rally detector**:
4      FastAPI + SQLite + ffmpeg + GPU CV/AI, with paid Gemini as the cloud AI. Sibling
5      repos: `sports-data-collector`, a VLC-based `rally-annotator`.
6
7      ## Read these first
8      - docs/README.md ? doc index. docs/CODE_MAP.md ? code navigation + data
9        contracts (the cold-start map; read before touching code).
10     - Strategy (post-scaffolding): docs/COMMERCIALIZATION.md,
11     docs/PMF_MARKET_RESEARCH.md,
12     docs/PLATFORM_ARCHITECTURE.md, docs/DEFERRED_HARDENING_PLAN.md. Also
13     docs/ROADMAP.md
14     (offline-segmenter narrative) and docs/BACKLOG.md.
15     - History: docs/archives/ ? ADRs (decisions/DECISIONS.md), research, and
16     checkpoints/ (latest: 2026-06-strategy-foundation/). Archive policy: only move
17     terminal-state (DONE/REJECTED) work; live docs stay at docs/ top level.
18
19     ## Current state (June 2026)
20     - Scaffolding complete; the post-scaffolding strategy foundation is laid and
21     merged
22     (#52/#53/#55/#57/#58). No platform/owned-model implementation has started.
23     - Next slice = async job model (DEFERRED_HARDENING_PLAN.md #16), single-tenant.
24     Do not start implementation without explicit owner approval.
25
26     ## Architecture in one breath
27     - Pluggable registries (ABC + Registry singleton): segmenters / providers /
28     validators
29     / sports / annotations ? see backend/pipeline/segmenters/base.py. Future:
30     StorageBackend + ComputeBackend registries (PLATFORM §3).
31     - video_id = MD5 of the file ? backend/utils/hashing.py::compute_video_id (one
32     helper).
33     - Typed config = backend/config/ (Pydantic). DB schema evolves via the PRAGMA
34     user_version migration ladder in backend/infrastructure/database.py.
35
36     ## Committed guardrails (non-negotiable)
37     - Paid LLMs (Gemini/OpenAI/Anthropic) = inference / serving / fallback ONLY ? never a
38     training data source (terms/copyright). Training teachers/bases = open Apache-2.0
39     only.
40     - Owned-model base: Apache-2.0, video-capable (Qwen-VL / Gemma 4); reject Gemma
41     3
42     (viral license). Exclude minors from the training pool; per-user training data
43     needs a
44     separate explicit opt-in.
45     - v1 = rally + highlight + history + correction loop; gait/biometrics strictly
46     future
47     (GDPR Art. 9).
48     - Generic data schema; coach ? athlete is an optional overlay, not baked in.
49
50     ## Workflow conventions
51     - Branch off master; ONE PR per work item; NO PR stacking. Don't open a PR unless
52     asked.
53     - Tests: python run_all_tests.py (offline, fully mocked). Python 3.8+; ffmpeg
54     required.
55     - Never push to the default branch directly; descriptive commit messages.
56     - ?? This dev container lacks GPU / torch / cv2 / numpy ? the full pipeline can't run
57     here.
58     Verify pure-logic paths and hand machine-side checks (real video, GPU/WSL suite) to
59     the owner.
60
61     47
```

3. user (2026-06-10T05:59:34.185Z)

```
1      # docs/
2
3      Live documentation for the badminton-highlight-indexer.
4
5      ## For new contributors / ramp-up (read these first)
6      - [CODE_MAP.md](CODE_MAP.md) ? Complete navigation map of the codebase and data
contracts.
7      - [COMMERCIALIZATION.md](COMMERCIALIZATION.md) ? Living tracker for the commercial
dimension (licensing, codec/output economics, hosting, vendoring, packaging) + PMF risks and the
quality/data-phase priorities. Supersedes the archived `COMMERCIAL_READINESS_REVIEW.md`.
8      - [PMF_MARKET_RESEARCH.md](PMF_MARKET_RESEARCH.md) ? Cited market study: sizing,
customer segments, competitor pricing, and the recommended beachhead (feeds COMMERCIALIZATION
§6).
9      - [PLATFORM_ARCHITECTURE.md](PLATFORM_ARCHITECTURE.md) ? Strategy for the multi-user
"assistant coach" platform: decoupled layers + pluggable storage/compute (BYO), multi-tenant
data model, orchestration, and a phased roadmap.
10     - [DEFERRED_HARDENING_PLAN.md](DEFERRED_HARDENING_PLAN.md) ? Design (pending approval)
for the larger deferred items: packaging, train/eval guardrails, and the async job model (#16,
the first infra slice toward the platform).
11     - [BACKLOG.md](BACKLOG.md) ? Deferred items with rationale and revisit triggers.
12     - [ROADMAP.md](ROADMAP.md) ? High-level four-phase product direction.
13     - [VIDEO_RECORDING_GUIDELINES.md](VIDEO_RECORDING_GUIDELINES.md) ? Capture best
practices (referenced by reports).
14     - [HOW_RALLY_DETECTION_WORKS.md](HOW_RALLY_DETECTION_WORKS.md) ? Mental model of the
two-layer detection system.
15     - [EVALUATION.md](EVALUATION.md) ? How we measure and improve quality.
16
17     ## Practical / ongoing
18     - [TRACKNET_WSL_SETUP.md](TRACKNET_WSL_SETUP.md)
19     - Golden set docs (`GOLDEN_SET`)
20     - Evaluation harness and calibration drivers (in `backend/eval/`)
21
22     ## Historical documents
23     Completed plans, old research, ADRs, and checkpoints live in [`archives/`](archives/).
24
25     See `archives/README.md` for the umbrella policy and `archives/past_projects/README.md`
for classification of small tech debt / starter projects.
26
```

4. user (2026-06-10T05:59:34.907Z)

```
1      # Next steps (live) ? owned-model + multisport
2
3      **Updated:** 2026-06-09 (end of the quality-iteration + multisport-recon session).
**Authoritative roadmap:**
4      `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md`. **Latest blow-by-blow:** memory `current-
state.md` / `session-handoff.md`.
5      This file = the prioritized "what to do next," refreshed each session.
6
7      ## The recommendation (one paragraph)
8      Build the **quality-first extraction harness** before anything else: a learned **TAL
head** (ActionFormer/ASFormer, MIT)
9      over trajectory features is the robust path ? it **decouples from the multi-court
"background games" confound** that the
10     unsupervised heuristic (and audio, and spatial court-gating) can't escape (all three
tested + failed; it's a *temporal*
11     rally-vs-non-rally problem). **Framework = ms-swift, base = Qwen3-VL** (Gemma-4
AMBER/bench) come **later** for the
12     conversational/grounding upgrade ? start with the head. **On-device is deferred**
(export-clean ops now so it stays
13     reachable). Everything stays **MIT/Apache-2.0/BSD** (code + data + weights). Bottleneck
= **data + per-sport trajectory
14     detectors**, not model architecture.
```

```

15
16     ## Latest findings (2026-06-09)
17     - **6-match badminton LOVO:** learned prototype **0.583** vs heuristic **0.480**
    (learned leads since n=3, AUC 0.83?0.92,
18     decoupled from multi-court). The "contrast" metric (in-rally÷dead-time visibility)
    predicts heuristic F1 monotonically.
19     - **?? Multisport (Roora pickleball + TT):** labels **ingest cleanly** (`curate convert-
    cvat` ? 6 rallies/sport, sane).
20     **WASB-transfer SURPRISE (directional, n=6 each ? TINY):** the badminton shuttle
    detector segments **TABLE TENNIS at
21     F1 0.909** out-of-the-box (!) but **pickleball only 0.308**. ? **The per-sport-
    detector blocker is SPORT-DEPENDENT:** TT may
22     be servable with WASB short-term; pickleball needs a proper (MIT/Apache) ball
    detector. Confirm/deny with more videos.
23     - **Owner field notes (2026-06-09):** camera height/distance **varied on purpose**
    (great generalization data); **multi-court
24     adjacent-court SOUND dominates the signal** ? empirically reconfirms audio-foreground
    isolation won't work + the
25     robustness-first learned-model path. Multi-court is the realistic (academy)
    environment, so this *is* the target distribution.
26
27     ## Prioritized next steps
28     1. **[owner greenlight] Phase-0 (M0.x), $0/local:**
29     - **M0.4 ? Gen-0 TAL harness** (first-class): productionize `rally_seq_proto.py` ?
    one-command train?eval?***ship-gate** over
30     ActionFormer/ASFormer, reusing `metrics.match_intervals`. **Define the ship-gate
    target** (explicit F1@tIoU; floor = beat
31     heuristic LOVO 0.480) ? "very high P/R" must be operationalized, not just "beat the
    floor."
32     - **M0.1 ? corpus spine + the per-video event-index contract** (the conversational
    seam: extract ? event store ? text-to-SQL).
33     - **M0.3 ? compliance:** purge Gemini-derived training labels
    (`training.label_candidates`); WASB **C3** own-weights/clean-swap (gates ship).
34     2. **Multisport thread:**
35     - Ingest the Roora pickleball/TT rally CSVs into the corpus (collector `import-
    local`).
36     - **Get more pickleball + TT videos** ? re-run the WASB-transfer eval to confirm/deny
    TT?0.91, pickleball?0.31 (n=6 is too few).
37     - Once each sport has ?3 matches, run a **merged prototype LOVO across badminton + TT
    (+ pickleball)** ? the "how would a
38     merged model do" experiment. Identify a clean MIT/Apache ball detector for sports
    where WASB doesn't transfer (pickleball).
39     3. **Housekeeping:** **rename repo off "badminton" (multisport); confirm collector =
    corpus/event-store system-of-record.
40
41     ## Pointers / artifacts
42     - Roadmap: `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` · Study:
    `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md`
43     (2026-06-09 banner) · Quality: `docs/archives/quality-iterations/2026-06-multivideo-
    lovo/`.
44     - Eval corpus: `output/human_lovo_manifest.json` (badminton n=6). Roora CSVs:
    `%TEMP%\roora\{pickleball,tabletennis}_rallies.csv`;
45     WASB-transfer trajectories: `%TEMP%\{pickleball,tabletennis}_traj.csv`. Code:
    `backend/eval/{eval_partial_labels,rally_seq_proto,calibrate_local}.py`.
46     - ? No owned-model implementation until owner greenlights a milestone. Committed next
    PLATFORM slice = async job model (single-tenant).
47

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5. user (2026-06-10T05:59:40.761Z)

```

1     # Owned-model implementation plan (roadmap) ? DRAFT for owner review
2
3     **Status:** PLAN / roadmap. No owned-model *implementation* starts until the owner
    greenlights a specific
4     milestone. Paid Gemini stays the always-ready inference/fallback throughout (never a

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training source).
5      **Grounded in:** `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md`, the n=6 LOVO +
learned prototype
6      (`docs/archives/quality-iterations/2026-06-multivideo-lovo/`), and a verified strategy
workflow (2026-06-09).
7
8      ## Design principle (owner directive): *envision now, build later*
9      Every deferred piece (conversational-QA, on-device, multisport, the VLM) must be
**architecturally envisioned
10     now** so we never hit a mammoth refactor. We **design the seams** (data contracts,
export boundaries, the
11     event store) up front and **implement behind them** when each milestone is greenlit.
Punting a *feature* is fine;
12     punting its *seam* is not.
13
14     ## North Star (reframed per PR #61 review)
15     - **Immediate deliverable:** a **training framework + harness** that produces a model
with **very high precision/
16     recall for extracting highlights + rallies from a clip**. *Quality is the first-order
objective.*
17     - **Eventual product:** a **single, sport-agnostic, conversational** video model ?
upload a clip, then ask
18     contextual questions ("longest rally", "make a clip of all service faults", "rallies
over 20 shots").
19     - **On-device is LATER** (a model-compression goal so power users save upload
cost/hassle) ? definitely on the
20     roadmap, but **not** the immediate target. Don't over-index the strategy on it now.
21     - **The moat = the consented dataset + the eval harness, not the weights.**
22
23     ## Hard guardrails (non-negotiable ? "never corrupted")
24     - **Commercial-clean licensing for EVERY dependency ? code, data, AND model weights: MIT
/ Apache-2.0 / BSD only.**
25     (Ratifies the prior "Apache-2.0 only" to include MIT/BSD.) Reject viral/NC/custom-
restrictive: Gemma 3 (viral),
26     **Gemma 3n** (Gemma Terms follow derivatives ? even cleaner reject), Llama 4 (700M-MAU
cap), Commons-Clause repos.
27     - **Never train on paid-LLM (Gemini/OpenAI/Anthropic) outputs.** Also forbidden: public
grounding-CoT datasets that
28     were Gemini-generated (e.g. TVG-R1's 56K) ? we must originate our own CoT supervision
(a real, binding cost).
29     - **Exclude minors; per-user training data needs a separate explicit opt-in; split data
by match.**
30     - ? **Base-model pretraining provenance is unauditatable for *every* open VLM (incl.
Qwen3-VL).** Our "no paid-LLM
31     training" rule is enforceable at *our fine-tuning data* layer, not the base's
pretraining ? an **explicit owner
32     risk-acceptance** is required to use any open VLM. (Governed by the base's license,
not our pipeline.)
33
34     ## The model strategy (verified 2026-06-09)
35     **Framework ? base** (the Gemma-4-vs-ms-swift clarification): a *framework* is the forge
(runs SFT/GRPO, emits
36     weights); a *base* is the metal you forge. You pick one framework and feed it any clean
base.
37     - **Framework: ms-swift (Apache-2.0)** ? the one forge that does native **video + audio
+ timestamp-grounding +
38     GRPO/RFT with a custom video reward** in one stack. Keep **unsloth** (Apache-2.0 core;
*avoid its AGPL Studio UI*)
39     as the 8 GB-local SFT/compression tool for the deferred on-device goal. ? Point-in-
time snapshot ? re-verify at impl.
40     - **Base (when we reach the VLM): Qwen3-VL (Apache-2.0, uniform across sizes)** ? long-
video context, purpose-built
41     **timestamp emission** (second-level localization), multi-turn video chat. Re-confirm
the exact size's LICENSE
42     pre-release. **Gemma-4 stays AMBER and bench-only** ? its Apache grant is likely but

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its Prohibited-Use posture is

43 unconfirmed (counsel needed), *and* its video/timestamp capability is disputed; not the primary extractor.

44 **InternVL3 (MIT/Apache)** = clean fallback if a Qwen blocker appears. ? base swap is cheap *only* within

45 timestamp-capable Apache bases* (timestamp data-format + grounding-CoT are base-specific).

46 - **START WITH THE TAL HEAD, NOT THE VLM** (the decisive call): fine-tuned VLMs **miss strict temporal boundaries**

47 (best RL-post-trained video VLM ? R@0.7 50% on open-domain benchmarks; "coarse regions, not accurate boundaries").

48 **Boundary accuracy IS the product.** The head (`rally\_seq\_proto.py`) is already de-risked on our corpus (held-out

49 AUC 0.83?0.92), trains in minutes at ~4.5 GB on the 2070 Super (\$0), is ~1M params (trivially compressible later),

50 and **doubles as a pseudo-label teacher + the honest baseline the VLM must beat.** So: **head now ? VLM later**,

51 strictly sequenced. When the VLM comes, prefer **two-stage** (VLM coarse-proposes, head refines boundaries) over

52 wholesale replacement. *(Open: the VLM-ceiling number is open-domain; our amateur footage may differ ? re-measure.)*

53

54 ## Conversational video-QA ? build the bookkeeping NOW, punt the model

55 **Punt the conversational *feature*** until extraction quality is solid (current LOVO ~0.583 ? "very high"; a chat UI

56 over a weak extractor exposes the weakness). **Reject end-to-end-VLM-answers-each-turn** (expensive; bad at exact

57 counting; the strong teachers are license-forbidden). **But build the *seam* now** (the brief requires it):

58 - **Architecture = "extract once, query many":** EXTRACTION (TAL head ? later VLM) emits structured timestamped

59 spans ? a durable **per-video EVENT STORE** ? a **structured QUERY layer** (text-to-SQL + ffmpeg clip-cut). The

60 cited queries ("longest rally", "all service faults", "rallies > 20 shots") are **structured-numeric ? SQL, not a

61 VLM job** (SQL counts exactly where VLMs hallucinate).

62 - **Designed-now seam:** make the **per-video event-index schema a first-class data contract** alongside the M0.1

63 corpus spine: per rally `{video_id(MD5), rally_ordinal, sport, start, end, shot_count, ending_reason/fault_tags,

64 derived_clip_ref, provenance, confidence}`. \*\*Reserve the event/fault-tag taxonomy (e.g. `service_fault`) up front**

65 even if unpopulated. ? This is a real DB delta (a `PRAGMA user\_version` migration): `play\_segments` today has

66 `video\_id/start\_time/end\_time/duration/server/receiver/metadata` only; `highlights` isn't a table yet ? so it's an

67 *extension with new columns + a highlights table*, not a no-op. **Punt** the NL parser/agent until quality clears the bar.

68 - Paid Gemini may answer genuinely open-ended edge questions **at inference** over our structured index (allowed ?

69 it reasons over our data, it is not trained on its outputs).

70

71 ## Multisport (single sport-agnostic model)

72 - **Model/framework: UNCHANGED.** One shared TAL head + one Qwen3-VL base, fine-tuned across sports (sport = an input

73 feature / per-sport calibration). The collector + indexer are already sport-generic. ? **ACTION: rename the repo off

74 "badminton" ASAP** (owner emphatic) ? consistent with this.

75 - **What multisport changes = DATA + DETECTORS (scale up, the accepted "good problem"):** each new sport needs its own

76 labeled matches (split by match) + a **clean trajectory detector** to feed the head's features. **The binding blocker

77 is per-sport detectors:** WASB (badminton) has the unresolved **C3** checkpoint-provenance issue, and **clean MIT/

78 Apache/BSD shuttle/ball detectors for tennis/TT/pickleball/padel are not yet

```

identified.** Sequence by public-data
79     richness: **badminton ? tennis/TT ? pickleball/padel.**
80     - ? **Single-model-multisport-temporal-generalization is an OPEN HYPOTHESIS**, not
proven ? our corpus is badminton-only,
81     and the evidence once cited for it (RacketVision) was *refuted on verification* (it's
ball-tracking, not temporal). The
82     concept is sound on first principles (rally signal = a racquet-sport invariant) but
must be validated once a 2nd-sport
83     corpus exists. Don't cite RacketVision as evidence.
84
85     ## Language ? Python everywhere; on-device runtime is a deferred, layer-local export
shell
86     - **Training/eval = Python-locked, and that's fine** (PyTorch/ms-swift/ActionFormer);
offline/batch, no latency hot path.
87     - **Platform = no Python hot path to escape** ? the heavy bytes/GPU already run out-of-
process (ffmpeg, WASB-in-WSL, cloud
88     GPU per-job); the control plane is I/O-bound. **Reject a Go/Rust hot-path split** ?
single-digit-ms wins dwarfed by
89     ~40-min GPU jobs, and it would fracture the one codebase holding the QA-state + eval
harness + provider registry.
90     - **On-device (deferred) = the only place a 2nd language appears, and it doesn't dictate
our dev language:** train in
91     Python ? **export across the boundary** (ONNX/Core ML/ExecuTorch/GGUF) ? a thin device
shell (Swift/Kotlin/C++/Rust)
92     runs the artifact with zero Python. **The one concrete constraint now: train with
export-clean ops** so the compression
93     target stays reachable without re-architecting. (On-device base stays on the
Qwen3-VL-4B Apache lineage ? *not* Gemma-3-270M.)
94     - Escalation order if a real Python CPU hot path ever appears: free-threading ?
numpy/vectorize ? Cython/PyO3 ? only then a
95     separate service. **None exists today.**
96
97     ---
98
99     ## Phase 0 ? Foundation (start now, per-milestone greenlight; $0, local, no cloud/DPA)
100
101     ### M0.1 ? The labeled-corpus spine **+ the event-index contract** (the moat) .
*greenlight item*
102     Canonical JSONL row per rally (corpus) **and** the queryable per-video event-index
schema (the conversational seam)
103     ? same row, wired to a store. Fields: `{video_id(MD5), rally_ordinal, sport, start, end,
ending_reason/fault_tags,
104     shot_count, label_type, source(human|pseudo|vendor), consent/training_opt_in, split(BY
MATCH), trajectory_ref,
105     labeled_window, derived_clip_ref, confidence}`. 3 transcoders (heuristic/LogReg . TAL
ActivityNet-JSON . VLM clip?QA),
106     one scorer spine (`metrics.match_intervals`). Authored in `sports-data-collector`
(system-of-record; extends PR #13).
107     **Reserve fault/event taxonomy now.** Acceptance: round-trip test; split-by-match
enforced; **no Gemini-sourced rows in any training view**.
108
109     ### M0.2 ? Grow the corpus . **DONE (n=6), running** . *owner action + my scoring loop*
110     6 distinct matches labeled (3 singles incl. Mahadevpura, 1 doubles, 2 multi-court); per-
match + LOVO recorded; the
111     "contrast" data-quality metric validated. Owner adding 1 more varied multi-court-club
match ? margin. **Bottleneck =
112     data + calibration, not method.** Keep growing per new sport.
113
114     ### M0.3 ? Compliance cleanup (**hard blocker ? do before any training run**) .
*greenlight item*
115     - **(i) PURGE the Gemini-derived TRAINING labels** in `training.label_candidates` (over
Gemini CSVs) ? a **live
116     no-paid-LLM-training violation** that bakes in the moment a training run precedes it.
Retarget to human + open-teacher
117     (own-model/WASB) labels; Gemini = eval/reference/fallback only.

```

```

118 - **(ii) WASB C3 ? first-class action item (owner):** the academic-data checkpoint
provenance is a **commercial-ship
119 blocker** that a clean head does NOT launder. Resolve via **own-trained weights** or a
clean detector swap; **multiplies
120 per sport.** Carry as a tracked blocker that **gates SHIP** (not Gen-0 research).
121
122 ### M0.4 ? Gen-0 TAL harness (**FIRST-CLASS ACTION ITEM** per owner) · *greenlight item*
123 Productionize `rally_seq_proto.py` ? a **one-command train?eval?ship-gate** harness over
**ActionFormer (MIT, 1:1
124 rally=instance)** and/or **ASFormer (MIT, TAS)** (use `sj-li/MS-TCN2` if MS-TCN++, *not*
the Commons-Clause repo),
125 reusing `metrics.match_intervals`. **Define the ship-gate target up front** (open item):
a FLOOR of "beat the honest n=6
126 heuristic LOVO **0.480**" is necessary but not sufficient ? set an explicit **F1@tIoU**
target for "very high P/R" (e.g.
127 F1@tIoU=0.5 for highlights, tighter for precise rally cuts) before declaring victory.
Trains on the GPU/WSL box.
128
129 ---
130
131 ## Phase 1 ? Gen-0 ship decision (gated on n?5 + M0.4)
132 Run the A/B; if the learned Gen-0 clears the ship-gate with per-video calibration, it
becomes the rally/highlight
133 extractor (local, MIT). If not, the gap (more data/features) is the next signal ? no
overfitting to declare victory.
134
135 ## Phase 2 ? Gen-1 / Gen-2 (gated on platform scaffolding P0?P4 + healthy corpus +
explicit go)
136 - **Gen-1:** concatenate fully-local cue channels onto the head. ? **Audio is NOT a
promising cue here** ? our E1 probe
137 (PR #35) found it weak (~2.2× discrimination, AUC ~0.6) and the foreground-audio idea
also failed this session; keep it
138 parked, not "in reserve." Pose/court is the more credible #2 cue (license/cost-vet
first).
139 - **Gen-2 (VLM):** ms-swift fine-tune of **Qwen3-VL** (SFT cold-start ? GRPO/RFT with a
temporal-IoU-vs-golden reward,
140 our own CoT supervision). **Two-stage** (VLM proposes, head refines). Cloud train on
Modal (no-access) / RunPod-Secure.
141 - **Hard pre-serving gates (unmeasured, load-bearing):** the **$/call crossover** (self-
hosted VLM vs Gemini) *** a real
142 VRAM/token + cost benchmark** on rented A100/L40S (ms-swift multimodal-GRPO examples
use 6?8 GPUs ? **budget is the
143 largest open risk and is currently unquantified**). Don't stand up an owned serving
stack until these clear.
144
145 ---
146
147 ## Decisions / status
148 | # | item | status |
149 |---|---|---|
150 | 1 | Greenlight Phase-0 milestone(s) ? rec: **M0.1 corpus+event-index**, then **M0.4
harness** | awaiting owner |
151 | 2 | WASB C3 ? first-class action item (own-weights / clean swap) | **owner: yes** ? |
152 | 3 | Ratify licensing **MIT / Apache-2.0 / BSD** (code+data+weights) | **owner: yes** ?
|
153 | 4 | Collector = corpus + event-store system-of-record | **owner: confirm?** |
154 | 5 | Rename repo off "badminton" | **owner action** ? |
155 | 6 | Conversational-QA: build seam now, punt model | **rec accepted?** |
156
157 ## Open risks / unknowns (none block Phase-0 research)
158 Budget/cost to reach Gen-2 **unquantified** (largest risk) · ship-gate "very high P/R"
target undefined · per-sport clean
159 trajectory detectors not yet identified · base-model pretraining provenance unauditale
(risk-acceptance) · real VLM
160 VRAM/token cost · single-model-multisport-temporal generalization unproven · DPA + no-

```



```
train (minors-excluded) for cloud
161     training · on-device 4B latency on 8 GB unmeasured.
162
163     ---
164     *Design-only and reversible. Nothing starts until you greenlight a milestone. Sequencing
discipline holds: the committed
165     next platform slice remains the async job model (single-tenant); only the M0.1?M0.4
foundation is safe to start.*
166
```

6. user (2026-06-10T05:59:41.097Z)

```
1     # Roadmap ? from annotated data to a self-improving, offline rally segmenter
2
3     This document is the narrative thread: how the indexer evolves from an
4     AI-API-dependent prototype into a self-contained tool that learns rally detection
5     from its own commercially-licensed, annotated video corpus. It explains how we
6     got here, why each step matters, and how it shapes the final tool. Decisions
7     are recorded as ADRs in [DECISIONS.md](DECISIONS.md); the scoring mechanics live
8     in [EVALUATION.md](EVALUATION.md).
9
10    ## The thesis
11
12    The indexer detects badminton rallies in long recordings. Its strongest mode
13    (`yolo_hybrid`) is a two-stage pipeline: cheap permissive person-detection
14    player-motion filtering (Stage 1, local ? torchvision Faster R-CNN + ByteTrack
15    tracking) followed by Gemini semantic refinement (Stage 2, paid API). That
16    works, but it costs money, needs the network, and never improves on its own.
17
18    > Naming note: the `yolo_hybrid` registry key is retained for back-compat, but
19    > AGPL YOLOv8 was removed and replaced with a BSD/MIT-licensed torchvision +
20    > ByteTrack stack ? see ADR-005. "YOLO" in this doc means the local
21    > player-motion gating stage, not the Ultralytics library.
22
23    A sibling project, sports-data-collector, curates sports videos with
24    per-rally `[start, end)` annotations and a commercial-license flag. The thesis of
25    this roadmap: use that annotated, commercially-usable data to (1) measure the
26    pipeline objectively and (2) train an offline replacement for the paid Gemini
27    stage ? turning the tool into one that improves as more data is curated, runs
28    air-gapped, and generalizes across sports.
29
30    Why it matters: it removes the per-run API cost and network dependency, gives us
31    a defensible quality metric instead of eyeballing, and creates a flywheel ?
32    more curated matches ? a better offline segmenter ? better highlights.
33
34    ## The four phases
35
36    ### Phase 1 ? Bridge the repos (DONE)
37    A sport-agnostic `curate export` in the collector emits the indexer's `start,end`
38    ground-truth CSVs plus a `manifest.json` pairing each CSV with its video, gated on
39    the `is_commercial` flag. ? Why: one clean producer/consumer contract that works
40    for all five supported sports (badminton/tennis/cricket/pickleball/table-tennis) alike.
41    See ADR-002.
42
43    ### Phase 2 ? Acquire full-resolution video (DONE)
44    The corpus was 256x144 (the original crawl used `worstvideo`). The `fetch` command
45    re-acquires every video at best available resolution (1080p here), in place
46    (preserving annotations), via authenticated Firefox Premium cookies and yt-dlp's
47    `default` client. ? Why: YOLO/shuttle detection needs real pixels; and the
48    journey surfaced non-obvious traps (SABR streaming ? 144p; Chrome DPAPI; rate
49    limiting) now encoded in the tooling. 22/22 fetchable matches are now 1080p.
50    See ADR-003.
51
52    ### Phase 3 ? Evaluate + tune (IN PROGRESS)
53    `run_phase3_eval.py` walks the manifest, processes each video, and scores detected
```

rallies against the ground truth ? per video and corpus-aggregate (precision/recall/
F1, mean IoU, boundary error), for both the `candidates` (raw detector) and `final`
(refined) stages. ? *Why:* the candidates-vs-final split tells us *where* error
lives ? a **detection** problem (candidates miss rallies) vs. a **segmentation**
problem (candidates catch them, refinement is loose). That diagnosis decides where
effort goes, and establishes the baseline any trained model must beat.

Phase 4 ? Learn an offline, API-free segmenter (NEXT)
Replace the paid Gemini refinement with a **lightweight classifier trained on the
rally annotations**, consuming features the pipeline already computes and stores
(`candidate\_segments.stats`). No external API, no new labeling. ? *Why:* this is the
payoff ? a free, offline, self-improving rally mode. The **substrate was chosen from
data**: the ADR-004 update (2026-06-02) measured the YOLO/player-motion candidate
stage at **recall ~0.05 @ IoU 0.5** on `shuttle-set\_8` (median best coverage of a
rally only ~20%) ? too low a ceiling to build on. So the offline segmenter's
substrate is **WASB shuttle-motion** (ADR-001), not player-motion. See **ADR-004**.

Complementary local cues (beyond the segmenter swap)

Once the substrate is WASB shuttle-motion, the next lever is **fusing additional
fully-local cues** to raise rally recall/precision without reintroducing a paid
API. The first one explored is **audio "thwack" hit-detection** (**ADR-007**,
accepted as a direction 2026-06-07): a pure-numpy spectral-flux onset detector in
`backend/pipeline/audio/`. The **E1 GO/NO-GO probe** has run on real GoPro audio ?
verdict **AMBER / leaning-negative**: recall signal is real (hit-clusters recover
WASB-missed rallies) but the classical-onset precision ceiling (~2.2x) is too weak
to fuse cleanly, and a band-pass doesn't rescue it. The only path past that is a
learned timbre classifier ? which, like Phase 4, is **gated on golden labels**.
See `AUDIO\_E1\_FINDINGS.md`. Pose/court serve-gating is parked as the #2 cue.

The recurring blocker across Phases 3?4 and the audio work is the same: **more
golden labels** is the single highest-leverage move (see the cross-cut note below).

How this shapes the final tool

- **Offline & free by default.** The production rally mode becomes a local model;
Gemini is kept only as an upper-bound reference, not a dependency.
- **Self-improving.** Every newly curated, annotated match is more training signal;
re-training is a local, deterministic step.
- **Measurable.** Every change is scored against the golden set, so "better" is a
number, not a vibe.
- **Sport-generic.** The label ? feature ? classifier machinery and the export
bridge already span sports; adding a sport is data, not new architecture.
- **Commercially clean.** Only `is\_commercial` data flows through the bridge, so the
trained model and any derived product rest on a licensable foundation.

Status at a glance

Phase	What	Status
1	Export bridge (`curate export` ? manifest + CSVs)	? Done
2	1080p acquisition (`fetch`, in-place, cookies)	? Done (22/22)
3	Eval + tuning loop (`run_phase3_eval.py`)	? In progress
4	Offline learned segmenter (WASB shuttle-motion substrate)	? Next (substrate decided: WASB, ADR-004)
+	Audio hit-detection cue (`backend/pipeline/audio/`, ADR-007)	? E1 probe run ? AMBER, gated on golden labels

> **Cross-cutting:** golden-set ground truth (which powers Phases 3?4 and on-demand
regression testing)

> is produced via a vendor process + pluggable annotation provider ? see
GOLDEN_SET_VENDORING.md and **ADR-006**. As of 2026-06-02
the annotation

> guideline is authored (collector `docs/annotation/`) and **Roora** (Bengaluru) is
selected for an

```

114 > initial **3-sport pilot** (badminton, pickleball, table tennis). Input-capture quality
requirements
115 > that bound achievable precision/recall are tracked in
[VIDEO_RECORDING_GUIDELINES.md](VIDEO_RECORDING_GUIDELINES.md).
116 >
117 > **Observability:** every processed video now emits an always-on per-video diagnostics
118 > report (spec-driven warnings/errors with FAQ refs, soft-dependency on `obsreport`), so
119 > pipeline regressions and footguns surface without re-running eval ? see
120 > [CODE_MAP.md](CODE_MAP.md) §2 (reporting/observability).
121

```

7. user (2026-06-10T05:59:48.511Z)

```

1 # Deferred Hardening Plan ? Train/Eval Guardrails, Async Jobs, Packaging
2
3 **Status:** ? Design for review ? **no code yet**. Owner-requested follow-up to the
4 "config & identity consistency" tech-debt pass. Each item below becomes its own PR
5 only after this doc is approved.
6
7 **Why these three:** they were cataloged as out-of-scope during the consistency
8 pass because each is larger/design-heavy or needs the owner's machine. They are the
9 remaining high-value hygiene items not already done (config/structure/Result/logging/
10 reports) or in flight (detector abstraction, PR #49).
11
12 **Recommended sequence:** #7 Packaging (lowest risk, unblocks clean installs/CI) ?
13 #13 Train/eval guardrails (correctness of the data flywheel) ? #16 Async jobs
14 (largest, product-facing).
15
16 ---
17
18 ## 1. Packaging modernization (debt #7)
19
20 ### Problem
21 - No `pyproject.toml` on `master`. (PR #49's description lists packaging as "done",
22 but it isn't present ? reconcile with the owner.)
23 - The root `setup.py` is a **diagnostics/dependency-checker**, not a PEP 517 build
24 script ? its name collides with setuptools semantics and will confuse `pip`/`build`.
25 - Deps live only in `requirements.txt`; `obsreport` is a commented soft dep
26 (BACKLOG) and `supervision<0.30.0` is pinned (debt #14).
27
28 ### Approach
29 - Add `pyproject.toml` (PEP 621): project metadata + runtime deps mirroring
30 `requirements.txt`, with optional-dependency groups:
31 - `[project.optional-dependencies] dev` ? `pytest`, coverage tooling.
32 - `reporting` ? `obsreport` pinned to its git tag (`v0.1.3`, per BACKLOG).
33 - `gpu` ? torch/torchvision (note: the cu124 `--extra-index-url` is **not**
34 expressible in standard PEP 621 deps; document the install line or use a
35 `[tool.uv]/pip-conf` approach rather than faking it).
36 - Console entry points: `indexer = backend.main:main` (so `indexer --server` works),
37 optionally eval drivers.
38 - **Rename the diagnostic `setup.py`** (e.g. `scripts/doctor.py`) so it no longer
39 shadows the build system; update README/`run_all_tests.py`/CI references.
40 - Choose a build backend (hatchling recommended) and confirm the `backend/` import
41 root stays valid (keep current layout; no src/ move required).
42
43 ### Risks
44 - `setup.py` rename touches docs and any muscle-memory invocations ? grep first.
45 - Torch CUDA wheels can't be pinned cleanly in PEP 621; must document the
46 `--extra-index-url` install path (the current `setup.py` logic can move into docs).
47 - CI must switch to `pip install -e .[dev]`.
48
49 ### Verification
50 - `pip install -e .[dev]` in a clean venv ? `run_all_tests.py` green.
51 - `indexer --help` resolves via the entry point.
52

```

```

53     ---
54
55     ## 2. Train/eval separation guardrails (debt #13)
56
57     ### Problem
58     Separation is by convention only. `backend/eval/calibrate_local.py` and
59     `calibrate_wasb.py` do honest leave-one-video-out (LOVO), but nothing *enforces*
60     that a video used to train the learned segmenter
61     (`backend/eval/training.py::build_training_set` ? `distill_local`) is absent from the
62     eval/regression set. Leakage would silently inflate the very metrics the product
63     story rests on (? in BACKLOG).
64
65     ### Approach
66     - **Split unit = match, never clip** (per `GOLDEN_SET_PHASE1_VIDEOS.md`: split by
67       match, hold ~? of each sport as eval).
68     - Add a `split: "train" | "eval"` field to the collector's `curate export`
69       manifest (cross-repo contract, ADR-002 lineage) ? or a separate eval manifest.
70     - New `backend/eval/splits.py`:
71       - `load_split(manifest) -> (train_ids, eval_ids)` keyed by **match id** (not just
72         file MD5 ? re-encoded duplicates share a match but differ in
73         `compute_video_id`, so group at match level).
74       - `assert_disjoint(train_ids, eval_ids)` ? raise on any overlap.
75     - Wire the guard into the training entrypoint (`distill_local`/`training`) and into
76       `run_regression.py` so the baseline is computed strictly on the eval split.
77
78     ### Risks
79     - Requires a collector-side manifest change (coordinate the cross-repo contract).
80     - Small corpus ? tiny held-out set ? high metric variance; the guardrail is correct
81       but the numbers will be noisy until the golden set grows.
82     - MD5-only dedup is insufficient (re-encodes evade it) ? must track match identity.
83
84     ### Verification
85     - Unit test: overlapping train/eval ids ? `assert_disjoint` raises.
86     - Integration: a deliberately-leaked manifest fails the training entrypoint.
87
88     ---
89
90     ## 3. Async job model (debt #16)
91
92     ### Problem
93     `POST /api/process` runs validate ? segment ? report **inline**, blocking the
94     request for the full duration of a long video. Unworkable as a hosted/product API.
95
96     ### Approach (minimal first slice ? single self-hosted box)
97     - `POST /api/process` returns **202 + `job_id`**; add `GET /api/jobs/{job_id}`
98       returning `{status, progress, result, reports}`.
99     - Run the pipeline on a background worker ? start with a `ThreadPoolExecutor`
100       (segmenters already use threads internally for AI handoff, so this fits); leave a
101       real queue (RQ/Celery) for later if multi-process is needed.
102     - Persist job state in the existing SQLite DB via the migration ladder
103       (`backend/infrastructure/database.py::_init_db`, bump `PRAGMA user_version` 1 ? 2,
104       add a `jobs` table). The observability report becomes the job's artifact.
105     - Update the static UI to poll `GET /api/jobs/{id}` instead of blocking.
106
107     ### Risks
108     - **SQLite thread-safety:** verify `check_same_thread`/connection-per-thread;
109       `get_connection` already re-issues `PRAGMA foreign_keys=ON` per connection ?
110       confirm it's safe under the executor.
111     - Cancellation/timeout and crash recovery (a job in RUNNING after a restart).
112     - In-process store is not horizontally scalable ? acceptable for the current
113       single-box self-hosted target; call it out so it isn't mistaken for production-HA.
114
115     ### Verification
116     - Submit a job ? poll ? DONE with the same segments the sync path produced.
117     - Server restart mid-job leaves a recoverable/clearly-FAILED job, not a hang.

```

```

118
119 ---
120
121 ## Out of this doc
122 - **Detector/WSL abstraction (#6):** in flight (PR #49).
123 - **Unify hybrid segmenters (#7 in LOW_REGRET):** owner-deferred.
124 - **ByteTrack `supervision<0.30.0` migration + real-footage validation (#14/#15):**
125     need a GPU/WSL box and real video ? owner-machine work, tracked separately.
126

```

8. user (2026-06-10T05:59:49.113Z)

```

1      # Backlog ? deferred items to revisit
2
3      Single tracker for work we consciously deferred, so nothing gets lost between sessions.
Each item records **what**, **why it's deferred**, and **the trigger to revisit**. Every entry
traces to a real decision/PR/doc (no speculative work).
4
5      > Maintained as a version-controlled doc for now. Say the word and I'll mirror any/all
of these to **GitHub issues** instead (or in addition).
6      > Status legend: ? blocks something · ? do when triggered · ? nice-to-have/cleanup.
7
8      _Last updated: 2026-06-07._
9
10     ---
11
12     ## Monetization / licensing
13     *(source: `archives/MONETIZATION_AUDIT.md` §7, ADR-005)*
14
15     - ? **Ultralytics Enterprise pricing** ? get a quote **only if** the
torchvision/ByteTrack detector (PR #8) proves insufficient on real footage. Currently *not
needed* (swap done). Quote-only pricing. **Revisit:** if detector quality forces reconsidering
YOLOv8.
16     - ? **Codec patent thresholds (AVC/HEVC/AAC)** ? confirm exact Via-LA (AVC) / Access
Advance (HEVC) royalty rates & free thresholds **with legal counsel** before relying on the
"small SaaS is under threshold" reading. Mitigation already chosen: emit royalty-free AV1/Opus
output. **Revisit:** before commercial launch.
17     - ? **Per-checkpoint weight/dataset licenses** ? if we borrow pretrained weights (YOLOX,
ActionFormer, WASB, TrackNetV3), the *code* license ? the distributed *weights/dataset* terms.
Either train our own or confirm each checkpoint. **Revisit:** when adopting any borrowed
weights.
18     - ? **Real GPU $/video & latency** ? benchmark the three LLM-step options (offline
ActionFormer vs self-hosted Qwen2.5-VL-7B vs Gemini paid-tier) on the target VM before pricing
the product. **Revisit:** when choosing the production LLM step.
19
20     ## Detector swap follow-ups
21     *(source: PR #8, ADR-005)*
22
23     - ? **`supervision` ByteTrack deprecation** ? `sv.ByteTrack` is deprecated as of
supervision 0.28.0 and **removed in 0.30.0** ; `requirements.txt` is pinned `<0.30.0` as a
stopgap. Migrate to `ByteTrackTracker` from the separate `trackers` package before lifting the
cap. **Revisit:** before bumping supervision past 0.30.
24     - ? **Real-video validation of the swap** ? the torchvision + ByteTrack Stage-1 is unit-
tested with mocks but **not yet validated on actual footage** for detection/tracking quality.
(ADR-004 separately found player pixel-motion is a weak rally proxy on broadcast video, so
expect tuning.) **Revisit:** when a real test clip + the pipeline are run together.
25     - ? **`yolo_hybrid` naming** ? the registry key, `yolo_` config keys, and `yolo-
hybrid-` DB detector tags keep the "yolo" name for back-compat despite no YOLO remaining.
Optional rename. **Revisit:** if/when a clean break is worth the migration.
26
27     ## Golden-set subproject
28     *(source: `docs/GOLDEN_SET_IMPLEMENTATION.md`, `docs/GOLDEN_SET_VENDORING.md`)*
29
30     - ? **Annotation guideline** ? **DONE (2026-06-02):** authored as the Roora brief +
illustrated guide in the collector repo `docs/annotation/` ? rally CSV schema; ignore dead time

```

```

/ include every rally (incl. faults & lets) / shot = one contact; `ending_reason`
(forced/unforced-error taxonomy, shared across net-separated sports); singles & doubles for
badminton & pickleball.
31 - ? **Internal gold key** ? hand-annotate 5?10 licensed clips to near-perfect as the
vendor answer key. **Unblocked.** **Revisit:** anytime.
32 - ? **GS-3 ? `HumanVendorProvider`** ? concrete adapter for the chosen annotation vendor
(submit/poll/fetch + DPA/secure-link controls). **Vendor selected for the initial pilot: Roora**
(Bengaluru; iMerit / Taskmonk / TELUS remain the fallback shortlist). **Revisit:** once the
Roora pilot confirms quality, then build the adapter.
33 - ? **GS-4 oracle model** ? `AutomatedProvider` interface is stubbed (this work) with a
deterministic fake. Wire a *concrete* fast-tier auto-labeler later. **Oracle rule:** it MUST be
a different model lineage than production, or the regression test is blind to shared failures.
**Revisit:** when a production LLM step is chosen (pick the oracle's lineage to differ).
34 - ? **GS-5 ? regression triggers** ? wire `regress()` into CI (pre-merge gate), on-
demand CLI (the "surprise" case), and a scheduled run; include a deliberate re-baseline path.
**Revisit:** after GS-4.
35 - ? **Vendor pilot (Roora)** ? run the paid pilot with **Roora** as a **single-vendor
pilot first** (not a multi-vendor bake-off): assess quality-per-rupee against the $4 bar;
escalate to the iMerit/Taskmonk fallback only if it misses. **Revisit:** the pilot doubles as
calibration once the gold key exists.
36 - ? **Phase-1 video collection** ? gather the round-1 golden corpus per
`docs/GOLDEN_SET_PHASE1_VIDEOS.md`: **static self-recorded** footage (mirror deployment, not
broadcast), 3 sports incl. singles & doubles, across the diversity matrix; ~12 clips/sport with
**? held out by match**. **Unblocked.**
37 - ? **Train/eval separation guardrails** ? train & eval videos live in **separate
catalogs** today, kept apart by convention only. Add **hard guardrails** (permissions / explicit
approval / re-confirmation at ingest & export) so a video can **never** land in both train and
eval. **Revisit:** dedicated design session with the user.
38
39 ## Recording best practices
40 *(source: `docs/VIDEO_RECORDING_GUIDELINES.md`)*
41
42 - ? **Validate recording hypotheses** ? empirically test which capture conditions
(resolution, fps, camera angle/height, lighting) actually move precision/recall, then promote
the winners into user-facing "how to record" guidance. Only the static-camera rule is currently
evidence-backed (ADR-004). **Revisit:** once a controlled corpus + the eval loop can measure
per-condition deltas.
43
44 ## Code & test organization
45 *(source: owner review comments on PR #35, 2026-06-07)*
46
47 - ? **`tests/` reorganization to reduce fragmentation** ? owner wants to isolate test
components on a grouping
48 metric (by subsystem/component) so `tests/` is less fragmented, mirroring the new
`backend/pipeline/audio/` +
49 `backend/eval/audio/` layout (e.g. audio tests grouped under `tests/audio/`, eval
tests under `tests/eval/`).
50 The exact grouping **metric is the owner's design choice** (consolidate small files
vs. add per-area dirs ? the
51 "lesser fragmentation" goal slightly tensions with adding dirs), so this is left as a
**dedicated refactor PR**,
52 not guessed here. **Revisit:** when the owner specifies the grouping scheme. (The
`backend/eval/audio/` half of
53 the request ? comment 1 ? was done in PR #35.)
54
55 ## Commercial Readiness & Low-Regret Moves ? Approved Decisions Checkpoint (2026-06-07)
56
57 *(source: Commercial Readiness Review + structure/sequencing discussion with Grok as
second-in-command)*
58
59 - ? **Docs-only checkpoint PR** ? This update + the refreshed
`docs/LOW_REGRET_MOVES_PLAN.md` formally record the approved path before any code changes. The
initial review docs (PR #37) introduced the findings; this entry locks the execution decisions.
60 - ? **Approved directory structure** ? Adopt a layered approach under `backend/` for
professionalism and future growth (no flat dumping of new modules, no explosion of 15+ single-

```

```

file dirs):
61     - `backend/config/` (package ? start with this)
62     - `backend/api/` (FastAPI app, routes, static UI)
63     - `backend/pipeline/` (validators + segmenters + stitcher + audio + detectors)
64     - `backend/infrastructure/` (database + external adapters)
65     - Keep `backend/eval/` prominent as a first-class concern.
66     - Logical grouping for providers, sports, annotations, utils, and a small `core/` only
when needed.
67     - ? **Sequencing (Config first, then atomic refactor)** ? Implement Deliverable 1
(Config) **first**, introducing the `backend/config/` package immediately. Perform the full
directory structure refactor as a **separate high-pri atomic PR** afterwards. This minimizes
conflicts, keeps PRs focused, and gives Claude a clean window with no concurrent moves.
68     - ? **Deliverable 1 (Config)** ? `backend/config/` package (`models.py` + `loader.py`)
instead of a single `config.py`. Typed `AppConfig`, central loader with built-in defaults + file
override, `extra="allow"` for migration. Update `setup.py` + `main.py`. Add tests. Land this
before the layout move.
69     - ? **Subsequent items** ? See the refreshed `LOW_REGRET_MOVES_PLAN.md` for the full
ordered list (Error contracts, logging, packaging, detector abstraction, etc.). The structure
refactor is now explicitly tracked as its own high-pri step right after Config.
70
71     **Status legend reminder:** ? = approved / in progress tracking, ? = do when triggered.
72
73     ## Observability reports ? shipped; intersections with the low-regret items (2026-06-07)
74     *(source: per-video observability feature ? indexer PR #41/#42/#43, collector PR
#11/#12; shared lib `obsreport`)*
75
76     - ? **Low-regret items 1 (Config pkg, PR #39) + 2 (directory restructure, PR #40) are
DONE.** Remaining = items 3?9 in `LOW_REGRET_MOVES_PLAN.md`.
77     - ? **Per-video observability reports shipped in BOTH tools** (a TECHNICAL report + a
CUSTOMER-friendly summary), via the **soft-dependency** lib `obsreport`
(github.com/avidullu/sports-obsreport, MIT, v0.1.3). Each run writes `**Item 8 (Input-Quality UX) is LARGELY SUBSUMED** by these reports (validation stage
+ footgun flags + verdict already surface quality; a clean customer summary already exists).
**Do NOT build a parallel mechanism** ? instead EXTEND the report: add an audio-readiness
stage/flag to `backend/reporting/` (harvest) + the obsreport spec. **Revisit:** when item 8 is
picked up.
79     - ? **Item 7 (unify hybrid segmenters) should land before the report's real AI-provider
timing.** The report records AI timing as `not_captured`; once `wasb_hybrid`/`tracknet_hybrid`
are unified there is ONE place to thread a recorder callback through `provider.analyze_video` ?
the report. **Revisit:** with item 7 (Claude to add the timing hook there).
80     - ? **Item 3 (Error/Result contract) synergizes with the report.** Today the report
INFERS failures (e.g. `silent_provider_noop` = 0 segments + no API key). A structured
Result/error from providers/segmenters would let the report read the real reason as a first-
class flag. **Revisit:** with item 3.
81     - ? **`obsreport` is a SOFT/commented dep** in `requirements.txt` (works locally via
editable install; pipeline no-ops if absent). **Pin the `git+ssh@v0.1.3` tag** in both repos
once SSH/CI access to the private lib repo exists.
82     - ? **`run_all_tests.py`** added at repo root ? run the full suite (clear PASS/FAIL,
pytest exit code) before sending PRs.
83     - ? **Config-migration follow-ups (DONE ? config & identity consistency pass)** ? the
two typed-config bugs from
84     #39/#40 are fixed: (1) **`/api/compile` now honors `--output-dir`** and (2)
**`output_dir` is a real field on
85     `OutputConfig`** (default `"output"`); the CLI overrides it on the typed model in
`main()`, retiring the
86     write-only `_runtime_overrides` hack. Same pass also unified `video_id` MD5 derivation
into
87     `backend/utils/hashing.py::compute_video_id` and migrated all four `run_*.py` scripts
off raw `json.load` onto
88     `backend.config.load_config`. See `CODE_MAP.md` §2.0/§6 (footguns retired).
89

```

```

1      # Commercialization Tracker
2
3      **Status:** ? LIVING ? track this in parallel with product/quality work.
4      **Owner:** avidullu . **Last refreshed:** 2026-06-07
5
6      **Purpose.** Single source of truth for the **commercial** dimension of the project:
7      licensing/legal posture, codec & output economics, hosting unit-economics, the
8      golden-set vendoring spend, packaging/distribution, and the product-market-fit
9      risks that gate monetization. Updated as decisions land ? see the
10     [Changelog](#changelog) at the bottom.
11
12     **Merged from / supersedes:**
13     - `archives/MONETIZATION_AUDIT.md` (2026-05-30 deep-research licensing & economics
14       audit) ? kept as the **research-of-record** (per-claim verification transcript,
15       primary sources). This doc carries its *current* status forward.
16     - `archives/COMMERCIAL_READINESS_REVIEW.md` (code-health + PMF assessment) ?
17       archived; its commercial findings + the quality-phase priorities live here now.
18
19     **Premise (unchanged, verified):** the product is a **closed-source hosted API** ?
20     users upload videos to a cloud NVIDIA GPU VM and get highlight reels back. We never
21     distribute code/binaries, so GPL/LGPL *distribution* copyleft is not triggered ? but
22     **AGPL §13 (network clause)**, **metered-API terms**, and **video-codec patent
23     royalties** all still apply.
24
25     ---
26
27     ## 1. Commercialization dashboard
28
29     Legend: ? done . ? in progress / watch . ? blocker / needs action . ?? needs legal
30     counsel . ? future
31     Priority: **P0** (do first) . **P1** . **P2** (lowest). Rows are grouped **Pending ? In-
32     flight ? Completed**, each sorted by priority. (C# IDs are stable identifiers ? they don't imply
33     order.)
34
35     | # | Workstream | Pri | Status | Where it stands | Next action / owner decision |
36     |---|---|---|---|---|---|
37     | | **? PENDING (needs action / not started)** | | | |
38     | C5 | **Output codec royalties** | **P0** | ? | Highlights are still encoded **H.264 +
39     AAC** (`libx264` in `backend/pipeline/stitcher/compiler.py`) ? encode-side AVC patent exposure
40     remains. | Switch output to **AV1 + Opus** (royalty-free) on an **Ada-class GPU (L4/L40)** for
41     HW encode. See §4. Decision: provision Ada-class GPU? |
42     | C9 | **Packaging / distribution** | **P0** | ? | No `pyproject.toml`; the diagnostic
43     `setup.py` shadows the build system. | Design in `DEFERRED_HARDENING_PLAN.md` (#7) ? **awaiting
44     your approval** before code. |
45     | C12 | **SaaS foundations (async/auth/tenancy)** | **P1** | ? | Sync `/api/process`
46     blocks on long videos; no auth/tenancy. | Multi-user/platform strategy in
47     `PLATFORM_ARCHITECTURE.md` (pluggable storage/compute, tenancy, orchestration); async slice =
48     `DEFERRED_HARDENING_PLAN.md` (#16). |
49     | | **? IN-FLIGHT (in progress / watch)** | | | |
50     | C4 | **Runtime AI strategy (A/B/C + owned model)** | **P0** | ? | Live = **(C) Gemini
51     2.5 Flash, paid tier**. Long-term = **own a self-hosted generative video-QA model** (fine-tuned
52     open VLM) as primary, **paid Gemini always-ready as fallback** via the provider registry. | Make
53     "provider registry + paid fallback" a core principle; build the owned model **after** the
54     scaffolding/flywheel are solid. See `PLATFORM_ARCHITECTURE.md` §5A. |
55     | C13 | **Beachhead validation** | **P0** | ? | Market research
56     (`PMF_MARKET_RESEARCH.md`) recommends **India badminton academies** as the #1 beachhead.
57     Unvalidated. | Cheapest tests: 10?15 academy/coach discovery interviews; rally-detection P/R
58     benchmark on 100+ real amateur clips; fake-door pricing (B2C ~$99/yr, B2B academy ~$600/yr). See
59     §6. |
60     | C8 | **Golden-set vendoring spend** | **P1** | ? | ADR-006: **Roora (Bengaluru)**
61     selected for an initial paid pilot; licensed/owned video only (no user uploads to vendors). |
62     Confirm quality-per-dollar from the pilot; then GS-3 `HumanVendorProvider`. See
63     `GOLDEN_SET_VENDORING.md`. |
64     | C3 | **Shuttle-tracking weights provenance** | **P1** | ? ?? | WASB-SBDT **code is MIT
65     (clear)**; distributed **checkpoints are academic-data-trained** ? not covered by the MIT code

```


grant. MonoTrack/YOLO-NAS banned (noncommercial). | Train-own weights (Phase-4 roadmap) ****or**** confirm per-checkpoint terms ?? before commercial deploy. |

43 | C14 | ****Owned-model base licensing + training-data policy**** | ****P1**** | ? ?? | Owned video-QA model (PLATFORM §5A) must build on ****Apache-2.0-clean, video-capable**** bases ? ****Qwen2.5-VL-7B/32B**** (Apache; 3B/72B are ***not***), ****Qwen3-VL-4B/8B****, or ****Gemma 4**** ? to keep weights private, gate, and monetize. ****Reject Gemma 3**** (viral derivative clause ? permanent Google PUP), Llama Vision (naming + 700M-MAU), ****gpt-oss**** (text-only). ****Hard rule: never train on paid-LLM (Gemini/OpenAI/Anthropic) outputs**** (competing-model clause). The moat is the ****consented dataset + eval harness****, not the model. | Lock to Apache-2.0; ****open-model teachers only****; separate explicit ****training opt-in****; ****exclude minors**** from the training pool; per-clip consent ledger. See PLATFORM §5A/§7. |

44 | C6 | ****Inbound HEVC decode**** | ****P2**** | ? ?? | Decoding users' GoPro H.264/****HEVC**** uploads is unavoidable; HEVC pools are fragmented + highest-uncertainty. | Low/uncertain residual; confirm decode-side exposure with counsel before launch. |

45 | C7 | ****Hosting unit-economics**** | ****P2**** | ? | Per-video cost is ***cheap*** for Gemini (~\$0.05 input for a 10-min clip); risk is volume × latency, not unit price. Self-host/offline costs unmeasured. | ****Benchmark \$/video & latency on the target VM**** for A/B/C before pricing. See §5. |

46 | C10 | ****Value layer (product surface)**** | ****P2**** | ? | Minimal exports shipped (EDL/JSON/CSV + ****basic_stats****, PR #51). | Deepen: per-rally tags/notes that feed golden, per-player/match stats, clip gallery. See §6. |

47 | C11 | ****Input-quality enforcement**** | ****P2**** | ? | Per-video observability reports surface a customer verdict + footgun flags incl. audio-present (PR #50). | Turn more ****VIDEO_RECORDING_GUIDELINES**** dimensions into validators/report rules. |

48 | C2 | ****supervision**** ByteTrack pin | ****P2**** | ? | Pinned ****<0.30.0**** (deprecation ahead). | Migrate before lifting the pin (BACKLOG). Not a legal risk, a maintenance one. |

49 | | ****? COMPLETED**** | | | |

50 | C1 | ****AGPL / copyleft removal**** | ? | ? | ADR-005 shipped: AGPL ****ultralitics**** YOLOv8 replaced with ****torchvision Faster R-CNN (BSD) + supervision**** ByteTrack (MIT). No AGPL code/weights in the stack. | Done. Keep new deps permissive. |

51

52 ---

53

54 ## 2. Licensing & dependency posture (refreshed risk table)

55

56 Verdicts updated against the current stack (****requirements.txt**** + code). Original

57 2026-05-30 verdicts and the 25-claim verification transcript are in

58 ****archives/MONETIZATION_AUDIT.md****.

59

60 | Dependency | License | SaaS obligation? | Current verdict | Notes |

61 |---|---|---|---|---|

62 | ****ultralitics**** + ****yolov8n.pt**** | AGPL-3.0 | Yes (§13) | ? ****REMOVED**** | Replaced per ADR-005 ? no longer in the stack. |

63 | ****torchvision**** detectors (Faster R-CNN/RetinaNet) | BSD-3 | No | ? LIVE | The chosen replacement; person = COCO class (see ****yolo_hybrid.py**** ****_PERSON_CLASS_ID****). |

64 | ****supervision**** (ByteTrack) | MIT | No | ? LIVE | Pinned ****<0.30.0**** ? migrate before bump (C2). |

65 | ****WASB-SBDT**** (shuttle) | MIT (code) | No | ? code clear | Weights provenance open (C3). |

66 | ****MonoTrack / YOLO-NAS**** | noncommercial | ? | ? BANNED | Do not use. |

67 | ****Gemini API**** (paid tier) | service terms | Paid: no training/review | ? LIVE WATCH | Must bill the paid tier to keep user video private (C4). |

68 | H.264 / HEVC / AAC codecs | patent pools | royalties on encode/decode | ? ?? | Separate from ffmpeg's license. Encode-side open (C5); decode-side residual (C6). |

69 | ffmpeg/ffprobe, google-genai, torch, opencv, numpy, fastapi/uvicorn/pydantic/jinja2/dotenv | LGPL/Apache/BSD/MIT | No | ? CLEAR | Invoked as subprocess / imported; all permissive. |

70 | ****obsreport**** (reports) | permissive (private) | No | ? soft dep | Pin the ****git+ssh@vx**** tag once the repo is published (BACKLOG). |

71

72 ****Bottom line:**** the two hard blockers from the audit (Ultralytics; banned weights)

73 are resolved/avoided. Remaining commercial-legal items are ****codec royalties (C5/C6)****

74 and ****weights provenance (C3)****, plus the metered-API privacy posture (C4).

75

76 ---

```

77
78 ## 3. Runtime AI strategy ? A/B/C
79
80 The semantic rally-boundary step is the one external/metered dependency. Three paths
81 (from the audit), with current standing:
82
83 - *(A) Eliminate ? offline learned segmenter.* ActionFormer (MIT) / our own learned
84 classifier on `candidate_segments.stats`. **Best steady-state margin** (no per-call
85 cost, no external dep) and matches the Phase-4 roadmap (substrate = WASB
86 shuttle-motion, decided in ADR-004). **Most engineering.** ? *target*
87 - *(B) Self-host VLM.* Qwen2.5-VL-7B/32B (Apache-2.0) or InternVL2.5-8B (MIT). No
88 per-call fees; pays in GPU VRAM (~16?24 GB for 7B). ?? Qwen 3B is noncommercial.
89 - *(C) Keep Gemini 2.5 Flash (paid).* Fastest, cheap per video, video-native;
90 ongoing metered cost + external dependency. **? current live path.**
91
92 Hosted alternatives for (C) comparison: Anthropic Claude (no training on
93 inputs/outputs; customer owns outputs; **no native video** ? weaker here);
94 OpenAI (no training by default; ZDR available); Gemini remains strongest video-native.
95
96 ---
97
98 ## 4. Codec / output economics (the live gap ? C5)
99
100 **The trap:** ffmpeg being free (LGPL) does **not** grant the *patent* rights to the
101 codecs it implements. Royalties attach to the act of encode/decode.
102
103 - **Encode side (we control it):** highlights currently emit **H.264/AAC**. AVC (Via LA)
104 is first 100k units/yr royalty-free with a permanent "free to end-users over the
105 internet" provision ? a small SaaS likely falls under thresholds, but ?? confirm.
106 - **Mitigation:** emit **AV1 (SVT-AV1, BSD + AOMedia royalty-free patent grant) + Opus
107 (royalty-free)**. NVENC AV1 HW-encode needs **Ada-class GPUs (L4/L40)**; Ampere/Turing
108 must use **SVT-AV1 (CPU)**.
109 - **Decode side (unavoidable, C6):** inbound GoPro **HEVC** decode ? fragmented pools,
110 highest uncertainty; low residual.
111
112 **Recommendation:** provision an **Ada-class GPU (L4/L40)** and standardize highlight
113 output on **AV1 + Opus** ? removes the encode-side royalty question almost entirely.
114 *Implementation lives in `backend/pipeline/stitcher/compiler.py` (codec args).*
115
116 ---
117
118 ## 5. Hosting unit-economics (C7)
119
120 Reference workload: ~30-min 1080p match on a cloud NVIDIA GPU VM.
121
122 - *(C) Gemini 2.5 Flash (paid):** input $0.30/1M tokens, output $2.50/1M; video ~263
123 tok/s ? 10-min clip ? 158k input tokens ? **~$0.05 input**. Cheap per unit; the
124 economic risk is **volume x latency**. Flash-Lite is cheaper; Pro ~10x.
125 - *(B) Self-host Qwen2.5-VL-7B:** $0 API; cost = GPU-hours / throughput; needs ?24 GB
126 VRAM (L4/A10). $/video vs Gemini is **open ? measure**.
127 - *(A) Offline ActionFormer/learned:** cheapest at inference; cost = one-time training
128 + light GPU. **Best steady-state margin.**
129
130 > The A/B/C $/video comparison depends on GPU instance + batch throughput ?
131 > **benchmark on the target VM before pricing the product.**
132
133 ---
134
135 ## 6. Product-market fit & the quality/data phase
136
137 ### Beachhead (from market research, 2026-06-07 ? see
138 [`PMF_MARKET_RESEARCH.md`](PMF_MARKET_RESEARCH.md))
139 The static-camera constraint is a **B2B filter**: the buyer must already record from a
140 fixed angle, which points away from the median casual player and toward coaches/clubs.
141 Ranked beachheads:

```

141 1. ****Badminton coaching academies & serious players, India**** ***(strongest)*** ? 20M+
 142 players,
 143 formalizing academy ecosystem (Khelo India), genuine fixed-angle recording, coaches
 144 who already monetize, badminton whitespace vs incumbents. Sell to the ****academy****,
 145 not
 146 the individual; needs INR-tuned pricing.
 147 2. ****US/global pickleball players & clubs**** ? fastest-growing sport (19.8M US, +45.8%
 148 YoY)
 149 but ****SwingVision** already owns single-camera tennis/pickleball AI** (USD 179.99/yr) ?
 150 head-to-head, not greenfield.
 151 3. ****Badminton academies/clubs in East/SE Asia + Denmark**** ? highest density; China
 152 needs
 153 local hosting/partners; Denmark/EU small but high-WTP, easy to pilot.
 154
 155 Pricing anchors ***(estimate)***: software-only BYO-camera ****B2C USD 60?180/yr****, ****B2B**
 156 academy/club USD 300?1,500/yr** per location (undercuts hardware-bundled Veo/Pixellot/
 157 Spiideo; comparable to SwingVision software-only).
 158
 159 **### PMF risks (from the readiness review)**
 160 - ****Capture quality is the gate.**** Accuracy is capture-conditional; broadcast/handheld
 161 footage is a failure mode. The wedge (static, tripod, full-court) is defensible but
 162 narrows the addressable market ? enforce/guide it (C11, `VIDEO\_RECORDING\_GUIDELINES`).
 163 - ****Value beyond raw segments is table stakes.**** Time-saved editing alone is weak;
 164 per-rally tags, stats, and a correction loop are what make it sticky (C10).
 165 - ****Golden labels are the recurring blocker**** across the offline segmenter, audio
 166 fusion, and measurable quality ? the highest-leverage investment (C8).
 167
 168 **### Refined priority order (quality/data phase ? "focus on quality once new videos
 169 land")**
 170 Carried from the readiness review; commercialization items above are tracked in
 171 parallel.
 172
 173 1. ****Golden-set expansion + data-flywheel UX**** (highest with new labeled videos):
 174 ingest the new batch, expand calibration/CV/regression baselines, make label
 175 contribution + safe retraining easy and non-contaminating (eval/training separation).
 176 2. ****Hybrid unification + observability hooks**** (deferred item 7, now higher): extract
 177 shared logic from `tracknet\_hybrid`/`wasb\_hybrid` before heavy AI timing/metrics
 178 land.
 179 3. ****Detector/windowing quality iteration**** with new data: re-tune velocity/merge/
 180 min-duration + `rally\_gate`; validate WASB on real video; trial audio fusion
 181 (ADR-007).
 182 4. ****Deeper value layer**** (beyond minimal exports, C10): per-rally tagging/notes ?
 183 `candidate\_segments`, per-player/match stats, clip gallery, human-correction loop.
 184 5. ****Input-quality enforcement & guidance polish**** (C11): more guideline dimensions ?
 185 validators/report rules; "your video is good enough" UX.
 186 6. ****Multi-sport maturation + recording best practices.****
 187 7. ****SaaS / multi-user / async foundations**** (C12) ? lower until monetization is firm.
 188 8. ****Polish & distribution**** (C9): installer story, first-run path, codec/output,
 189 economics.
 190
 191 ****Recommendation:**** with the mid-week labeled-video batch, spend ~80% on #1 + #3
 192 (flywheel + quality iteration on real data), then unification (#2) and deeper value
 193 (#4).
 194 ---
 195
 196 **## 7. Open questions needing owner / counsel sign-off ??**
 197
 198 1. ****Ultralytics Enterprise price**** ? only relevant if a permissive detector ever proves
 199 insufficient (fallback). Quote-only.
 200 2. ****Per-checkpoint / dataset licenses**** for WASB / TrackNetV3 / any VLM ? train-own or
 201 confirm each (C3).
 202 3. ****Exact Via LA (AVC) / Access Advance (HEVC) rates & free thresholds**** ? confirm live
 203 figures with counsel before relying on the "small SaaS under threshold" reading
 204 (C5/C6).

```

195 4. **Real per-video GPU cost & latency** of self-hosted VLM vs Gemini paid at expected
196 throughput (C7).
197 5. **Provision Ada-class GPU (L4/L40)** for HW AV1 encode? (C5 ? gates the codec
switch.)
198
199 ---
200
201 ## Changelog
202
203 - **2026-06-07** ? Added the market-research-of-record (`PMF_MARKET_RESEARCH.md`):
204 ranked beachhead (India badminton academies #1), pricing benchmarks, and a C13
205 "beachhead validation" workstream. Wired the summary into §6.
206 - **2026-06-07** ? Created this living tracker. Merged the 2026-05-30 monetization
207 audit + the commercial-readiness review. **Refreshed since the audit:** C1 AGPL
208 removal now ? shipped (was the #1 blocker); added C5 as a ? (output is still H.264,
209 the audit's AV1 recommendation is not yet implemented); C9 packaging tied to
210 `DEFERRED_HARDENING_PLAN.md`; reflected PR #50/#51 (input-quality reports, value-layer
211 exports). Archived `COMMERCIAL_READINESS_REVIEW.md`.
212
213 ---
214
215 ## See also
216 - `PMF_MARKET_RESEARCH.md` ? market-study-of-record (sizing, segments, competitors,
beachhead; cited).
217 - `archives/MONETIZATION_AUDIT.md` ? licensing/economics research-of-record (sources +
verification transcript).
218 - `archives/COMMERCIAL_READINESS_REVIEW.md` ? superseded assessment (history).
219 - `archives/decisions/DECISIONS.md` ? ADR-005 (permissive stack), ADR-006 (vendor
process).
220 - `ROADMAP.md` ? Phase-4 offline segmenter (the C4 "(A) eliminate" path).
221 - `DEFERRED_HARDENING_PLAN.md` ? packaging (C9), async (C12) designs (pending approval).
222 - `GOLDEN_SET_VENDORING.md` ? vendor process + Roora pilot (C8).
223 - `BACKLOG.md` ? `supervision` migration (C2), `obsreport` pin, codec follow-ups.
224

```

10. user (2026-06-10T06:00:04.534Z)

```

1  # Platform Architecture ? Multi-User "Assistant Coach"
2
3  **Status:** ? Strategy / research ? for review. No code yet.
4  **Date:** 2026-06-07 . **Owner:** avidullu
5
6  **Purpose.** Map the path from today's **single-user, local** tool to a **multi-tenant
7  "assistant coach" platform**: users upload videos, get highlights **stored and
8  browsable by history**, and choose **where their data lives** (Google Drive / cloud
9  bucket / local) and **where the compute runs** (our GPU / their VM). The organizing
10 principle is to **decouple the layers** ? identity, ingestion, storage, orchestration,
11 execution/compute, the detection+highlight pipeline, and data-serving ? behind small
12 **pluggable interfaces**, reusing the registry pattern the codebase already uses for
13 segmenters/providers/validators/sports/annotations.
14
15 **Product framing (the north star *and* the v1 line).** This is an **assistant coach**,
16 not just a clipper ? but the strategy is **phased and unambiguous about scope**:
17 - **v1 (concrete, near-term):** **rally detection ? highlight-reel compilation ? per-
user
18 history ? a coach/user correction loop that feeds the golden set** ("assistant-coach
19 *lite"). Essentially today's pipeline made multi-user ? this is the real v1
deliverable.
20 - **Mid-term:** **rich rally/shot semantic metadata** (shot type, rally outcome) and the
21 **owned video-QA model** (§5A; §6 P4?P5).
22 - **North-star vision (strictly future ? *not* v1):** **gait / biomechanics, per-athlete
23 identifying templates, and deep coaching feedback under a supervising coach** (§6 P6).
24 Inspiring, but explicitly deferred ? and the part that raises the **biometric**
privacy
25 bar (§7).

```



```

84         ? Storage (local upload | Google Drive | S3 | ?
85         ? GCS | Azure) ? raw video + highlight outputs ?
86         ?????????????????????????????????????????????
87     ...
88
89     The two new pluggable seams are Storage and Compute/Execution; everything
90     else either exists (pipeline) or is net-new infra (identity, orchestration, serving). A
91     model/training layer ? the owned generative video-QA model ($5A) ? sits atop these,
92     fed by Storage (training corpus), Data (labels), and Compute (training + inference), and
93     is delivered through the existing AI-provider registry with a paid fallback.
94
95     ## 3. Pluggable abstractions (extend the existing registry pattern)
96
97     The codebase already proves the model: an `ABC` + a `Registry` singleton with
98     `register/get/list_available` (e.g. `SegmenterRegistry`). Add two more registries with
99     the same shape so they're plug-and-play and testable in isolation.
100
101     ### 3.1 `StorageBackend` registry
102     Replace the bare `video_path: str` that flows through the pipeline with a
103     StorageRef (a backend id + a key/URI), resolved by a backend.
104
105     ```python
106     class StorageBackend(ABC):
107         name: str
108         def open_read(self, ref: StorageRef) -> BinaryIO: ... # stream large video
109         def download_to_local(self, ref: StorageRef, dst: str): ... # for ffmpeg/cv2/WSL
110         def upload(self, local_path: str, ref: StorageRef): ... # persist
111         def signed_url(self, ref: StorageRef, ttl: int) -> str: ... # hand to LLM / browser
112         def stat(self, ref: StorageRef) -> StorageStat: ...
113         def list(self, prefix: StorageRef) -> list[StorageRef]: ...
114     ...
115
116     - Recommended implementation: build the backend over fsspec (one matrix ?
117     local / S3 (`s3fs`) / GCS (`gcsfs`) / Azure (`adlfs`) + Google Drive via `gdrivefs`,
118     Dropbox via `dropboxdrivefs`), reaching for smart_open as the robust large-MP4
119     byte-streaming primitive. Storage becomes one more registry. (Treat Drive/Dropbox as
120     add-ons ? less battle-tested than `s3fs`/`gcsfs`; cover with integration tests.)
121     - Backends: `local` (upload dir), `gdrive`, `s3`, `gcs`, `azure`. BYO-storage:
122     the tenant registers their own bucket/Drive; raw video never leaves their account.
123     - Auth ? never persist users' long-lived keys. For buckets, have the user create an
124     IAM role you AssumeRole into, scoped to one bucket/prefix, and mint presigned
125     URLs so raw video moves browser ? the user's bucket directly, never through our
126     servers. For Drive, OAuth with the narrow drive.file scope; for academics, a
127     service account with domain-wide delegation. Store any per-tenant creds encrypted.
128     - Why it matters for the pipeline: ffmpeg/OpenCV/WSL need a local file, while
129     the LLM provider can take a signed URL. The backend exposes both `download_to_local`
130     (for compute) and `signed_url` (for the provider/browser), so the pipeline stops
131     caring where the bytes live.
132     - Migration path: introduce StorageRef alongside the current local path; a local
133     backend wraps today's behavior so nothing breaks; adapters land incrementally.
134
135     ### 3.2 `ComputeBackend` / `ExecutionBackend` registry
136     Where the pipeline actually runs for a given job.
137
138     - local ? in-process / local subprocess (today's behavior; dev + self-host).
139     - hosted ? our GPU VM pool (the SaaS default).
140     - byo_worker ? the tenant runs a pull-based worker (a small agent) on their
141     own VM/GPU. Critical security property: the worker pulls jobs scoped to its tenant
142     from the control plane, pulls input from the tenant's own storage, runs the
143     pipeline, and writes artifacts + results back ? so the orchestrator never holds the
144     user's cloud credentials and needs no inbound access to their network. This

```

mirrors the **self-hosted CI runner** pattern (GitHub Actions runners use a 50s outbound-only HTTPS long-poll). This pull-based runner is the **cheapest, most credential-safe v1** for BYO-compute.

- For users who want the platform to **provision GPUs in their own cloud account on demand** (an academy with an AWS account but no standing GPU box), add **SkyPilot** (BYOC: everything launches in the user's accounts/VPCs; creds stay with them). **Modal** is the managed fallback for users wanting zero infra (accepting that raw video then flows through Modal).
- **Operational / support burden (a real adoption risk).** For the primary beachhead ?

Indian badminton **academies/coaches** ? asking them to **run and maintain a worker on their own VM/GPU** is a meaningful ongoing load (updates, monitoring, firewall/NAT, the "it stopped after our IT change" ticket). So: **default academy users to `hosted`**, and

treat ``byo_worker`/SkyPilot` as a **power-user / enterprise / privacy-sensitive tier**, not the academy default. If/when BYO ships, plan for **auto-updating, self-healing agents + health heartbeats + clear worker status in the UI**, and budget the support cost.

- **Middle tier to consider ? "zero-infra BYO":** a managed-but-isolated path where the **storage** stays in the user's bucket (signed URLs) while compute runs on **our GPUs** in a **strictly isolated, per-tenant, ephemeral sandbox** (nothing persists our side). This gives the BYO **privacy** benefit (raw video never persists on our infra) **without** the user running a worker ? a pragmatic default between full-hosted and full-BYO.

The pipeline code is unchanged; only **where it's invoked** and **how it gets bytes** vary.

3.3 Reuse, don't reinvent

The existing ``segmenter` / `provider` / `validator` / `sport` / `annotation`` registries are the proven precedent and stay as the **pipeline** layer. The ``annotation`` registry + ``candidate_segments`` table are also the natural seam for the **coach-correction / golden-set** loop in the assistant-coach phase (\$6, P4).

4. Multi-tenant data model & serving layer

4.1 Entities ? generic core, with the coach model as an optional overlay

The foundational schema is deliberately **domain-neutral** so we don't overfit to the coach mindset ? the coach/athlete framing is a **market-fit hypothesis** that will iterate

(see C13 / PMF), not a load-bearing assumption. **Generic core:**

```

User ??< VideoCollection ??< Video ??< Highlight ??< HighlightMetadata
      ???< PlaySegment ??< Event
Video / Highlight ??< Job (processing record: status, compute backend, timings)

```

- **Generic & re-targetable:** a ``VideoCollection`` groups a user's videos (a season, an event, a "folder") with no assumption of a coach or academy ? a power user, a coach, or a parent are all just ``User`s` with collections.
- **Highlights are first-class** (today compile only writes a file): a ``highlights`` table (segment ids + output ``StorageRef`` + params + created_at) powers "see past videos and highlights by history," and ``HighlightMetadata`` carries the **rich semantic tags** (shot type, rally outcome, etc. ? see \$6 metadata stage) plus titles / ratings / notes.
- **`video\_id`** stays the content MD5 (dedup within a tenant), scoped by ``tenant_id``.
- **Coach overlay (optional, additive):** for the academy/coach GTM, layer ``Org/Academy ? Coach ? Athlete`` **on top** (e.g. a ``Membership`/role` table + an ``athlete_id`` tag on collections/videos) **without** baking it into the core tables ? so the schema stays usable for solo power users and the coach model can evolve, or be dropped, as PMF dictates.
- **History/serving API:** list collections/videos/highlights per user (and, with the

```

199 overlay, by athlete/coach/date); **query highlights by `HighlightMetadata`** (e.g.
shot
200 type / outcome); re-stitch from stored segments without reprocessing.
201
202 ### 4.2 Tenancy isolation ? options & recommendation
203 | Model | Isolation | Ops cost | Best for |
204 |---|---|---|---|
205 | **Shared DB + `tenant_id`** (row-level, ideally Postgres **RLS**) | logical | low |
**Hosted SaaS, many small tenants** ? *recommended default* |
206 | Schema-per-tenant | medium | medium | tens?hundreds of mid tenants |
207 | **DB-per-tenant** (incl. **SQLite-per-tenant**) | strong/physical | higher at scale |
**self-hosted single-academy**, or high-isolation enterprise |
208
209 **Recommendation:** for the hosted product, **migrate SQLite ? Postgres with shared-DB +
210 `tenant_id` + Row-Level Security**; keep **SQLite-per-tenant** as the **self-hosted /
211 single-academy** deployment (it reuses today's code almost verbatim). Either way,
212 **abstract DB access behind a repository layer now** so the engine is swappable, and
213 graduate the `PRAGMA user_version` migration discipline to **Alembic** under Postgres.
214
215 ## 5. Orchestration / async jobs
216
217 This is the serving layer that makes multi-user real. It is the **multi-user superset of
218 [`DEFERRED_HARDENING_PLAN.md`](DEFERRED_HARDENING_PLAN.md) #16** (async job model) ?
build
219 that first, single-tenant, then generalize.
220
221 - **Job record** (`jobs` table): `id, tenant_id, video_id, type (index|compile|analyze),
222 status (queued|running|done|failed), progress, compute_backend, result_ref, error,
223 timestamps`. `/api/process` returns **202 + job_id** ; `/api/jobs/{id}` polls.
224 - **Idempotency** keyed on `(tenant_id, video_id, pipeline_version)` ? reuse the
225 content-hash discipline already in `compute_video_id`.
226 - **Dispatch:** the orchestrator enqueues; a worker (hosted or `byo_worker`) claims jobs
227 for its tenant, resolves input via the tenant's `StorageBackend`, runs the pipeline,
228 writes artifacts + DB rows. **Observability reports** (already per-video) become the
229 job artifact surfaced in history.
230 - **Queue/worker backend ? a decision to prototype in #16, *not* a default here.** The
231 *requirements* are firm ? **idempotency**, **resumability** (a 40-min GPU job mustn't
232 restart on a worker blip), **progress states** ? which rule out fire-and-forget
Celery/RQ
233 for the long jobs. The *choice* should be made by prototyping in the
234 [DEFERRED_HARDENING_PLAN.md](DEFERRED_HARDENING_PLAN.md) #16 slice, weighing:
235 - **Start simplest:** **Arq** (async-native, natural FastAPI fit) or even a **plain
236 Postgres-backed queue + worker** ? likely sufficient for the *first* single-tenant
237 async slice, with the lowest operational complexity and the easiest debugging.
238 - **Durable-execution upgrade if/when needed:** **Hatchet** (Postgres-backed, per-step
239 retries/timeouts/concurrency; pairs neatly with the pull-based worker) ? attractive
240 because it adds **no new datastore** once we're on Postgres, but it *is* a new
241 dependency + operational/debugging surface; adopt only when long-job resumability/
242 observability demands it.
243 - **Heaviest:** **Temporal** ? most mature durable workflows, operationally heavy;
244 reserve for genuinely complex multi-stage pipelines at scale.
245
246 Record the decision criteria in #16: operational complexity, maturity, **debugging
247 experience for long GPU jobs**, and whether durability is worth a new dependency.
248 **Bias: start with the simplest thing that gives retries + status; graduate only on
249 evidence.**
250
251 ## 5A. Owned generative model (the video-QA assistant) & the data flywheel
252
253 The north-star capability: an **owned, self-hostable generative video model** ?
254 fine-tuned on sports video (including **consented, user-contributed** footage) ? that
255 **answers questions about rallies, shots, and technique**. This is the conversational
256 brain of the "assistant coach." Crucially, it does **not** replace the known stack; it
257 is introduced **behind the same abstraction, with the paid API as an always-ready
258 fallback**.

```



```

259
260 > **The moat is *not* the model.** Anyone can download Qwen-VL/Gemma. The defensible
261 > assets are **(a) the consented, badminton-specific video?QA dataset** and **(b) the
262 > eval harness**. Build and message *those* ? not "we own an LLM."
263
264 > **? Committed guardrail (legal, non-negotiable).** We respect the licensing findings
in
265 > full. **Paid frontier models (Gemini / OpenAI / Anthropic) are used for *inference /
266 > serving / fallback* only ? never as a *training* data source.** Harvesting their
outputs
267 > to train our owned model would breach their "no competing model" terms (a
terms/copyright
268 > violation), so it is prohibited in both engineering and legal process. **Training
teachers
269 > are open, Apache-2.0 models only** (Qwen-VL / InternVL / Gemma 4). Detail in the
270 > distillation constraint below.
271
272 ### Core principle: AI is a pluggable provider with a paid fallback
273 The codebase already abstracts the cloud AI behind `provider_registry` /
274 `AIVideoProvider` (`backend/providers/`). Make that a **first-class platform
principle**:
275 - Every generative capability routes through the **provider registry**.
276 - The **owned model** plugs in as a new provider (`owned_vlm`); **Gemini (paid, via
277 Vertex AI)** stays registered as the **fallback** (failover on error/low-confidence,
278 or as the quality ceiling/teacher reference).
279 - This lets us **ship today on the known paid stack** and **swap in the owned model when
280 its quality clears a measured bar** ? with zero architectural churn and the paid path
281 always one config flip away. (Mirrors the COMMERCIALIZATION C4 A/B/C strategy: the
282 owned model is the long-run "(A) eliminate the per-call dependency"; paid is "(C)".)
283
284 ### The model / training layer (new sub-layer, fed by the scaffolding)
285 ```
286 Data curation      Training      Eval      Serving
287 (consented user vid (fine-tune an open, (extend the existing (self-host inference
288 + golden set +      commercially-usable  eval harness to QA   as a provider /
289 rally/segment/event base VLM via      quality + the        ComputeBackend;
290 annotations ?      LoRA/QLoRA)          regression gate)     paid fallback)
291 video-QA pairs)
292 ```
293 This layer **reuses every other layer**: **Storage** holds the training corpus;
294 **Data** supplies labels (the `annotation` registry + `candidate_segments` +
295 golden set are already the labeling seam); **Compute/Orchestration** runs training and
296 inference jobs (even **BYO-compute** can serve inference on the tenant's GPU); the
297 **eval harness** already in `backend/eval/` extends to grade QA quality. **In other
298 words, the rock-solid scaffolding is exactly what makes the model flywheel possible** ?
299 which is why scaffolding comes first (see §6).
300
301 ### Base model & training (Apache-2.0 only ? licensing is decisive)
302 **Recommended near-term base: `Qwen2.5-VL-7B-Instruct`** ? Apache-2.0, native long-video
303 (dynamic-FPS sampling, temporal mRoPE), mature LoRA tooling. Upgrade path
304 **Qwen3-VL-4B/8B (Apache-2.0)**; **Gemma 4 multimodal (Apache-2.0, 2026)** is a clean
305 Google-stack alternative to evaluate.
306 - ?? **Lock to Apache-2.0-clean bases** so you can fine-tune, **keep weights private,
307 gate behind subscriptions, and never owe open-sourcing or branding**. (Note Qwen2.5-VL
308 is Apache **only at 7B/32B** ? 3B/72B use the custom Qwen license.)
309 - ? **Reject for the flagship:** **Gemma 3** ? custom terms with a **viral derivative
310 clause** (anything fine-tuned on it, *or even trained on its synthetic outputs*, stays
311 permanently bound to Google's *updatable* Prohibited-Use Policy); **Gemma 4's switch
to
312 Apache-2.0 removes this**. **Llama 3.2 Vision** ? custom license forces a "Llama-?"
313 model name + "Built with Llama" badge and a 700M-MAU ceiling. **`gpt-oss`** ?
Apache-2.0
314 but **text-only**, so usable only as an optional text-reasoning layer, never the video
315 encoder.
316 - **Training method:** **QLoRA** (? r=64, ?=128) on the 7B ? fits one prosumer GPU and

```

```

317     avoids catastrophic forgetting of general video skills. Build the dataset by
converting
318     the existing rally/shot/segment annotations + golden set into templated **clip?QA
319     pairs**, expand synthetically with **open-model teachers only** (see constraint
below),
320     and hold out a **golden QA eval** that gates every fine-tune. A few hundred gold pairs
?
321     **5k?20k** curated is a realistic band for this narrow domain.
322 - **Serving economics:** a 7B VLM ? 14 GB VRAM (FP16) / ~3.5 GB (INT4); a ?16 GB GPU is
a
323     comfortable serve target (vLLM for throughput). Self-hosting wins **only** once you
have
324     sustained video-QA volume **** a defensible consented dataset **** real per-call API
325     cost pain ? below that, the paid API is cheaper than a standing GPU and the second
stack
326     is overhead. Hence: **paid-primary now ? owned-primary / paid-fallback later.**
327
328     Base-model licensing + training-data policy are tracked as **COMMERCIALIZATION C14**.
```

329

```

330     ### Training backend ? managed service vs self-managed (the fine-tune "forge")
331     *Where* you run the LoRA/RL fine-tune. Weigh on **video support, weight
export/ownership,
332     smallest servable model, RL support, no-train DPA, and infra burden**.
333
334     **Decisive finding (researched 2026-06):** managed fine-tuning services are
335     **image-based** (JSONL + base64 images) ? for video you pre-sample frames and feed them
336     as image sequences. **Only the self-managed `ms-swift` explicitly advertises *native*
337     video (+audio) training.** So for true video-QA, a self-managed framework is the
338     strongest technical fit ? and it also lets you fine-tune the **small 7B** (cheap to
339     serve) and keep full ownership. Managed services trade some of that for zero infra.
340
341     | Option | Type | Video / VLM support | Smallest VL | RL | Export weights | Best for |
342     |---|---|---|---|---|---|---|
343     | **ms-swift** (ModelScope) | Self-managed (rented/own GPU) | **Native video+audio**;
```

Qwen3-VL/Omni, 300+ MLLMs | any incl **7B** | ? DPO/GRPO | ? full ownership | **? video-QA;

slots into BYO-compute (\$3.2)** |

```

344     | **LLaMA-Factory** | Self-managed | Qwen3-VL + 100+ VLMs; video datasets | any incl 7B
| ? | ? | popular, well-documented |
345     | **Unsloth** | Self-managed (1-GPU optimized) | Qwen3-VL LoRA (vision+lang; frames) |
any incl 7B | ? GRPO | ? | cheapest first runs |
346     | **Tinker** (Thinking Machines) | Managed API (low-level) | Qwen3-VL **30B-A3B+ only**;
```

image input (video **unverified**) | 30B-A3B MoE | ? PPO/GRPO, custom loss | ? adapters + ckpt;

no-train ToS | zero-infra RL/research at MoE scale |

```

347     | **Together AI** | Managed (tune + serve) | Qwen3-VL / Gemma-3 / Llama-4; image data |
verify | partial | ? adapters downloadable | managed Qwen3-VL + serving |
348     | **Fireworks AI** | Managed (tune + serve) | Full Qwen2.5-VL (3B?72B SFT; LoRA@32B +
LoRA-upload) | **7B** | partial | ?? verify export (lock-in risk) | managed *small*-VL + serving
|
349     | **Predibase** (Rubrik) | Managed *** VPC self-host** | VLM instruct-tune; **LoRAX**
multi-adapter | verify | ? (VPC) | ? **RFT/GRPO** flagship | RL "creativity" + multi-adapter +
privacy |
350     | ~~OpenAI GPT-4o / Gemini tuning~~ | Closed | ? | ? | ? | ? no export | **non-fit ?
violates "own the model"*** |
351
352     *(Enterprise clouds ? Databricks Mosaic / SageMaker / Vertex on open models ? give full
353     control but are only worth it if you're already on that stack.)*
354
355     **Recommendation:**
356     - **Early iteration / true video-QA ?** self-managed **`ms-swift`** (or LLaMA-Factory /
357     Unsloth) on a **rented GPU** (RunPod / Lambda / Modal / SkyPilot-to-your-cloud):
native
358     video, any size incl. **7B** (cheap to serve), full ownership, and it drops into the
359     **ComputeBackend / BYO-compute** seam ($3.2). **Likely a better fit than Tinker** for
360     the video requirement *and* serving economics.
361     - **Managed, no GPU ops ?** **Together AI** (Qwen3-VL + downloadable adapters + serving)
```

```

362 or **Fireworks** (full Qwen2.5-VL incl. 7B + serving) ? confirm frame/video support
and
363 adapter export.
364 - **If RL "coaching creativity" + per-sport multi-adapter is central ?** **Predibase**
365 (RFT + LoRAX, VPC self-host for the biometric/privacy posture) or **Tinker** (lowest-
366 level RL primitives).
367 - **Tinker** remains a strong zero-infra RL-research option with clean weight export ?
just
368 not the cheapest path to a *small video* model (its VL floor is 30B-A3B).
369 - **Portability is the hedge:** you own the **data + recipe + exported weights**, so the
370 backend is swappable ? same ethos as the provider/registry pattern. Don't get locked
in.
371
372 **Verify before committing (any backend):** (1) **native video / frame-sequence**
373 ingestion + max frames per example; (2) **weight export** + a **commercial DPA**
covering
374 consented, minors-excluded video (biometric, §7); (3) the **inference plan** for the
size
375 you train (a self-managed **7B** is far cheaper to serve than a 30B-A3B MoE).
376
377 ### Training harness ? make it agent-runnable (nanochat-style operability)
378 The goal: **an agent (or CI) can run a fine-tune on a fresh batch of golden-labeled
video
379 and get a trustworthy ship / no-ship verdict, unattended.** Karpathy's **nanochat** is
the
380 reference for that *operability* ? one MIT repo, one `speedrun.sh` that runs
381 tokenizer?train?eval?serve, a single complexity dial, and an automatic **report-card**
382 score. It is **not** a usable engine here (text-only, from-scratch GPT-2-class), but its
383 five properties are exactly what make training agent-runnable, and we copy them:
384 ** (1) ** one command, end-to-end; ** (2) ** one complexity dial (sane auto-derived
defaults);
385 ** (3) ** an automatic **report card** ? a deterministic pass/fail vs a target (the crux);
386 ** (4) ** reproducible + deterministic + cheap; ** (5) ** minimal/readable.
387
388 **Engine (don't reinvent):** **`ms-swift`** already delivers this for VLMs ? CLI-driven,
389 **native video**, Qwen3-VL support, an eval backend (EvalScope), and reproducibility via
390 `args.json` + `full_determinism`. (LLaMA-Factory is the close second; HF **`nanoVLM`** ?
391 ~750 lines ? is the minimal reference to *learn* VLM training, not the production path.)
392
393 **The harness = thin glue over components we already have** ? one declarative,
394 agent-invokable job (call it `train_speedrun`):
395 1. **Build dataset** from golden labels ? clip?QA pairs (reuse the `annotation` registry
+
396 `candidate_segments` + golden set ? the labeling seam in `backend/`).
397 2. **Fine-tune** via ms-swift on a **ComputeBackend** GPU (§3.2), run as an
orchestration
398 job (§5 / DEFERRED #16): one config, idempotent, logged.
399 3. **Auto-eval + report card** ? extend the existing `backend/eval/` regression harness
to
400 **QA quality**; emit a `report.md` pass/fail vs the **held-out golden QA eval**.
401 4. **Register** the result as the `owned_vlm` provider (Gemini paid fallback stays)
**only
402 if it beats the bar**.
403
404 So: **golden labels ? ms-swift train (on a compute backend) ? eval gate ? register
405 **provider**, behind one command an agent or CI runs. nanochat is the inspiration for the
406 wrapper; ms-swift is the engine; the **golden QA eval is the report card**.
407
408 > **Prerequisite (the crux):** the harness is only as trustworthy as the **golden QA
eval
409 > **set**. nanochat's speedrun is meaningful *because* its eval (CORE score) is automatic
and
410 > trustworthy; ours is the golden QA eval ? **build that first**, or an agent-run loop
411 > optimizes blind. (Same "moat = dataset + eval", scaffolding-first principle.)
412

```

```

413     ### ?? Teacher / distillation constraint (the single biggest legal landmine)
414     You **cannot** train the owned model on outputs from the paid frontier APIs ? a general
415     video-QA assistant is squarely a "competing model": **Gemini** ("may not use the
416     Services to develop models that compete"), **OpenAI** ("may not use Output to develop
417     models that compete"), **Anthropic** (bars using Outputs to train competing models). Two
418     safe lanes: ** (1) ** generate synthetic QA with **open-model teachers only** (Qwen-VL /
419     InternVL / Gemma 4 ? Apache-clean); ** (2) ** keep the paid API strictly as a **runtime
420     fallback** ? serving answers is fine, *harvesting its outputs to train the model that
421     replaces it is not*. Write a hard ***"no training on competitor-API outputs"*** rule into
422     eng + legal process. (This makes the ROADMAP's "Gemini as offline teacher only" precise:
423     the teacher is *open models*, not the paid APIs.)
424
425     ### Data flywheel, personalization & consent tiers
426     Two complementary loops, each with its **own** opt-in (training consent is separate from
427     processing consent ? §7):
428
429     - **Global model (shared ? raises the floor).** Train the base owned model on an
430     **explicitly-contributed, licensed** corpus ? *not* on whatever sits in a tenant's
431     bucket
432     (preserves "your video, your storage"). Flywheel: *more consented, labeled matches ? a
433     better shared model ? better highlights/answers ? more reason to contribute.* Higher
434     consent bar (pooled data, biometric/minor care ? §7); **minors excluded by default**.
435
436     - **Per-user adapters (personalized ? raises the ceiling) ? owner hunch, under
437     exploration.**
438     A small **per-user (or per-tenant) LoRA adapter, fine-tuned *only on that user's own
439     consented videos*** on top of the shared base, served via **multi-LoRA** (LoRAX /
440     Together / Predibase). Why it's compelling:
441     - **Precision:** adapts to *that user's* court, camera angle, lighting, and players'
442     styles ? materially better detection/highlights for them.
443     - **Cleanest consent story:** *"improve **your** detection using **your** data"* is
444     far
445     lower-friction and lower-risk than contributing to a global model ? the data
446     improves
447     **only the user's own adapter**, with **no cross-user mixing**, sidestepping much of
448     the
449     pooled-corpus biometric/minor entanglement. Reinforces the differentiator: *your
450     video,
451     your storage, your compute ? and now **your model**.* With BYO-compute the adapter
452     can be
453     trained/served on the user's own GPU and **never leave their boundary**; **deletion
454     =
455     drop the adapter** (clean erasure).
456
457     - **Open questions (still a hunch):** cold-start (need enough of the user's labeled
458     data
459     before a personal adapter beats the base ? fall back to the shared model meanwhile);
460     a
461     **per-user held-out eval** to prove the adapter helps (don't ship a regression);
462     base +
463     optional global-adapter + per-user-adapter stacking; per-user training cost at
464     scale.
465     Worth a spike once the harness + golden eval exist.
466
467     **Two consent tiers, then:** ** (a) ** "improve **my own** detection with **my** data"
468     (per-user, low friction) and ** (b) ** "contribute to the **shared** model" (global,
469     higher
470     bar). **Default for both is off.**
471
472     ## 6. Phased roadmap
473
474     | Phase | Deliverable | Build / reuse |
475     |---|---|---|
476     | **P0** | **Async job model, single-tenant** (DEFERRED #16) | `jobs` table via
477     `user_version` ladder; worker; 202+status endpoints |
478     | **P1** | **Identity + multi-tenant data model + upload + highlight persistence +

```

```

history** | auth (build vs Auth0/Clerk/Supabase ? decide); Postgres + `tenant_id`/RLS;
`highlights` table; repository layer; history API + UI |
464 | **P2** | **`StorageBackend` registry + BYO-storage** | new ABC+registry;
`local`?`gdrive`/`s3`/`gcs`; replace `video_path` with `StorageRef`; encrypted per-tenant creds
|
465 | **P3** | **`ComputeBackend` registry + BYO-compute** | `local`/`hosted`/`byo_worker`;
pull-based worker agent; control/data-plane split |
466 | **P4** | **Rich rally/shot semantic metadata & tagging** *(the next value stage after
solid rally detection + highlights)* | shot type (smash / backhand drive / net), rally
**outcome** (winner / forced / **unforced error**), context (e.g. **open court but error**) ?
query-API filters + `HighlightMetadata` + a **human correction/review loop** feeding the golden
set via the `annotation` registry + `candidate_segments` |
467 | **P5** | **Owned generative video-QA model** ($5A) | global flywheel **per-user
adapters** ; P4 metadata is training signal ? fine-tune an open VLM (Qwen-VL) via LoRA ? serve as
`owned_vlm` **provider with Gemini paid fallback** ; extend the eval harness to QA quality |
468 | **P6** *(visionary)* | **Gait / biomechanics under a supervising coach** |
deliberately **back-burner** ; technique feedback; highest Art. 9 bar ? gate behind explicit
biometric + parental consent ($7) |
469
470 P0?P1 deliver the headline ask (per-user upload + stored highlights + history). P2?P3
471 deliver the "plug-and-play storage/compute" ask. **P4 (rich semantic metadata) is the
472 immediate next value stage once rally detection + highlights are solid** ? e.g.
**"rallies
473 that ended in a smash or backhand drive,"* **rallies where the court was open but the
474 player made an unforced error."* **P5** owns the model; **P6 (gait) is the visionary
475 horizon, intentionally deferred.**
476 **Sequencing principle (owner directive):** make the scaffolding (P0?P4) **rock-solid
477 first**, keep the **known paid stack as an always-ready fallback**, and only then shift
478 focus to **owned-model quality** (P5). The provider registry keeps the paid path one
479 config flip away.
480
481 ## 7. Privacy & legal (gait raises the bar ? design for it now)
482
483 - **The load-bearing nuance:** biometric data is special-category **only when used to
484 uniquely identify* a person** (GDPR Art. 9). **Highlight/shot detection on players a
485 coach already knows is general analytics ? generally *not* Art. 9.** But
486 **gait/biomechanics that builds a per-athlete identifying template tips into Art. 9**
487 and needs **explicit, granular consent** referencing biometric data. Engineer gait
488 with that boundary (consent gate, purpose limitation, no cross-account
identification);
489 decide whether it runs **locally/on-tenant only** (strong posture) vs our cloud. (EDPB
490 Guidelines 3/2019 on video devices apply.)
491 - **Minors (e.g. academies).** Parental/guardian consent and safeguarding are first-
order,
492 not an afterthought. Model consent explicitly per relationship (user?subject,
coach?athlete
493 where the overlay applies, parent?minor).
494 - **BYO-storage is the central privacy/liability lever:** if raw video stays in the
495 tenant's own bucket/Drive and we process via signed URL or their compute, we minimize
496 what we hold ? shrinking breach surface, residency, and erasure obligations.
497 - **Paid-LLM no-train (C4):** use **Gemini via Vertex AI** (contractual no-training),
498 **not the consumer/free tier** ; set short/zero retention, document it in the DPA, and
499 surface "your footage is never used to train AI" as a trust feature for academies. For
500 the gait phase, prefer keeping biometric inference off third-party APIs entirely.
501 - **Training consent is a distinct, higher bar (for the owned model, $5A).** Training on
502 user video needs a **separate, explicit, opt-in training licence** ? not bundled with
503 processing consent ? and it **compounds** the biometric (Art. 9) and minor-consent
504 issues above. Train **only on explicitly contributed/licensed video** ; never train on
505 what merely passes through a tenant's bucket. **Exclude minors' video from the
training
506 pool by default** , keep the training corpus a **separate, firewalled, opt-in sub-
pool** ,
507 and log **per-clip consent/provenance** for revocation + audit. Incentives must not
make
508 consent non-free (the product stays fully usable without contributing). And the

```

509 ****teacher-distillation rule**** (§5A): never train on paid-LLM outputs. ****Two consent**
510 tiers
511 (§5A):**** (a)* "improve **my own** detection with **my** data" ? a per-user adapter**
512 trained
513 only on the user's own video (lowest friction; no cross-user mixing; deletion = drop
514 the
515 adapter); ***(b)* "contribute to the **shared** model" ? pooled corpus, higher bar.**
Default
516 for both is ****off****.
517 - ****Right to erasure / retention:**** per-tenant, per-athlete delete must cascade across
DB +
518 storage artifacts (the `ON DELETE CASCADE` discipline already in `database.py` extends
here).
519 **## 8. Key decisions for the owner (the forks)**
520 1. ****Hosted-first vs BYO-first.**** Recommended: ****hosted control-plane first**** (P0?P1)
with
521 ****BYO-storage soon after**** (P2); ****BYO-compute later**** (P3) as a power-
user/enterprise
522 option ? it's the highest-complexity, highest-security item.
523 2. ****Database now or later.**** Move to ****Postgres at P1**** for the hosted path, or stay
524 SQLite-per-tenant if self-hosted-single-academy is the first GTM. (Abstract DB access
525 regardless.)
526 3. ****Auth: build vs buy**** (Auth0 / Clerk / Supabase Auth / Cognito).
527 4. ****Orchestration tech**** (see research recommendation) ? bias to the simplest that
gives
528 retries + status.
529 5. ****Gait processing location**** ? local/on-tenant (privacy-max) vs cloud (capability-
max).
530 6. ****How far to push BYO-compute**** given its security/ops complexity.
531 7. ****Owned-model base + licensing (§5A).**** Which commercial-OK video VLM to build on
532 (Qwen-VL Apache-2.0 vs Gemma 3 custom license vs ?) and confirm a gated commercial
533 derivative is permitted ? see COMMERCIALIZATION C14.
534 8. ****Training-data sourcing & consent.**** Opt-in contribution terms, incentives, and the
535 teacher-distillation constraint (whether paid-LLM outputs may seed the corpus).
536 9. ****When to start the owned model**** ? strictly after the scaffolding + flywheel are
537 solid (the explicit directive), with the paid provider as the standing fallback.
538 ****Risks:**** scope explosion (sequence strictly P0?P5); multi-tenant data-leak bugs (RLS +
539 tests); BYO-compute security; ****biometric/minor liability**** (get this right before any
540 gait feature ships); ****training on user video without proper opt-in/licence****, and
541 ****base-model / teacher-distillation licence violations****; cost of hosted GPU (training
542 *and* inference); ****premature model-building**** before data/eval are ready (mitigated by
543 keeping the paid fallback and gating P5 behind the flywheel).
544 **## 9. Sources**
545 External best-practices research (2026-06-07) backing the recommendations above. Each
546 underlying claim in the research was tagged fact-vs-synthesis; figures are as-of and
547 should be re-validated before committing budget.
548 ****Storage abstraction:**** [fsspec](https://fsspec.com/) ·
549 [cloudpathlib](https://github.com/drivendataorg/cloudpathlib) ·
550 [smart_open](https://github.com/piskvorky/smart_open) · [Apache
551 Libcloud](https://libcloud.apache.org/) · [gdrive-fsspec](https://github.com/fsspec/gdrive-
fsspec) · [dropboxdrivefs](https://github.com/fsspec/dropboxdrivefs) · [AWS S3 presigned
URLs](https://docs.aws.amazon.com/AmazonS3/latest/userguide/using-presigned-url.html) · [token-
based S3 with STS](https://oneuptime.com/blog/post/2026-02-12-token-based-access-s3-sts/view) ·
[signed URLs across clouds](https://medium.com/@shaikhsarwan49/signed-urls-in-aws-gcp-and-azure-
a-beginner-friendly-guide-e2e549425865)
552 ****Multi-tenancy:**** [Bytebase: multi-tenant DB
553 patterns](https://www.bytebase.com/blog/multi-tenant-database-architecture-patterns-explained/)
· [Hunchbite: RLS vs schema-per-tenant](https://hunchbite.com/guides/multi-tenant-saas-

architecture) · [PlanetScale: tenancy in Postgres](https://planetscale.com/blog/approaches-to-tenancy-in-postgres) · [DEV: DB-per-tenant vs shared schema](https://dev.to/young_gao/multi-tenant-architecture-database-per-tenant-vs-shared-schema-1n2e)

554

555 ****BYO-compute / orchestration:**** [Pinecone BYOC](https://www.pinecone.io/blog/byoc/) · [Nuon: control/data-plane](https://nuon.co/blog/byoc-control-plane-data-plane-architectures) · [GitHub self-hosted runners (ARC)](https://docs.github.com/en/actions/hosting-your-own-runners/managing-self-hosted-runners-with-actions-runner-controller/about-actions-runner-controller) · [SkyPilot](https://github.com/skypilot-org/skypilot) · [SkyPilot AI control plane](https://blog.skypilot.co/ai-job-orchestration-pt2-ai-control-plane/) · [Modal (AWS Marketplace)](https://aws.amazon.com/marketplace/pp/prodview-j727623xqhh2k) · [Northflank: Anyscale alternatives](https://northflank.com/blog/anyscale-alternatives-for-ai-ml-model-deployment) · [DevPro: Python task queues 2025](https://devproportal.com/languages/python/python-background-tasks-celery-rq-dramatiq-comparison-2025/) · [Judoscale: choosing a task queue](https://judoscale.com/blog/choose-python-task-queue) · [Hatchet vs Temporal](https://hatchet.run/versus/hatchet-vs-temporal) · [Hatchet](https://github.com/hatchet-dev/hatchet)

556

557 ****Privacy / legal:**** [GDPR Art. 9](https://gdpr-info.eu/art-9-gdpr/) · [GDPR Local: biometric compliance](https://gdprlocal.com/biometric-data-gdpr-compliance-made-simple/) · [Legiscope: Art. 9](https://www.legiscope.com/blog/gdpr-article-9-special-categories.html) · [EDPB Guidelines 3/2019 (video)](https://www.edpb.europa.eu/sites/default/files/files/file1/edpb_guidelines_201903_video_devices_en_0.pdf) · [Fox Rothschild: EDPB video guidance](https://www.foxrothschild.com/publications/video-surveillance-under-gdpr-edpb-issues-final-guidance) · [ICO: children & UK GDPR](https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/childrens-information/children-and-the-uk-gdpr-old/what-are-the-rules-about-an-iss-and-consent/) · [EDPB Guidelines 05/2020 (consent)](https://www.edpb.europa.eu/sites/default/files/files/file1/edpb_guidelines_202005_consent_en.pdf) · [AI data-retention compared](https://ax-sentinel.com/blog/ai-data-retention-policies-compared) · [Xenoss: enterprise LLM platforms](https://xenoss.io/blog/openai-vs-anthropic-vs-google-gemini-enterprise-llm-platform-guide)

558

559 ****Comparable products:**** [Veo](https://www.veo.com/) · [Veo Help Center](https://support.veo.com/hc/en-us/articles/4459159526929-Overview-of-cam-veo-co-Record-manage-and-upload-footage-with-your-Veo-Cam-1) · [Latterly: Hudl competitors](https://www.latterly.org/hudl-competitors/) · [Apple: Behind the Design ? SwingVision](https://developer.apple.com/news/?id=0pg4dthn) · [SwingVision FAQ](https://swing.vision/faq) · [Sportsvisio: coach video-analysis comparison](https://www.sportsvisio.com/stories/video-analysis-software-for-coaches-complete-comparison)

560

561 ****Training backends & harness (\$5A):**** [karpathy/nanochat (harness reference)](https://github.com/karpathy/nanochat) · [HF nanoVLM (minimal VLM)](https://github.com/huggingface/nanoVLM) · [nanoVLM blog](https://huggingface.co/blog/nanovlm) · [Tinker (GA + vision)](https://thinkingmachines.ai/blog/tinker-general-availability/) · [Tinker model lineup](https://tinker-docs.thinkingmachines.ai/model-lineup) · [Tinker ToS (weights/no-train)](https://thinkingmachines.ai/legal/tos/) · [Fireworks VLM fine-tuning](https://fireworks.ai/blog/vlm-tuning) · [Together AI fine-tuning](https://www.together.ai/fine-tuning) · [Predibase: fine-tune & serve VLMs](https://predibase.com/blog/how-to-fine-tune-and-serve-vlms-in-predibase) · [Predibase RFT/GRPO](https://docs.predibase.com/user-guide/fine-tuning/grpo) · [ms-swift (native video/audio)](https://github.com/modelscope/ms-swift) · [LLaMA-Factory (VLMs)](https://github.com/hiyouga/LlamaFactory) · [Unsloth (Qwen3-VL)](https://qwen.readthedocs.io/en/latest/training/unsloth.html) · [Fine-tuning tools 2026 (LoRA/QLoRA)](https://www.index.dev/blog/top-ai-fine-tuning-tools-lora-vs-qlora-vs-full)

562

563 ****Owned model & licensing (\$5A):**** [Qwen2.5-VL](https://qwenlm.github.io/blog/qwen2.5-vl/) · [Qwen2.5 license split](https://qwenlm.github.io/blog/qwen2.5/) · [Qwen3-VL](https://github.com/QwenLM/Qwen3-VL) · [Gemma 3 (multimodal)](https://huggingface.co/blog/gemma3) · [Gemma Terms of Use](https://ai.google.dev/gemma/terms) · [Gemma license-risk analysis](https://wcr.legal/google-gemma-license-risks/) · [Gemma 4 / Apache-2.0](https://www.mindstudio.ai/blog/what-is-gemma-4-apache-2-license-commercial-ai-deployment) · [Llama 3.2 license (700M MAU + naming)](https://www.llama.com/llama3_2/license/) ·

[gpt-oss (text-only, Apache-2.0)](<https://openai.com/index/introducing-gpt-oss/>) · [Gemini API terms (compete clause)](<https://ai.google.dev/gemini-api/terms>) · [OpenAI terms](<https://openai.com/policies/row-terms-of-use/>) · [Anthropic: outputs & training](<https://support.claude.com/en/articles/12326764-can-i-use-my-outputs-to-train-an-ai-model>) · [QLoRA practical](<https://apxml.com/courses/fine-tuning-adapting-large-language-models/chapter-4-parameter-efficient-fine-tuning/practice-qlora-fine-tuning>) · [VLM LoRA cookbook](https://huggingface.co/learn/cookbook/en/fine_tuning_vlm_trl) · [GDPR Art. 9](<https://gdpr-info.eu/art-9-gdpr/>) · [athletes & biometric data](<https://www.architectureandgovernance.com/applications-technology/biometric-data-and-athletes-privacy-law-and-compliance-implications/>) · [federated learning + DP](<https://research.google/blog/federated-learning-with-formal-differential-privacy-guarantees/>) · [self-hosted LLM VRAM/cost](<https://www.gpu-mart.com/guides/self-hosted-llm>) · [Gemini API pricing](<https://ai.google.dev/gemini-api/docs/pricing>)

564

565 ## See also

566 - [``COMMERCIALIZATION.md``](COMMERCIALIZATION.md) ? C12 (SaaS foundations), C4 (paid-LLM), economics.

567 - [``DEFERRED_HARDENING_PLAN.md``](DEFERRED_HARDENING_PLAN.md) ? #16 async job model (the P0 infra slice), #7 packaging.

568 - [``PMF_MARKET_RESEARCH.md``](PMF_MARKET_RESEARCH.md) ? academy/coach beachhead (who this serves).

569 - [``CODE_MAP.md``](CODE_MAP.md) ? current architecture + the registry extension points.

570 - [``ROADMAP.md``](ROADMAP.md) ? offline self-improving rally detection (the pipeline's own trajectory).

571

11. user (2026-06-10T06:00:15.571Z)

1 # PMF & Market Study ? AI Highlight-Reel Tool for Self-Recorded Racket Sports

2

3 > **Research snapshot ? 2026-06-07.** Figures are cited as-of with source + year;

4 > validate live before relying on them for investment/pricing decisions. This is the

5 > market-research-of-record behind the PMF section of the living

6 > [``COMMERCIALIZATION.md``](COMMERCIALIZATION.md) tracker (§6). Product framing:

7 > a closed-source hosted API/SaaS (plus local mode) that ingests long-form,

8 > self-recorded, **static-camera, full-court amateur** racket-sports video and

9 > auto-detects rallies and compiles highlight reels. Primary sport: badminton;

10 > sport-generic architecture roadmapped to tennis, pickleball, table tennis, cricket.

11

12 > **Confidence convention.** Figures from named industry reports, federations, or

13 > company pricing pages are tagged with source + year. Inferred items are tagged

14 > **(estimate)** / **(low confidence)**. Emerging-category market sizes vary widely

15 > between vendors; ranges are shown rather than false precision.

16

17 ---

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19 ## 1. Executive summary

20

21 The product sits at the intersection of three favorable trends: (a) a fast-growing

22 **sports analytics market** (~USD 5?6B in 2025, ~18?22% CAGR to USD 24?36B by ~2034

23 depending on vendor); (b) a structural shift to **amateur self-recording** on action

24 cameras and phones (action-camera market ~USD 6.5B in 2024, sports the largest

25 segment); and (c) **strong racket-sport participation** ? 220M+ badminton players,

26 106M tennis players (2024), and a near-vertical pickleball curve (~19.8M US players in

27 2024, +45.8% YoY).

28

29 The central tension is the **static-camera constraint**: the tool works best on fixed,

30 full-court amateur footage and **fails** on broadcast/panning footage. This is both a

31 moat (most competitors target team sports / broadcast) and a filter on the addressable

32 user ? the customer must already set a camera on a tripod behind the court. That

33 behavior is concentrated in **coaches/academies, clubs, and serious enthusiasts**, not

34 the median casual player.

35

36 **Recommended beachhead (strongest single):** **badminton coaching academies and**

37 **serious club/competitive players in India** ? a large, rapidly formalizing academy

ecosystem (Khelo India tailwind), genuine fixed-angle recording behavior, existing willingness to pay for coaching, an underserved tooling gap (incumbents are team-sports/broadcast), and a direct-but-immature competitor field. The product is *native badminton-first* ? a genuine edge here. Secondary: ***(2) US/global pickleball players & clubs*** (capital-rich, fastest-growing, but SwingVision already strong); ***(3) badminton academies/clubs in East/SE Asia + Denmark*** (highest density).

****Pricing**** clusters into hardware-bundled auto-filming (Veo, Trace, Pixellot, Spiideo) and software-only phone AI (SwingVision USD 179.99/yr; BadPro+ pay-per-use). A software-only BYO-camera model undercuts the hardware players; realistic anchors ***(estimate): B2C USD 60?180/yr; B2B academy/club USD 300?1,500/yr per location.***

****Top risks:**** (1) amateurs won't reliably set up a tripod ? TAM collapses to coaches/clubs; (2) free/near-free alternatives cap B2C WTP; (3) capture-quality variance is the core failure mode; (4) badminton's strongest markets (India, Indonesia, China) are the most price-sensitive and hardest to bill at Western SaaS prices.

2. Market sizing & trends

2.1 Sports analytics / sports-video market

- **2025 size:**** ~USD 5.47B (Precedence Research) to ~USD 5.79B (Fortune Business Insights); other vendors USD 5.6?6.0B. ***(2025)***
- **Growth:**** CAGR 15.7%?22.5%. Precedence ? ~USD 29.75B by 2034 (20.63%); Fortune BI ? USD 24.03B by 2032 (22.5%); MarketsandMarkets (narrower) USD 4.75B by 2030 (15.7%); Future Market Insights ? ~USD 36.2B by 2035 (22.1%). ***(reports 2024?2025)***
- **Video analytics is the leading segment**** within sports analytics (Grand View / Fortune BI) ? the directly relevant sub-market.
- **Interpretation (estimate):**** the *amateur/club auto-highlight* slice is a small but fast-growing fraction; most analytics revenue today is pro/broadcast/team-sports. The product targets the long tail the big numbers under-serve.

2.2 Action-camera / amateur recording adoption

- **2024 size:**** ~USD 6.5B (GMIInsights) ? ~USD 16.9B by 2034 (~10% CAGR). ***(2024?2025)***
- **Sports is the largest end-use segment**** of action cameras (GMIInsights, 2024).
- **GoPro ~47% share, DJI ~20%**** (electroIQ/industry, 2024); phones + cheap tripods are the de-facto amateur rig.
- **Implication:**** the capture-side behavior the product depends on is already widespread and growing ? *for the segments that record* ? validating "BYO-camera, software-only."

2.3 Participation & popularity by sport and geography

Sport	Participation signal	Source (year)
Badminton	220M+ play regularly; 160+ countries BWF est. via BadmintonBites (2023)	
Badminton ? China	~80M registered, up to ~100M (largest base) industry compilations (2023)	
Badminton ? India	20M+ recreational by 2023; academy boom + Khelo India industry compilations (2023)	
Badminton ? Indonesia	National sport; ~1M club players industry (est.)	
Badminton ? Denmark	100,000+ registered club members (Europe #1) industry (2023)	
Badminton ? Japan/Korea	Major school-club + workplace sport industry (2023)	
Tennis	106M players (2024)**; +25.6% over 5 yrs (87M in 2021) ITF Global Tennis Report (2021, 2024)	
Tennis ? Asia	~1/3 of players (33M of 87M) ITF (2021)	
Pickleball ? US	19.8M players (2024)**; +45.8% YoY, +311% over 3 yrs; ~6.2M core SFIA / Pickleheads / The Dink (2024)	
Pickleball ? US courts	~70,000 courts, +55% places to play YoY SFIA / Pickleheads (2024)	
Pickleball ? global	IFP 78 member countries; UK +73% in 2024; ~282M played monthly per PPA claim *(low confidence)* various (2024?2025)	
Table tennis	30M+ players; 227 member associations ITTF (current)	

96 | **Cricket** | 2.5B+ fans; 500M+ in India (mostly **fans**, not self-recorders) | Cricket
 Rise / Nielsen (2024?2025) |

97

98 - **Pickleball market value:** ~USD 2.2B (2024) ? USD 9.1B by 2034 at 15.3% CAGR
 99 (Market.us/Custom Market Insights, 2024?2025).

100 - **Badminton equipment market:** ~USD 7.23B global (2023), Asia-Pacific dominant.

101

102 **Geographic read:** badminton density is overwhelmingly Asian + Denmark; tennis is
 103 broad/high-income; pickleball is US-dominant, capital-rich, globalizing fast; cricket
 104 is huge as **fandom** but its amateur self-recording + static-full-court fit is weak
 105 (large field, not a fixed-frame court). For **this product's static-camera fit**,
 106 badminton/pickleball/tennis/table-tennis are the core; cricket is a long-tail roadmap
 107 item, not a beachhead.

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111 **## 3. Customer segment evaluation**

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113 | Segment | Market-size signal | Willingness to pay | Accessibility | Static-camera fit
 | B2C/B2B | Verdict |

114 |---|---|---|---|---|---|

115 | **Individual amateur/enthusiast** | Largest raw TAM (100s of millions) | **Low** ?
 competes with free; ~USD 5?15/mo ceiling | High via app stores; high CAC/churn | **Mixed** ?
 most won't set a tripod | B2C | Volume play, weak economics; use as funnel |

116 | **Coaches & coaching academies** | India academy boom; thousands across Asia |
High ? already sell coaching; USD 300?1,500/yr plausible | Reachable via
 federations/distributors/social | **Strong** ? already film fixed-angle | B2B/B2B2C | **Best**
 fit ? buyer, recorder, payer aligned |

117 | **Clubs & leagues** | Denmark 100k+; thousands of clubs Asia/EU | Medium-high; USD
 500?3,000/yr | Medium ? committee sales cycle | **Strong** | B2B/B2B2C | Strong secondary;
 longer cycle |

118 | **Tournament/event organizers** | Many small/regional events | Medium ? per-
 event/seasonal | Medium ? episodic | **Strong** | B2B | Good per-event upsell; lumpy |

119 | **Schools & universities** | Large (esp. Asia/Japan clubs) | Low-medium ? budget-
 constrained | Low ? slow procurement | Strong | B2B | Slow; grant/Khelo-India dependent |

120 | **National/regional federations** | Few, high-value | Medium-high; procurement-
 heavy/political | Low ? long cycles | Strong | B2B | Lighthouse/credibility, not beachhead |

121 | **Broadcasters / streamers** | High value per deal | High | Low ? incumbents
 entrenched | **Poor** ? panning = failure mode | B2B | **Avoid** ? violates core constraint |

122

123 **Key dynamics:** the static-camera constraint is a **B2B filter** ? coaches/academies/
 124 clubs already record from a fixed angle, so the product fits existing behavior (cleanest
 125 PMF). B2C is a large low-WTP, high-CAC funnel best used to feed B2B. Broadcasters
 126 explicitly violate the constraint ? exclude.

127

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130 **## 4. Competitive & pricing benchmarks**

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132 | Product | Category | Model | Price | Notes (year) |

133 |---|---|---|---|---|

134 | **Veo (Cam 3)** | Team-sports auto-filming | Hardware + sub | Camera ~USD 1,533 (5G
 ~1,886) + ~£395?895/yr | 15,000+ clubs/80 countries; \$115M raised (2025) |

135 | **Trace** | Soccer auto-filming | **Free camera + sub** | USD 180/yr (Basic) ? 300
 (Pro) | Camera leased free with sub (2025) |

136 | **Pixellot (Air)** | Auto camera/streaming | Hardware + sub | Camera ~USD 949 + USD
 69?167/mo | Institutional/league; 10,000+ clubs (2025) |

137 | **Spiideo** | Auto-filming/analytics | Sub incl. camera | ~USD 3,800/yr (1 team) ?
 6,100/yr (club) | Institutional budgets (2025) |

138 | **Hudl** | Team video analysis | SaaS tiers | From ~USD 400/yr | Broad team-sports
 incumbent (2026) |

139 | **SwingVision** | **Tennis/pickleball AI, single iPhone** | Software freemium | **USD**
 179.99/yr; free 2 hrs/mo | Closest analog; 200k+ monthly users; \$10M+ raised (2025) |

140 | **BadPro** | **Badminton AI on phone** | **Pay-per-use, no sub** | Per-analysis (? "<
 one coaching session") | Direct badminton competitor; shot detection, heatmaps, auto-highlights

(2025) |

141 | Niche pickleball/badminton apps | Various | Mixed | Varies | Fragmented; few true
auto-highlight tools (2025) |

142

143 ****What buyers actually pay (synthesis):**** consumer software-only anchors at ~USD
144 100?200/yr (SwingVision 179.99); hardware-bundled USD 180?900/yr prosumer/club,
145 USD 3,800?6,100/yr institutional; badminton-specific BadPro+ uses *pay-per-use* ?
146 signaling a real badminton AI market but ****unproven subscription WTP**** (opportunity +
147 warning). ****Positioning:**** software-only + BYO-camera undercuts hardware players and
148 sells worldwide via SaaS. Realistic anchors **** (estimate):** B2C USD 60?180/yr; B2B
149 academy/club USD 300?1,500/yr** per location. The differentiators (cheap offline
150 self-improving detection, per-rally stats/tagging, clean licensing) support a ****B2B****
151 value story more than a B2C one.

152

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154

155 **## 5. Recommended beachhead**

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157 Ranked by ****WTP × participation density × existing recording behavior × static-camera**
158 **fit × competitive whitespace.****

159

160 **### #1 (strongest) ? Badminton coaching academies & serious players, **India****
161 20M+ recreational players (2023); rapidly ****formalizing academy ecosystem**** + Khelo
162 India tailwind; academies already record fixed-angle; coaches are buyers who monetize
163 instruction (high relative WTP); incumbents ignore badminton; the only direct badminton
164 AI (BadPro+) is early with unproven pay-per-use. Product is *native badminton-first*.
165 ****Watch-outs:**** price sensitivity (need INR-tuned tiers); sell to the ****academy****, not
166 the individual.

167

168 **### #2 ? Serious pickleball players & clubs, **US (then UK/Australia/Asia)****
169 Fastest-growing sport (19.8M US, +45.8% YoY, 2024); capital-rich; ~70,000 courts;
170 strong recording/sharing culture; pickleball is on the roadmap. ****Watch-out:****
171 ****SwingVision already owns single-camera tennis+pickleball AI**** at USD 179.99/yr (200k+
172 monthly users) ? head-to-head differentiation, not greenfield.

173

174 **### #3 ? Badminton academies/clubs, **East/SE Asia (China, Indonesia, Malaysia, Japan,**
Korea) + Denmark**

175 Highest global density (China 80M+; Denmark 100k+ structured club members; Japan/Korea
176 school-club & workplace systems), mature club infra, fixed-court recording.
177 ****Watch-outs:**** China needs local hosting/partners + is regulatory-complex; SE Asia
178 price-sensitive; Denmark/EU small but high-WTP and easy to pilot.

179

180 ****Bottom line:**** lead with ****India badminton academies**** (whitespace + behavior + buyer
181 alignment), use ****enthusiast B2C as the funnel****, validate a ****club/academy B2B2C**
182 **motion****, then expand to ****pickleball-US**** (out-build SwingVision) and
183 ****badminton-Asia/Denmark**** (density).

184

185 ---

186

187 **## 6. Risks & open questions (what would invalidate the thesis)**

188

189 1. ****Tripod/static-camera adoption among amateurs.**** If serious amateurs won't set up a
190 fixed camera, B2C TAM collapses to coaches/clubs. ***Validate:*** observe 3?5 academies +
191 50?100 enthusiasts on whether they already record fixed-angle.

192 2. ****Free / near-free alternatives**** (YouTube, free app tiers, BadPro+ pay-per-use) cap
193 B2C WTP. ***Validate:*** fake-door pricing at USD 60?180/yr.

194 3. ****Capture-quality variance = the failure mode**** (gym lighting, off-center angle,
195 occlusion, mixed resolutions). ***Validate:*** benchmark rally-detection P/R on 100+ real
196 amateur clips; define a "good-capture" spec + auto-reject (ties to
197 `VIDEO\_RECORDING\_GUIDELINES.md` + the observability reports).

198 4. ****Monetization mismatch in best-participation markets**** (India/Indonesia/China are
199 price-sensitive, hard to bill). ***Validate:*** local pricing/payments pilot; per-academy
200 INR licensing.

201 5. ****Competitive response**** (SwingVision or a team-sports incumbent adds racket sports).
202 ***Validate:*** ship a badminton-quality advantage incumbents can't trivially match

(offline self-improving detection, per-rally tagging).

6. ****B2B sales-cycle drag**** for clubs/federations/schools. ***Validate:*** single-coach/academy self-serve before institutional deals.

7. ****Clean-licensing positioning**** matters mainly to B2B buyers ? confirm it's a real purchasing criterion via 5?10 buyer interviews.

****Cheapest next steps:**** (a) 10?15 discovery interviews with Indian badminton academy owners/coaches; (b) a capture-quality benchmark on real amateur footage; (c) a fake-door pricing test across B2C (~USD 99/yr) and B2B academy (~USD 600/yr).

7. Sources

- Sports analytics market: Precedence Research; Fortune Business Insights; Grand View Research; MarketsandMarkets; Future Market Insights. ***(2024?2025)***
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 - <https://www.fortunebusinessinsights.com/sports-analytics-market-102217>
 - <https://www.grandviewresearch.com/industry-analysis/sports-analytics-market>
 - <https://www.marketsandmarkets.com/PressReleases/sports-analytics.asp>
 - <https://www.futuremarketinsights.com/reports/sports-analytics-market>
- Action camera market: <https://www.gminsights.com/industry-analysis/action-camera-market> (2024);
 - <https://www.precedenceresearch.com/action-camera-market> (2024);
 - <https://electroi.com/stats/action-camera-statistics/> (2024)
- Badminton participation/market: <https://badmintonbites.com/how-many-badminton-players-are-there-in-the-world/> ;
 - <https://stringsandpaddles.com/where-is-badminton-played-10-countries-where-its-popular/> ;
 - <https://en.wikipedia.org/wiki/Badminton> ;
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- Tennis: <https://www.itftennis.com/en/news-and-media/articles/itf-global-tennis-report-participation-hits-106-million-in-five-years/> ;
 - <https://sgbonline.com/itf-global-tennis-report-2021-shows-worldwide-participation-gains/>
- Pickleball: <https://sfia.org/resources/as-pickleball-continues-unprecedented-growth-in-every-age-group-and-region-for-third-straight-year-significant-investments-still-needed-for-court-and-facility-demand/> ;
 - <https://www.thedinkpickleball.com/report-pickleball-participation-surges-to-nearly-20-million-in-2024/> ;
 - <https://www.pickleheads.com/guides/pickleball-statistics> ;
 - <https://www.custommarketinsights.com/report/pickleball-market/> ;
 - <https://market.us/report/pickleball-market/>
- Table tennis: https://en.wikipedia.org/wiki/International_Table_Tennis_Federation
- Cricket: <https://cricketrise.com/cricket-fans-in-india/> ;
 - <https://www.nielsen.com/insights/2025/global-cricket-connecting-fans-driving-roi/>
- Competitors & pricing: <https://www.véo.com/en-us/pricing> ; <https://traceup.com/pricing> ;
 - <https://www.pixellot.tv/> ; <https://www.hudl.com/pricing> ;
 - <https://swing.vision/subscribe> ;
 - <https://www.badproplusapp.com/>

12. user (2026-06-10T06:00:21.449Z)

```

1      # Architecture Decision Records
2
3      ## ADR-001: Precise shuttle tracking via self-hosted TrackNetV4/WASB on WSL2
4
5      **Date:** 2026-05-25
6      **Status:** Accepted
7
8      ### Context
9

```

```

10 The pipeline uses an AI multimodal provider (Gemini) for *semantic* rally segmentation.
11 We need **precise frame-level detection of the moving shuttlecock** (and, generically,
12 balls for other sports) to produce higher-quality candidate rallies and a fine-tuning
13 signal.
14
15 We evaluated whether a commercially available model/API could do this. Findings:
16
17 - **Multimodal LLM APIs (Gemini, Claude, GPT-class) are the wrong tool for precise
18   localization.** They reason well about *what* is happening (rally vs. dead time) but
19   cannot reliably track a tiny, fast (~up to 400 km/h), motion-blurred shuttle frame by
20   frame.
21 - The precision leaders are **purpose-built trajectory models**, which are open source
and
22   self-hosted, not big-vendor APIs:
23   - **TrackNetV4** (TensorFlow) ? badminton-specific, heatmap + motion-attention,
24     highest reported shuttle accuracy; ships `src/predict.py` producing per-frame (X, Y)
25     coordinate CSV. https://github.com/AR4152/TrackNetV4
26   - **WASB** (PyTorch) ? multi-sport strong baseline (badminton/tennis/volleyball/?),
27     fits our `SportProfile` abstraction; needs a custom-video inference wrapper.
28     https://github.com/nttcom/WASB-SBDT
29   - True turnkey commercial APIs (Roboflow hosted models) exist but cap below a tuned
30     TrackNet; enterprise systems (Hawk-Eye, SkillCorner, TRACAB) are not API-accessible.
31
32 ### Decision
33
34 1. Adopt **self-hosted TrackNetV4** as the primary precise shuttle detector, with
**WASB**
35   as the multi-sport fallback.
36 2. Run these models under **WSL2 (Ubuntu)**, **not** native Ubuntu. The entire indexer
37   already runs on Windows; WSL2 keeps a single dev environment and quarantines the
38   TensorFlow/PyTorch dependencies. Native Ubuntu would require dual-boot and migrating
39   the whole project for marginal gain.
40 3. Integrate via a **subprocess adapter**: a detector module that shells out to the WSL
41   `predict.py`, reads the coordinate CSV, and converts trajectory ? candidate rallies
42   feeding the existing stitching flow. This keeps the TF dependency fully isolated from
43   the main Python environment.
44
45 ### Consequences
46
47 - **WSL I/O caveat:** `/mnt/c` access is slow. Stage match videos inside the WSL
48   filesystem (e.g. `~/clips/`) before inference rather than reading large mp4s across
the
49   Windows?WSL boundary.
50 - TrackNetV4 (TensorFlow) and WASB (PyTorch) live in **separate conda envs**.
51 - Next step after integration: manual video annotation ? fine-tune the detector.
52
53 ---
54
55 ## ADR-002: Use the sports-data-collector's rally annotations as a golden eval +
temporal-segmenter training set
56
57 **Date:** 2026-05-26
58 **Status:** Accepted
59
60 ### Context
61
62 A sibling project, **sports-data-collector**, curates sports videos and stores per-rally
63 `[start, end)` timestamps (plus stroke counts, scores) in an SQLite DB, with a
64 commercial-license flag per source. We asked whether this data could improve the
65 indexer's rally detection, and for which task.
66
67 Key findings:
68
69 - The annotations are **temporal-only** (when rallies start/end). They do **not**
contain

```

```

70     per-frame shuttle `(x, y)` coordinates, so they **cannot train the shuttle detector**
71     (TrackNetV4/WASB ? see ADR-001), which needs coordinate labels.
72     - They are an excellent **golden evaluation set** and a **training signal for the
73     *temporal* rally segmenter** (active vs. dead play over time) ? the stage that is
today
74     a hand-tuned heuristic, not a learned model.
75     - The collector's five sports tables (`badminton_rallies`, `tennis_points`,
76     `cricket_deliveries`, `pickleball_rallies`, `tabletennis_rallies`) all share a
77     `(video_id, ordinal, start_time, end_time)` shape, and the indexer already has
78     `sport_registry` / `segmenter_registry`. So a *generic* `(start, end)` bridge serves
79     every sport.
80     - The two repos key videos differently: the collector by a human `video_id`, the indexer
81     by **file MD5**. A re-encode/re-download changes the bytes, hence the MD5.
82
83     ### Decision
84
85     1. Treat the collector's rally timestamps as (a) the **golden eval set** for rally
86     *segmentation* and (b) the **labels** for training a temporal segmenter ? explicitly
87     **not** as shuttle-detector training data.
88     2. Bridge the repos with a sport-agnostic **export** (`curate export` in the collector):
89     per-video `start,end` CSVs (the indexer's eval format) + a `manifest.json` that pairs
90     each CSV with its video file. Gate on `is_commercial` so only commercially usable
data
91     flows through.
92     3. Honour the **MD5 keying** contract: evaluation/processing must run on the exact video
93     file recorded in the manifest; acquisition (ADR-003) therefore precedes processing.
94
95     ### Consequences
96
97     - The collector becomes the *producer* of labels; the indexer the *consumer* (processing
+
98     scoring + training). Neither takes a hard dependency on the other beyond the manifest.
99     - Scoring is free and deterministic (reads the DB; no API). See `docs/EVALUATION.md`.
100    - Only invest in frame-level coordinate labels (for the detector) **if** evaluation
shows
101    *detection*, not *segmentation*, is the bottleneck.
102
103    ---
104
105    ## ADR-003: Video acquisition ? always best resolution, upgrade in place, authenticated
cookies
106
107    **Date:** 2026-05-26
108    **Status:** Accepted
109
110    ### Context
111
112    The collector's initial crawl downloaded videos with `yt-dlp -f worstvideo` (256x144),
113    mislabelled as 1080p. The indexer needs 720p MP4 for meaningful YOLO/shuttle detection.
114    Re-acquiring full-resolution sources surfaced several hard-won realities:
115
116    - The ShuttleSet corpus **tops out at 1080p** (no 4K); "best available" resolves to
1080p.
117    - Authenticating with a **YouTube Premium** account is required for reliable,
unthrottled
118    downloads ? anonymous downloads are throttled (~1.3 MiB/s) and **non-deterministic**
119    (identical commands variously yielded 1080p DASH, 1080p HLS, or a 144p fallback).
120    - **Chrome** cookies can't be read on Windows (app-bound encryption / DPAPI, yt-dlp
121    #10927); **Firefox** cookies read cleanly even while open.
122    - The `web`/`web_safari` player clients are forced into **SABR streaming**, which strips
123    download URLs and silently drops to 144p (yt-dlp #12482). yt-dlp's `default` client
124    serves real downloadable 1080p DASH with auth cookies.
125    - Bulk bursts trip YouTube **rate-limiting** (a 22-video burst failed all at once,
though
126    each worked in isolation).

```

```

127
128     ### Decision
129
130     1. **Always fetch the best resolution available** (`bv*+ba/b -S res,...`, merged to
MP4);
131         ffprobe the result and store the *actual* width/height/fps ? never assume.
132     2. **Upgrade in place, never clean-slate:** the `fetch` command re-downloads a video and
133         updates only `local_path`/`resolution`/`fps`, preserving every rally annotation and
134         curation status (deleting would cascade-delete the valuable labels).
135     3. Use **authenticated cookies** (`--cookies-from-browser firefox` / `--cookies file`)
and
136         the **default` player client**; enable the EJS challenge solver (deno).
137     4. **Throttle bulk runs** (`--sleep` + per-video `--retries`/backoff) and treat a
138         below-min-height download as a **failure** (discard, don't commit) so a flaky low-res
139         result never overwrites a good entry.
140
141     ### Consequences
142
143     - All 22 fetchable ShuttleSet matches are now true 1080p (one match, shuttleset_12, has
no
144         source URL in the catalog and remains video-less but keeps its annotations).
145     - Acquisition is reproducible and resumable; the ~26 GB corpus is kept outside git.
146     - Future non-YouTube or 4K sources are supported by the same code (optional
`max_height`).
147
148     ---
149
150     ## ADR-004: Add an offline, API-free *learned* rally segmenter as a first-class mode
151
152     **Date:** 2026-05-26
153     **Status:** Proposed (substrate pending data)
154
155     ### Context
156
157     The default high-quality path (`yolo_hybrid`) refines YOLO candidate windows with
**Gemini**,
158     which costs money and requires network access. With a now-1080p, commercially-licensed,
159     rally-annotated corpus (ADR-002/003), we can learn the refinement step offline instead.
160
161     Crucially, the **API-free signal already exists and is already persisted**:
`yolo_hybrid`
162     Stage 1 is pure YOLO (player tracking ? per-window motion features: peak/mean velocity,
163     active-frame count, track-id count) stored in `candidate_segments.stats`. Gemini is only
164     the Stage-2 refiner. So the rally annotations can **train a classifier to replace
Gemini**,
165     with no external API and **no new labeling**.
166
167     Three candidate substrates were weighed:
168
169     - **Tune `motion` thresholds** on the annotations ? trivial, fully offline, but a low
170         ceiling (pixel motion can't separate rally from crowd/pan/replay). Use as a *floor*.
171     - **YOLO features ? trained classifier** ? reuses already-stored, semantically
meaningful
172         player-motion features; medium-high ceiling; moderate effort. *Recommended primary.*
173     - **WASB shuttle-motion ? rally detector** ? highest ceiling (the shuttle defines play),
174         fully offline but heaviest (needs the WSL2 WASB runner, ADR-001); can't train WASB
175         itself (no xy labels), only a rally layer on its output.
176
177     ### Decision
178
179     1. Build an **offline, API-free learned segmenter** as a new registry mode (e.g.
180         `yolo_learned`): label detector windows by overlap with GT rallies, train a
lightweight
181         classifier on the persisted features, emit final segments with **zero API calls**,
and

```

```

182     score it with the existing harness against `motion` and a Gemini reference.
183 2. **Choose the substrate from data, not assumption.** A learned refiner can only
sharpen
184     the windows the detector found, so the deciding metric is the detector's **candidate-
stage
185     recall** vs. the annotations: high recall ? build on YOLO features; low recall ?
bring in
186     WASB's shuttle signal to raise the ceiling.
187 3. Keep the Gemini-backed `yolo_hybrid` as an **upper-bound reference**, not a
production
188     dependency.
189
190 ### Consequences
191
192 - The final tool gains a **self-contained, offline, free** rally mode trained on its own
193   curated data ? runnable air-gapped, improvable as more matches are annotated.
194 - The approach is **sport-generic**: the same label?feature?classifier machinery extends
to
195   tennis/cricket once those have annotations.
196 - The full paid 22-video Gemini evaluation is **not** run; spend is limited to a single
197   reference video.
198
199 ### Update (2026-05-26): Phase 3 evidence ? substrate is WASB, not YOLO
200
201 The deciding candidate-recall measurement is in. On shuttleset_8 (64 GT rallies), pure
202 YOLO Stage-1 produced 181 candidate windows with:
203
204 - **Recall @ IoU 0.5 = 0.05**, but recall @ IoU 0.1 = 0.53 and @ IoU 0.01 = 0.72.
205 - **72%** of rallies have some candidate overlap, yet the **median best coverage of a
206   rally by any candidate is only ~20%.
207 - Post-hoc **merging adjacent candidates does not help** (recall stays ~0.05 and
degrades
208   as the merge gap grows ? windows overshoot rather than align).
209
210 Interpretation: YOLO isn't blind to rallies, but its windows are **badly misaligned**
?
211 because the **broadcast camera tracks the players**, their on-screen (pixel) velocity
dips
212 during rallies and spikes on pans/cuts/replays. Player-pixel-motion is therefore a
poor
213 rally proxy on broadcast footage (it was designed for static GoPro recordings). A
214 classifier that only filters these windows is capped at ~0.25 recall ? below any
useful
215 bar ? and no simple boundary post-processing recovers it.
216
217 **Decision update:** per this ADR's data-driven rule (low candidate recall ? bring in
the
218 shuttle signal), the offline segmenter's substrate is **WASB shuttle-motion** (ADR-001),
219 not YOLO player-motion. The shuttle's motion/presence is camera-invariant and directly
220 defines active play. The `motion` segmenter is likewise unusable here (0 TP ? moving-
camera
221 pixel motion is everywhere).
222
223 **Caveats / next:** this is one video; the mechanism (camera tracking) generalizes but
224 should be confirmed on 273 more via a free, local, candidates-only YOLO pass (no
Gemini).
225 Processing is also slow (~50 min per 1080p video) ? the detection pass likely needs
reduced
226 resolution and/or cached detector outputs to be practical across 22 matches. The
227 substrate-agnostic training-data + classifier layers already built still apply: they
will
228 consume **WASB-derived** windows/features instead of YOLO's.
229
230 ---
231

```



```

232     ## ADR-005: Permissive Dependency Stack for the Hosted Monetization Path
233
234     **Date:** 2026-05-30
235     **Status:** Accepted
236
237     ### Context
238
239     The project will be monetized as a closed-source hosted API (users upload videos to
a cloud GPU VM and get highlights). Because we never distribute code/binaries, GPL/LGPL
distribution copyleft is not triggered ? but AGPL §13 (network clause) and metered-API
terms are, and video-codec patent royalties are a separate concern from software licenses.
A deep-research audit (2026-05-30, 25 adversarially-verified claims against primary sources ?
see `docs/MONETIZATION_AUDIT.md`) identified the blockers and clean alternatives. This ADR also
resolves the open licensing question in ADR-001/ADR-004 about WASB.
240
241     ### Decision
242
243     1. Drop `ultralytics`. Both the library and the `yolov8n.pt` weights are
AGPL-3.0; §13 makes them a blocker for a closed-source hosted service. Swapping the codebase
while keeping the `.pt` does not escape ? the weights are AGPL too. Replace player/motion-
gating detection with a permissive detector: torchvision Faster R-CNN / RetinaNet (BSD,
default ? zero new deps, person = COCO class 0), escalating to YOLOX or RT-DETRv2 (Baidu
repo, Apache-2.0) if more speed/accuracy is needed. Ultralytics Enterprise License is a
paid fallback only (quote-only pricing).
244     2. Shuttle tracking stays permissive. WASB-SBDT and TrackNetV3 are MIT (CLEAR for
commercial use) ? this resolves the licensing unknown flagged in ADR-001/ADR-004. MonoTrack
(Adobe Research, noncommercial) and YOLO-NAS weights (Deci, noncommercial) are banned. Train
our own checkpoints (or confirm per-checkpoint terms), since distributed research
weights/datasets aren't covered by the MIT code grants.
245     3. LLM step ? keep all three strategies open: (A) eliminate via an offline temporal
segmenter (ActionFormer, MIT ? the Phase-4 direction), (B) self-host Qwen2.5-VL-7B/32B
(Apache-2.0) or InternVL2.5 (MIT), or (C) keep Gemini 2.5 Flash on the paid tier
(required ? the free tier trains on and human-reviews uploaded video). The google-genai SDK
itself is Apache-2.0 (clear).
246     4. Emit royalty-free output. Standardize highlight output on AV1 (+ Opus) to
avoid AVC/HEVC encode-side patent exposure; provision an Ada-class GPU (L4/L40) for hardware
AV1 (NVENC), falling back to SVT-AV1 (CPU, BSD-3-Clause-Clear) on T4/A10/A100.
247
248     ### Consequences
249
250     - Eliminates the only two hard legal blockers; the rest of the stack (torch, opencv,
fastapi, numpy, lapx, ffmpeg-as-subprocess, google-genai SDK) is already permissive.
251     - Detector swap is contained to `backend/indexer/yolo_hybrid.py` (the `from ultralytics
import YOLO` load + the `model.track(...)` call) plus `config.json` knobs and
`tests/test_yolo_hybrid.py` mocks. Stage-1 feature outputs (velocity/active-frame/track-id
stats) and Stages 2?4 are detector-agnostic and unchanged.
252     - We now own the frame loop + tracking (ultralytics did both internally). Use a
permissive ByteTrack (e.g. `supervision`, MIT, or the MIT ByteTrack repo) on top of `lapx`
(already a BSD dep).
253     - Residual items needing legal sign-off: exact codec-pool royalty thresholds (AVC/HEVC
decode of uploads), Ultralytics Enterprise pricing, and per-checkpoint weight licenses ? tracked
in `docs/MONETIZATION_AUDIT.md` §7.
254     - The `yolo_hybrid` registry key is now a slight misnomer (no YOLO); keep it for back-
compat or rename in a follow-up.
255
256     ---
257
258     ## ADR-006: Golden-Set Generation via Vendor Bake-off + a Pluggable Annotation Provider
259
260     **Date:** 2026-05-30
261     **Status:** Accepted (process v1; vendor shortlist pending research)
262
263     ### Context
264
265     The product needs golden-set ground truth (hand-verified rally `[start,end]`

```

boundaries) to (a) measure pipeline quality and (b) run **on-demand / surprise regression tests** against the production tool. Two forces shape the design: the hosted-API monetization goal raises a "who views the user's video" concern (ADR-005 / MONETIZATION_AUDIT.md), and the founder is **Bangalore-based** and prefers **local/India vendors**. We already own a deterministic interval scorer (``backend/eval/metrics.py``) and a growing collection of **licensed/owned** match videos.

266

267 Decisions locked with the owner: **licensed/owned video only** (no user uploads to vendors) . **rally `[start,end)` annotation depth** . **tiered turnaround** (fast automated + slow human) . **vendor bake-off** to choose . **Bangalore/India-local preference**.

268

269 **### Decision**

270

271 1. **One scorer, three jobs.** A vendor's ``[start,end)`` output is structurally identical to a prediction, so ``metrics.py`` grades product-vs-truth, vendor-vs-key, and annotator-vs-annotator (IAA) with no new scoring code.

272 2. **Select vendors by measured quality-per-dollar**, via a bake-off: write a precise rally-definition guideline, build a small internal gold key (510 licensed clips), send the same clips to 273 vendors (favoring Bangalore/India), and grade every submission with our own harness. The numbers are the decision.

273 3. **Pluggable `AnnotationProvider` registry** mirroring the existing registry pattern: ``FileProvider`` (today's manual/ShuttleSet flow), ``HumanVendorProvider`` (slow tier ? authoritative truth), ``AutomatedProvider`` (fast tier ? independent model oracle).

274 4. **Regression = existing harness + one new component.** ``regress(video)`` calls a provider for ground truth, runs the production pipeline, and scores with the unchanged ``metrics.py`` against a per-video ``baseline.json``.

275 5. **Privacy by construction.** Golden sets use only licensed/owned footage; all third-party video egress flows through the single provider boundary where DPA / no-reuse / delete-on-delivery / access-logging controls live.

276

277 **### Consequences**

278

279 - Eliminates the "human views user video" concern for golden-set/regression work, because user uploads never enter the pipeline.

280 - **The oracle problem is a first-class caveat:** the fast-tier ``AutomatedProvider`` must be a *different model lineage* than production (else shared blind spots make the test falsely green); fast-tier alarms escalate to the slow human tier before blocking a release.

281 - New docs: ``docs/GOLDEN_SET_VENDORING.md`` (the full process v1) and ``docs/VIDEO_RECORDING_GUIDELINES.md`` (a tracked draft: capture-quality thresholds to be derived empirically from precision/recall, since input quality bounds output quality ? cf. ADR-004's broadcast-camera finding).

282 - The vendor shortlist itself is **pending a primary-source research pass** (favoring India/Bangalore) and is intentionally left blank until verified ? no un-sourced vendor claims.

283 - **Honest limitation:** a licensed/owned corpus may not fully represent real user capture, so regression greens are strong evidence, not guarantees ? the recording-guidelines work aims to narrow that gap.

284

285 **### Update (2026-06-02): shortlist researched; Roora selected for the initial pilot**

286

287 - The **vendor shortlist** is filled in (GOLDEN_SET_VENDORING.md §6, primary-source pass 2026-05-30).

288 - The founder selected **Roora ML Pvt Ltd** (HSR Layout, Bengaluru) for the **initial paid pilot**, diverging from the doc's iMerit/Taskmonk default toward a Bangalore-local, startup-friendly shop. Roora's existence, image/video/temporal capability, and location are verified (rooragr.com); **pilot quality is still to be measured** ? run as a **single-vendor pilot first** (the pilot *is* the calibration), not a multi-vendor bake-off.

289 - **Step 0 ? the annotation guideline ? is authored** (the highest-leverage artifact): collector repo ``docs/annotation/`` ? ``ROORA_BRIEF.md`` (rally CSV schema), ``ANNOTATION_GUIDE.md`` (illustrated guide), plus intake-manifest and NDA templates.

290 - **Pilot scope is 3 sports** (badminton, pickleball, table tennis); the collector's ``import-local`` ingestion for all three is implemented (collector ADR-001).

291 - **Still pending:** the internal **gold key** (510 expert-labeled clips) and the vendor-dependent **GS-3 `HumanVendorProvider`** ? GS-1/GS-2/GS-4 (``FileProvider``, regression runner, automated-oracle scaffold) have already landed (see GOLDEN_SET_IMPLEMENTATION.md).

292

```

293     ---
294
295     ## ADR-007: Audio "thwack" hit-detection (fused with WASB) is the next rally-detection
cue; pose/stance parked
296
297     **Date:** 2026-06-07
298     **Status:** Accepted (direction); experiments gated on golden eval labels
299
300     ### Context
301
302     After building the WASB shuttle-trajectory substrate (ADR-001), a Gemini-teacher ?
local-student distillation
303     (ADR-004 lineage), and measuring both, two honest findings emerged: *(a)* threshold-
tuning the local
304     windowing ceilings around F1 ? 0.44 and the learned student *underperforms* the
threshold baseline at n=2
305     labeled videos; *(b)* the dominant failure is *candidate RECALL* ? WASB-derived
windows simply don't align
306     with (or miss) true rallies (IoU-0.5 recall ceiling ~0.43 on one clip). To decide
whether teacher?student was
307     even the right backbone, we ran a deep-research survey
(`docs/RALLY_DETECTION_RESEARCH.md` ? 23 sources, 25
308     adversarially-verified claims). Verdict: *teacher?student is sound but NOT uniquely
best*; the strongest
309     direction is a *complementary, fully-local multi-cue stack*, and the *#1 cheapest
lever is audio hit
310     detection fused with the trajectory** (audio+visual fusion measurably lifts rally
*recall* ? tennis HMM 83%
311     fused vs 54% visual / 63% audio). Pose/court serve-gating is the validated #2 but is
engineering-heavier.
312
313     ### Decision
314
315     1. *Adopt a fully-local AUDIO hit-detection cue* (shuttle "thwack" impacts) and *fuse
it with the WASB
316     trajectory* ? primarily to recover rallies WASB misses (recall) and secondarily as
features for the
317     distilled student. Detailed design: `docs/AUDIO_RALLY_DETECTION_PLAN.md`.
318     2. *Gate on an empirical GO/NO-GO probe (E1).* Shuttle impacts are quiet vs tennis/TT,
so before building
319     anything heavy, measure on *real GoPro audio* whether hit-clusters align with golden
rallies and recover
320     WASB misses. Build E2/E3 only if E1 passes.
321     3. *Keep Gemini as an OFFLINE teacher only* (never runtime) ? research confirms this
is the correct role for
322     a cloud VLM; it is not the deployed localizer.
323     4. *Park pose/court-geometry stance detection as the #2 cue* (serve-ready START +
flight-gradient END). It is
324     expected to be substantially harder to get right; revisit after audio (and a TBD
owner hunch).
325     5. *Make "audio always enabled" a capture requirement*
(`docs/VIDEO_RECORDING_GUIDELINES.md`), since the cue
326     is only as good as the recording.
327     6. Stay permissive (ADR-005): classical DSP (numpy + stdlib `wave`, BSD), no restrictive
weights; any later model
328     trained in-house.
329
330     ### Consequences
331
332     - New `backend/pipeline/audio/` module (extraction + onset/peak + confidence), feeding
`distill_local` and scored
333     by the existing `metrics.py` / golden loaders. *(As built: the detector is **pure
numpy + stdlib `wave`, no
334     scipy** ? the anticipated scipy dependency proved unnecessary.)*
335     - *All audio measurement is blocked on golden eval labels* ? which the VLC rally-

```

annotator (now a standalone
336 MIT, multi-sport open-source repo: <https://github.com/avidullu/rally-annotator>) lets
the owner self-produce,
337 and the in-repo `--labels` loaders / collector `import-local` ingest.
338 - Sport-generic: tennis/table-tennis hits are *louder*, so the cue should transfer at
least as well there.
339 - **Honest risk:** if E1 shows shuttle SNR is too poor on real amateur audio, audio is
deferred ? this ADR
340 commits to the *experiment and direction*, not to a guaranteed win.
341
342 ### Integration architecture: modular fusion now (Option A), an owned end-to-end model
later (Option B)
343
344 A recurring design question: do we **feed audio into WASB** so it refines its own
output, or **keep audio as a
345 separate signal fused in our code**? The answer today is the latter ? and understanding
why sets up the
346 long-term route.
347
348 **Why fusion lives in our rally layer, not inside WASB (Option A ? now).** WASB is a
**frozen, vision-only
349 shuttle *detector*** (image frames ? per-frame shuttle `(x, y, visibility)`; MIT weights
we *run*, not train).
350 It has no audio pathway and it does not decide rallies at all ? **rally detection is our
own layer** on top of
351 its trajectory (`trajectory\_to\_action\_windows` ? gate ? the distilled classifier).
Making WASB consume audio
352 would mean bolting on an audio encoder and **retraining the network**, which needs per-
frame
353 shuttle-coordinate-*plus*-audio labels we don't have (we only have rally `[start,end]`
labels) and would fork
354 the clean MIT model. It also conflates two different questions: WASB answers ***where*
the shuttle is**; audio
355 answers ***when* a hit happened**. Audio doesn't make WASB localize better ? it's an
independent rally-activity
356 cue. So fusion belongs where cues are *combined into rallies*: our layer.
357
358 ```
359 frames ?? WASB (frozen, MIT) ?? shuttle trajectory ??
360 audio ?? hit detector (ours) ?? hit events ?????????? RALLY LAYER (ours) ?? rallies
361 [pose ?? stance heuristic (ours; the parked #2) ??? (feature + decision fusion)]
362 ```
363
364 Two complementary fusion modes (both in `docs/AUDIO\_RALLY\_DETECTION\_PLAN.md`):
365 1. **Feature-level (primary):** audio features (hit_rate, inter-hit cadence/regularity,
time-since-last-hit)
366 join WASB-derived features (velocity, net-crossings) in the **single distilled
classifier** that emits
367 rallies ? "use audio to sharpen our rally metrics" (experiment E3).
368 2. **Decision-level recall-recovery:** where WASB proposed nothing but a dense hit-
cluster exists, propose a
369 rally there ? directly correcting WASB's recall misses.
370
371 Each cue (WASB, later the person/pose detector) stays a **swappable black box** ; every
signal is independently
372 testable ? which matters while we're label-starved.
373
374 **The long-term route (Option B ? the intended destination, once a healthy golden corpus
exists).** As we
375 accumulate enough golden eval/training videos (owner self-rating via the annotator +
vendor labels), the right
376 end-state is to **collapse the modular cues into a single OWNED multimodal rally
model**, trained/fine-tuned
377 end-to-end on (frames + audio + pose/stance + ?) ? rally boundaries ? the HitNet /
temporal-action-localization

378 direction surfaced in `docs/RALLY\_DETECTION\_RESEARCH.md`. Crucially, that lets us
 **fine-tune from our own code

379 and data instead of depending on WASB's pretrained weights** ? removing both the *recall
 ceiling* imposed by

380 WASB's candidate windows and the *licensing question* on its academic-trained weights.

381

382 **Why we don't jump to Option B now ? and why Option A is the path to it.** Option B
 needs data we don't yet

383 have, and it's far harder to debug as one black box. So we deliberately build **modular
 cues first** ? audio

384 (this ADR), then a stance heuristic (the parked #2), then other signals ? each cheap,
 debuggable, label-

385 efficient, and *each one generating the very golden labels an end-to-end model will
 need*. When the corpus is

386 healthy, we train the unified model. **Option A is not a detour from Option B; it is how
 we earn it.**

387

13. user (2026-06-10T06:00:29.432Z)

1 # Checkpoint: 2026-06 Strategy Foundation (post-scaffolding)

2

3 **Date:** June 2026

4 **STATUS: COMPLETED**

5 **Trigger:** Owner merged the post-scaffolding strategy docs ? #52 (config/identity
 cleanup), #53 (deferred-hardening design), #55 (commercialization tracker), #57 (PMF market
 study), #58 (platform architecture).

6

7 ## Purpose

8 Captures the state once the **strategic / direction-setting foundation** for the
 9 post-scaffolding phase was laid: the commercial dimension, market/PMF, the multi-user
 10 platform architecture (pluggable storage/compute + the owned-model layer), and the
 11 deferred-hardening designs. **Strategy only ? no implementation started.** Provides a
 12 regression-detection baseline for priorities/decisions as quality + implementation work
 begins.

13

14 ## Key living documents at checkpoint time

15 - **COMMERCIALIZATION.md** ? commercial tracker (dashboard C1?C14, grouped Pending ? In-
 flight ? Completed with a P0/P1/P2 column).

16 - **PMF_MARKET_RESEARCH.md** ? cited market study; beachhead = **India badminton
 academies**.

17 - **PLATFORM_ARCHITECTURE.md** ? multi-user "assistant coach" platform: pluggable
 `StorageBackend`/`ComputeBackend` (BYO), generic schema (coach overlay optional), owned video-QA
 model layer (\$5A), training-backend survey + nanochat-style harness, committed legal guardrail,
 per-user-adaptor direction, v1-vs-vision scoping.

18 - **DEFERRED_HARDENING_PLAN.md** ? packaging, train/eval guardrails, async job model
 (#16) ? *pending approval*.

19 - **CODE_MAP.md / BACKLOG.md / ROADMAP.md** ? updated for the config/identity cleanup.

20

21 See git history around #52/#53/#55/#57/#58 for exact snapshots.

22

23 ## Committed decisions / non-negotiables recorded here

24 - **Paid frontier models (Gemini/OpenAI/Anthropic): inference / serving / fallback only
 ? never a training data source** (competing-model terms). Teachers/bases = open Apache-2.0 only.

25 - **Owned-model base = Apache-2.0-clean, video-capable** (Qwen2.5-VL-7B/32B, Qwen3-VL,
 Gemma 4); **reject Gemma 3** (viral license clause); **minors excluded** from the training pool;
 per-user training needs a separate opt-in.

26 - **v1 = rally detection + highlight compilation + per-user history + coach/user
 correction loop** ("assistant-coach lite"); **gait/biomechanics strictly future** (Art. 9).

27 - **Generic data schema** (User ? VideoCollection ? Video ? Highlight ?
 HighlightMetadata); coach ? athlete is an **optional overlay** (a market-fit hypothesis).

28

29 ## What comes next (per owner)

30 - First implementation slice = **async job model (DEFERRED #16)** , single-tenant ?
 pending explicit go-ahead (no implementation started without approval).

```

31      - Quality iterations with new labeled videos; golden-set expansion + the **golden QA
eval set** (the prerequisite for the owned-model training harness).
32
33      When the first implementation slice or a quality milestone completes, add a new dated
34      checkpoint here (or under `quality-iterations/`).
35

```

14. user (2026-06-10T06:00:36.058Z)

```

1      # Owned-model training study (rally segmentation ? end-to-end multimodal)
2
3      **Date:** 2026-06-08 . **STATUS: DESIGN BLUEPRINT (no implementation until PLATFORM
P0?P4 + healthy corpus; owner sign-off required).**
4
5      > How to train our **OWNED** rally/highlight model **reusing the videos we label**,
aligned to the
6      > Apache-2.0 / no-paid-LLM-training / privacy-local guardrails. Produced by an 11-agent
research
7      > workflow (ground ? 5 web-research dimensions ? adversarial verification of
costs/licenses/privacy ?
8      > synthesis + completeness critic; 35 sources). Empirical results from THIS session
(LOVO + a learned
9      > prototype) are folded in and **resolve the study's biggest open unknown** ? see $"What
we proved today".
10     > Numbers from web sources are 2026 snapshots in a ±15?25% band; **re-quote live before
budgeting.**
11
12     ## Update (2026-06-09, post-PR-#61-review) ? authoritative roadmap is now
`docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md`
13     A verified strategy workflow + owner review refined this study. Key changes (the plan
supersedes where they differ):
14     - **VISION (foregrounded):** the goal is **rally/shot/player analytics + highlights for
ALL net-separated racquet
15     sports** (badminton/tennis/TT/pickleball/padel) ? eventually all sports, as **ONE
sport-agnostic, conversational
16     model**. This does **not** change the model/framework picks; it scales **data + per-
sport trajectory detectors** (the
17     accepted "good problem") and adds a licensing front (clean per-sport detectors).
Single-model temporal generalization
18     is an **open hypothesis** (corpus is badminton-only; the once-cited RacketVision
evidence was **refuted** ? it's
19     ball-tracking, not temporal).
20     - **Framework = ms-swift (Apache-2.0)** (only forge with native video+audio+timestamp-
grounding+video-GRPO).
21     **Base = Qwen3-VL (Apache-2.0, uniform sizes, timestamp emission)** ? primary,
**replacing the Qwen2.5-VL-7B framing**.
22     - ? **Gemma 4 stays AMBER and bench-only** (NOT "confirmed clean" as the table below
says): Apache grant likely but
23     Prohibited-Use posture unconfirmed (counsel) + video/timestamp capability disputed ?
not the primary extractor.
24     - **Start with the TAL head, not the VLM** (VLMs miss strict boundaries ? boundary
accuracy is the product). Head now ? VLM later.
25     - **Immediate goal = a high-P/R extraction *harness*, not on-device** (on-device
deferred). **Reframe supersedes the on-device-now framing.**
26
27     ## Thesis
28
29     Train the owned model in **three staged generations off ONE model-agnostic labeled
corpus** ? a label
30     collected once feeds every generation, so labels are never re-collected. **The moat is
the consented
31     dataset + the eval harness, never the weights.**
32
33     | Gen | When | Approach | Base / model | Compute |
34     |---|---|---|---|---|

```

35 | ****Gen-0**** | now ? unblocks at n?5 matches | tiny ****temporal action segmentation (TAS)**** head over a hand-built per-frame feature stack from the WASB trajectory we already produce; per-frame rally(1)/non-rally(0) ? merge into [start,end] | ****ASFormer (MIT, ~1.1M params)**** primary; ****ActionFormer (MIT)**** A/B challenger (overtakes once corpus = hundreds of matches) | ****local RTX 2070 Super****, trains in minutes, \$0 |

36 | ****Gen-1 / Option A**** | 6?12 mo | concatenate more local cue channels onto the SAME head: trajectory + audio-onset + pose/court serve-gating; fusion in our rally layer, cues swappable | same ASFormer/ActionFormer, wider input | local + occasional cloud burst (pseudo-labeling) |

37 | ****Gen-2 / Option B**** | 12?24+ mo, after scaffolding + healthy corpus | end-to-end multimodal VLM: (frames+audio+pose+trajectory) ? rally [start,end] (+semantics); SFT cold-start then ****GRPO/RFT with a temporal-IoU-vs-golden reward**** | ****Qwen3-VL-4B-Instruct**** (Apache-2.0, on-device) primary; ****Qwen2.5-VL-7B**** fallback; ****Gemma 4 E4B/12B-Unified**** (Apache-2.0 + native video+audio) end-state | rented 24?48 GB A100/L40S (does NOT fit 8 GB box) |

38

39 ## What we proved today (de-risks the blueprint)

40

41 The study's ****"dominant empirical unknown"***** was: *can a low-dim WASB trajectory (?3?4 raw dims) feed a

42 TAS head, given ASFormer's small-data result used 2048-d I3D features?* This session answered it.

43

44 - ****Cross-video LOVO of the current heuristic**** (`calibrate\_local`, Adarsh-96 + Kushagra-32, human labels):

45 mean held-out F1 ****0.443 ± 0.279****; Adarsh 0.722, ****Kushagra 0.163**** (its own ceiling ? *no* velocity-windowing

46 config does better). ****The ±0.279 spread is method variance, not config.**** ? This also ****corrects the study's**

47 ****"~-0.73 bar"*****: that was a fit-on-self single-match number; the honest cross-video heuristic bar is ****~0.44****,

48 which ****lowers**** the Gen-0 ship gate.

49 - ****A learned per-frame prototype**** (`backend/eval/rally\_seq\_proto.py`: 6 hand-built features ? visibility rate,

50 mean/max shuttle speed, ****net-crossing rate****, x/y spread ? into the repo's dependency-free LogisticRegression,

51 honest LOVO): ****per-frame AUC 0.878 (Adarsh) / 0.841 (Kushagra), held-out.**** On Kushagra ? the match the

52 heuristic ****cannot**** segment (0.163) ? a calibrated threshold reaches ****segment F1 0.578 (3.5x)****. The naive

53 cross-video threshold gives only 0.049, so ****the remaining gap is threshold/operating-point calibration at n=2,**

54 not representation.**

55

56 - ****n=3 update (doubles match added ? `LargeTestVideo`, 49 rallies):**** the learned prototype ****overtakes the**

57 heuristic.** Mean held-out LOVO F1 ? learned ****0.377 ? 0.615****, heuristic ****0.443 ? 0.544****. Kushagra's learned

58 held-out F1 jumped ****0.049 ? 0.417**** (one added match closed most of the calibration gap); the ****doubles**** match,

59 held out and trained only on the two singles, scored AUC ****0.914**** / F1 0.729 (generalizes across play type +

60 frame rate). The heuristic still cannot rescue Kushagra at any n (0.151, capped). ****The honest "bar to beat" is**

61 thus ~0.54 (n=3), not 0.73 ? and the learned path already clears it on the mean and keeps rising with n.**

62

63 ****Conclusion:**** the rally signal IS learnable from the WASB trajectory features, cross-footage (AUC ~0.85?0.91) ?

64 the heuristic just can't exploit it. This ****validates the Gen-0 ASFormer-over-WASB-features path****, shows the

65 hand-built feature stack the study calls for is the right lever, and confirms the bottleneck is ****data + per-video**

66 calibration (n?5 matches), not method.** The prototype is the seed of Gen-0, and the n=2?n=3 trend (learned mean

67 +0.24, overtaking the heuristic) is the empirical green light.

```

68
69     ## Platform ranking (privacy + cost lens; the fine-tune is a sub-24h, $10?50 job ?
privacy/ergonomics dominate)
70
71     1. **Modal ? PRIMARY.** Strongest *written* no-access contract (verbatim: "Modal will
never access? the inputs/outputs to your Functions, or any data you store"), ?7-day TTL,
SOC2-II, HIPAA-BAA; per-second scale-to-zero matches retrain-on-each-batch. A100 80 GB
~$2.50/hr, H100 ~$3.95/hr (corrected down from cited). Privacy is the load-bearing reason, not
price.
72     2. **RunPod (Secure Cloud tier ONLY) ? FALLBACK / Option-B scale-up.** Cheapest vetted-
DC self-serve; A100 80 GB ~$1.39?1.49/hr. Community/P2P tier is UNVETTED ? **never** touches
consented video. Confirm AES-256/DPA at booking.
73     3. **Vertex AI / GCP ? ON BENCH.** Only strong-privacy option with an India region, but
it's **capacity-gated** (asia-south1 high-end GPU is invitation-only, H100-class A3 not A100)
and **tuning Gemini yields NO portable on-device weights** ? only tuning an Apache-2.0 base in
Model Garden does. Use only if India residency-at-rest is mandated.
74     4. **Self-hosted RTX 2070 Super 8 GB** ? Gen-0 training + ALL inference + CI only.
**Cannot** QLoRA a 7B video VLM (~20?24 GB). Best privacy (nothing leaves the box).
75     5. **Fireworks** (managed, contractual no-train + ZDR) usable; **Together BLOCKED**
until a signed DPA + explicit no-train clause. AWS SageMaker / Azure ML = heavy for a solo
founder; India-residency only.
76
77     ## Label reuse ? one corpus, three transcoders (the actual moat)
78
79     Promote the rally CSVs from eval-only files to **one model-agnostic corpus**: JSONL, one
row per
80     `(video_id, rally_ordinal)` = `{start, end, ending_reason, sport, source
(human|pseudo|vendor),
81     annotator/provenance, consent/opt-in, label_type, split (BY MATCH not clip),
trajectory_ref}`. The
82     provenance/consent fields are the **enforcement point** for the legal guardrails. Three
deterministic
83     transcoders read it, zero re-labeling:
84     1. **Heuristic/LogReg view** ? already exists
(`backend/eval/training.py::label_candidates` via `metrics.match_intervals`).
85     2. **TAL view** ? ActivityNet JSON (`segment:[start,end]`, `label:'rally'`); our seconds
map 1:1.
86     3. **VLM clip?QA view** ? "list every rally [start,end] in this clip"; `ending_reason` ?
a second QA task.
87
88     **One scorer spine:** reuse `metrics.match_intervals` for heuristic-labeling, TAL eval,
AND the Gen-2 RFT
89     reward ? no metric drift. **Leakage guard:** split **by match, never by clip**.
**Amplification (all
90     guardrail-clean):** open-teacher pseudo-labels (WASB/own-model, confidence-gated ? never
paid-LLM),
91     boundary-preserving temporal augmentation, active learning to spend the Roora budget on
highest-uncertainty
92     matches. ? **The new partial-label windowing**
(`eval_partial_labels.restrict_to_window`) must flow into the
93     TAL/VLM views so prefix-labeled videos aren't mis-scored (a corpus that's partially
tagged is the norm).
94
95     ## Licensing & privacy guardrails (all respected; verify per-checkpoint)
96
97     - **Bases: Apache-2.0/MIT only.** APPROVED: Qwen2.5-VL-7B/32B (Apache-2.0),
Qwen3-VL-4B/8B (Apache-2.0),
98     Gemma 4 E2B/E4B/12B-Unified (Apache-2.0 + native video/audio), InternVL3 ?38B
(MIT/Apache), ASFormer/
99     ActionFormer/TriDet (MIT). **REJECTED:** Gemma 3 (viral Gemma Terms + PUP),
Qwen2.5-VL-3B (NC research),
100     Qwen-72B/InternVL3-78B (100M-MAU cap), Llama Vision (700M-MAU), DAMO VideoLLaMA3
weights (NC + paid-LLM-trained),
101     TimeChat/VTimeLLM/Momentor (NC/no-license = method-reference only), VideoMAE weights
(CC-BY-NC), `yabufarha/ms-tcn`

```


102 (Commons Clause ? use `sj-li/MS-TCN2` MIT). ? **Gemma 4 size names = E2B/E4B/26B-A4B/31B/12B-Unified ? there is no plain "4B".**

103 - **No paid-LLM training data** ? Gemini/OpenAI/Anthropic = inference/eval/teacher-reference only, **never** a

104 training signal. Gemini silver labels stay out of every transcoder.

105 - **Exclude minors by default + separate training opt-in;** per-user adapters trained only on that user's

106 consented video (deletion = drop the adapter). Enforced at the corpus-membership layer before transcoding.

107 - **Privacy lever = local inference** (Gen-0 ~1.1M params; Gen-2 4B quantized). Any cloud *training* needs a

108 signed DPA + no-train assurance; route only owned/consented footage, never user uploads.

109

110 ## Cost (spend on labels, not GPUs)

111

112 - **Gen-0:** ~\$0 (local 2070 Super; trains in minutes; ~4.5 GB VRAM).

113 - **Gen-2 LoRA:** ~\$5?40/run on a rented 24?48 GB GPU (RunPod cheapest; **VRAM/token cost on real 1080p/60fps

114 clips is UNVERIFIED ? 24 GB is a floor, re-benchmark**). Managed (Fireworks/Together) ~\$2?30 *if* text-priced

115 (video tokens can balloon). RFT = compute-heavy (multiple rollouts/sample).

116 - **Inference crossover (the unanchored number):** self-hosted 7B beats Gemini (~\$0.05/10-min clip) **only**

117 above sustained QA volume; below it, **paid Gemini is cheaper** ? paid-primary now, owned-primary later.

118 **Benchmark \$/call before committing** (COMMERCIALIZATION C7).

119

120 ## Open risks & critic caveats (read before acting)

121

122 - **?? WASB checkpoint provenance (C3) propagates.** The study calls WASB "MIT," but that's the *code*; the

123 distributed `wasb\_badminton\_best.pth.tar` is **academic-data-trained, provenance unresolved** (COMMERCIALIZATION

124 C3, still open). Every Gen-0/1 plan uses WASB features as input AND as a pseudo-label teacher ? a "clean" owned

125 model trained on provenance-unclear features **does not launder the provenance.**

ROADMAP Phase-4 "train own WASB

126 weights" is a **prerequisite, not an aside.**

127 - **Gemma 4 "clean + native-video" rests on post-cutoff secondary sources** ? keep AMBER until a raw HF LICENSE

128 revision is verified, given the Gemma-3 rejection. Its timestamp-emission recipe is less documented than Qwen's.

129 - **Existing distilled model was trained on Gemini silver labels** ? the (correct) eval-only policy now **forbids

130 this; that pipeline must be purged of Gemini-derived training labels** to comply.

131 - **Policy ratifications needed:** guardrail literally says "Apache-2.0 only" but the approved list includes MIT ?

132 ratify "Apache-2.0/MIT". The MAU-cap rejections (72B/78B) may be over-conservative at our scale.

133 - **On-device privacy claim is unbenchmarked** ? no tokens/sec for a quantized 4B VLM on 8 GB Turing at

134 1080p/60fps. **Measure before relying on it.**

135 - **Amplification is load-bearing for Gen-2** ? 128 raw rallies can't reach the ~5k?20k clip?QA band alone; if

136 augmentation + QA-multiplexing + pseudo-labels underdeliver, Option B is under-fed. Amateur 60fps may need more

137 labels than broadcast-trained literature implies.

138 - **Dropped option worth keeping:** InternVideo2 encoder (MIT) + light grounding head ? the *only* data-realistic

139 owned-video experiment available at n?128 today. And: a **one-time ~\$1?2k 24 GB local card** (e.g. used 3090)

140 would preserve the privacy lever for *training* and dissolve most of the cloud-DPA analysis ? uncoded here.

141

```

142     ## Open questions (need the GPU/WSL machine-side suite ? not runnable in this dev
container)
143
144     1. Does ASFormer/ActionFormer over the WASB feature stack beat the **honest ~0.44 LOV0**
bar at n?5 matches?
145     (Today's prototype says the per-frame AUC is there ? 0.84 ? so this is now *likely*;
confirm at segment level.)
146     2. Real VRAM + token cost of fine-tuning Qwen3-VL-4B / Qwen2.5-VL-7B on our actual
clip?QA frame counts?
147     3. Working Gemma-4 timestamp-emission recipe + exact checkpoint (E4B vs 12B-Unified)?
148     4. Measured self-hosted-7B $/call + latency vs Gemini on the target workload (the owned-
model ROI crossover)?
149     5. Will Together/Lambda/Azure sign a DPA with explicit no-train for consented video,
India-configurable?
150     6. Realistic clip?QA yield from 128 rallies via augmentation + QA-multiplexing + pseudo-
labels?
151
152     ## Sequencing (owner directive, non-negotiable)
153
154     This is a **design blueprint only.** It unblocks PLATFORM **P5** but starts **no
implementation** until P0?P4
155     scaffolding (async jobs, storage/compute backends) is rock-solid AND the golden corpus
is healthy. Paid Gemini
156     stays the always-ready primary/fallback. The only items safe to act on now ? **and even
these need owner
157     sign-off before code** ? are (a) the model-agnostic corpus format and (b) growing n?2 ?
n?5 matches (a small
158     Roora batch), which unblocks honest tuning immediately and is the **highest-ROI spend
(labels, not GPUs).**
159
160     *Full agent transcript + 35 sources: workflow `wf_49aab888-e10` (this session).*
161

```

15. assistant (2026-06-10T06:01:40.397Z)

Now let me verify repo reality against the doc claims.

16. user (2026-06-10T06:01:42.039Z)

Found 2 files
backend\eval\distill_local.py
backend\eval\training.py

17. user (2026-06-10T06:01:43.154Z)

```

config.json:26:         "default_segmenter": "gemini",
README.md:74:This outputs a validation summary and indexes all rallies directly into the local
SQLite database, using whichever segmenter is set as `default_segmenter` in `config.json`
(`yolo_hybrid` by default).
README.md:124:         "default_segmenter": "yolo_hybrid",
README.md:155:*   `indexing.default_segmenter` ? `yolo_hybrid` | `gemini` | `motion`.
README.md:215:   *Or set `"default_segmenter": "yolo_hybrid"` (or `"gemini"`/`"motion"`) under
`"indexing"` in `config.json`.*
README.md:219:[Omitted long matching line]
README.md:237:         "default_segmenter": "gemini",
run_process_and_stitch.py:72:         seg_name = cfg.indexing.default_segmenter
docs\CODE_MAP.md:29: -> seg_name = req.segmenter or global_config.indexing.default_segmenter
(fallback 'motion')
docs\CODE_MAP.md:86: - ? `--segmenter` argparse `choices` frozen at build time from
`segmenter_registry.list_available()` ? a lazily-registered segmenter wouldn't appear. CLI
default_segmenter fallback `motion`.
docs\CODE_MAP.md:96:[Omitted long matching line]
docs\CODE_MAP.md:102:- `@config.json` ? on-disk config; mirrors `AppConfig`. ?
**`indexing.default_segmenter='gemini'`** diverges from model default `yolo_hybrid` ? file

```

```

wins at runtime.
docs\CODE_MAP.md:177:[Omitted long matching line]
docs\CODE_MAP.md:238:- **Segmenter default:** single source of truth = model
`IndexingConfig.default_segmenter='yolo_hybrid`; `config.json` may override at runtime (ships
`gemini`). run *.py read the model value (no literals); `main.py` keeps the defensive `motion`
getattr fallback noted above.
docs\CODE_MAP.md:264:| Change a config default / add a config field |
`@backend/config/models.py` (single source of truth) -> regenerate via
`setup.py` / `save_default_config`; ? config.json may override (default_segmenter) |
backend\config\models.py:47:    default_segmenter: str = "yolo_hybrid"
backend\config\models.py:69:    @field_validator("default_segmenter")
tests\test_config.py:24:    assert cfg.indexing.default_segmenter == "yolo_hybrid"
tests\test_config.py:34:        "default_segmenter": "gemini",
tests\test_config.py:45:    assert cfg.indexing.default_segmenter == "gemini"
tests\test_config.py:58:        "indexing": {"default_segmenter": "yolo_hybrid"},
tests\test_config.py:66:    assert cfg.indexing.default_segmenter == "yolo_hybrid"
tests\test_config.py:78:    assert cfg.indexing.default_segmenter == "yolo_hybrid"
tests\test_config.py:122:    assert cfg.get("indexing", {}).get("default_segmenter") ==
"yolo_hybrid"
docs\TRACKNET_WSL_SETUP.md:71:Then select it as the segmenter (e.g. set
`indexing.default_segmenter` to
backend\reporting\harvest.py:210:    config_default = (config.get("indexing") or
{}).get("default_segmenter")
backend\reporting\spec.yaml:112:    The segmenter that ran differs from config.json's
default_segmenter, and no
backend\main.py:105:    default_seg = getattr(getattr(global_config, "indexing", None),
"default_segmenter", "motion")
backend\main.py:371:    seg_name = args.segmenter or getattr(getattr(global_config,
"indexing", None), "default_segmenter", "motion")
tests\test_reporting_harvest.py:46:    config={"ai_provider": "gemini", "ai_model": "gemini-
flash", "indexing": {"default_segmenter": "wasb_hybrid"}},
tests\test_reporting_harvest.py:98:        config={"indexing": {"default_segmenter":
"motion"}})
tests\test_reporting_harvest.py:107:    cfg = {"ai_provider": "gemini", "indexing":
{"default_segmenter": "wasb_hybrid"}}
tests\test_reporting_harvest.py:150:    config={"ai_provider": "gemini", "indexing":
{"default_segmenter": "wasb_hybrid"}},

```

18. user (2026-06-10T06:01:52.970Z)

```

1      """Build a training set for the offline, API-free learned rally segmenter.
2
3      Phase 4 of the collector->indexer bridge (see docs/ROADMAP.md, ADR-004). The idea:
4      a detector (YOLO player-motion today, optionally WASB shuttle-motion later)
5      produces candidate rally windows with features; the collector's ground-truth
6      rally intervals tell us which windows are real. A lightweight classifier learns
7      that mapping, replacing the paid Gemini refinement with a local model.
8
9      This module is the *data layer* ? pure and unit-tested:
10
11      - `label_candidates` ? generic: label windows positive/negative by whether they
12        match a ground-truth rally (same IoU matching the evaluator uses, so training
13        labels are consistent with how we score).
14      - `extract_yolo_features` ? YOLO-specific feature vector from a candidate row's
15        persisted `stats`. Pluggable: a WASB substrate supplies its own feature_fn.
16      - `build_training_set` ? glue: rows + GT -> (X, y, intervals).
17
18      No model training or I/O here; that lives in the trainer/segmenter once the
19      substrate is chosen from the Phase 3 candidate-recall result.
20      """
21
22      from typing import Callable, Dict, List, Tuple
23
24      from backend.eval.metrics import Interval, match_intervals
25

```

```

26     # YOLO Stage-1 persists these per candidate window in `candidate_segments.stats`
27     # (see HybridYoloSegmenter). Duration is derived from the window bounds.
28     YOLO_FEATURE_NAMES = [
29         "duration", "peak_velocity", "mean_velocity", "active_frame_count",
30         "track_id_count",
31     ]
32
33     def extract_yolo_features(row: Dict) -> List[float]:
34         """Feature vector for one YOLO candidate row (dict from query_candidate_segments).
35
36         `stats` is already JSON-deserialized by the DB layer. Missing fields default
37         to 0.0 so a sparse/old row still yields a well-formed vector.
38         """
39         stats = row.get("stats") or {}
40         duration = float(row["end_time"]) - float(row["start_time"])
41         return [
42             duration,
43             float(stats.get("peak_velocity", 0.0)),
44             float(stats.get("mean_velocity", 0.0)),
45             float(stats.get("active_frame_count", 0)),
46             float(stats.get("track_id_count", 0)),
47         ]
48
49
50     def label_candidates(candidate_intervals: List[Interval], gt_intervals: List[Interval],
51                         iou_threshold: float = 0.5) -> List[int]:
52         """Label each candidate window 1 (real rally) or 0 (spurious).
53
54         A window is positive iff it is matched one-to-one to a ground-truth rally at
55         `iou_threshold` ? the *same* matching the evaluator uses for true positives,
56         so the classifier is trained on exactly what we score. Substrate-agnostic:
57         works for YOLO or WASB-derived windows alike.
58         """
59         mr = match_intervals(candidate_intervals, gt_intervals, iou_threshold)
60         positive = {m.pred_index for m in mr.matches}
61         return [1 if i in positive else 0 for i in range(len(candidate_intervals))]
62
63
64     def build_training_set(
65         candidate_rows: List[Dict],
66         gt_intervals: List[Interval],
67         iou_threshold: float = 0.5,
68         feature_fn: Callable[[Dict], List[float]] = extract_yolo_features,
69     ) -> Tuple[List[List[float]], List[int], List[Interval]]:
70         """Turn one video's candidate rows + ground truth into (X, y, intervals).
71
72         `feature_fn` is pluggable so a WASB substrate can supply shuttle-motion
73         features without changing the labeling logic. Returns the feature matrix `X`,
74         binary labels `y`, and the candidate intervals (handy for downstream merging).
75         """
76         intervals: List[Interval] = [
77             (float(r["start_time"]), float(r["end_time"])) for r in candidate_rows
78         ]
79         y = label_candidates(intervals, gt_intervals, iou_threshold)
80         X = [feature_fn(r) for r in candidate_rows]
81         return X, y, intervals
82

```

19. user (2026-06-10T06:02:10.234Z)

```

1     """Distill Gemini (teacher) into a LEARNED local rally classifier (student).
2
3     Threshold-tuning the local knobs hit a low ceiling (see calibrate_local). This goes
4     further: learn "is this WASB candidate a real rally?" from Gemini silver labels,

```

```

5     using a few resolution/fps-INVARIANT local features, so the model transfers across
6     videos and runs 100% local at inference (no Gemini, privacy lever intact).
7
8     Flow (all local at runtime):
9         WASB trajectory -> recall-oriented candidate windows (propose many; classifier
10         filters) -> per-candidate features [duration, peak_velocity, mean_velocity,
11         active_ratio, net_crossings] -> LogisticRegression(prob >= threshold) -> merge
12         -> rally windows.
13
14     Training labels come from Gemini full-context labels via the SAME IoU matching the
15     evaluator uses (training.label_candidates). Honest leave-one-video-out: train on the
16     other videos' candidates, predict the held-out one, score vs its Gemini labels, and
17     compare against the threshold baseline. Reuses classifier.py, training.label_candidates,
18     rally_gate, metrics, calibrate_wasb.load_golden_gts, gemini_refine.merge_overlaps.
19
20     Run:
21         python -m backend.eval.distill_local --manifest videos.json --model-out
output/local_rally_model.json
22     """
23     from __future__ import annotations
24
25     import json
26     import argparse
27     from typing import Dict, List, Tuple
28
29     import numpy as np
30
31     from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
32     from backend.pipeline.detectors import rally_gate
33     from backend.eval.calibrate_wasb import load_golden_gts
34     from backend.eval.training import label_candidates
35     from backend.eval.classifier import LogisticRegression
36     from backend.eval.metrics import score
37     from backend.eval.gemini_refine import merge_overlaps
38
39     Interval = Tuple[float, float]
40
41     FEATURE_NAMES = ["duration", "peak_velocity", "mean_velocity", "active_ratio",
"net_crossings"]
42     # Recall-oriented proposal: deliberately permissive so real rallies are among the
43     # candidates (the classifier removes the junk). (velocity_thresh, merge_gap, min_dur)
44     PROPOSAL = (0.005, 1.0, 0.5)
45     BASELINE_LOCAL = (0.02, 2.0, 2.0) # today's production-ish windowing, no learning
46
47
48     def build_candidates(trajectory_csv: str, fps: float, frame_width: float,
49                         proposal: Tuple[float, float, float] = PROPOSAL) ->
Tuple[List[Interval], List[List[float]]]:
50         """WASB trajectory -> (candidate intervals, fps/resolution-invariant feature
rows)."""
51         points = TrackNetRunner.parse_trajectory_csv(trajectory_csv)
52         vt, mg, mwd = proposal
53         wins = TrackNetRunner.trajectory_to_action_windows(
54             points, fps=fps, frame_width=frame_width,
55             velocity_thresh=vt, merge_gap=mg, min_window_duration=mwd)
56         intervals = [(w["start"], w["end"]) for w in wins]
57         gated = rally_gate.gate_windows(points, intervals, fps, min_crossings=0) # for net-
crossings
58         X: List[List[float]] = []
59         for w, g in zip(wins, gated):
60             dur = max(1e-6, w["end"] - w["start"])
61             win_frames = max(1.0, dur * fps)
62             X.append([
63                 dur,
64                 # rally length (sec) ?
65                 fps-invariant

```

```

64         float(w["peak_velocity"]),          # already normalized by
frame_width
65         float(w["mean_velocity"]),
66         float(w["active_frame_count"]) / win_frames,    # active-shuttle ratio ?
fps-invariant
67         float(g.crossings),                  # net crossings (count) ?
the rally signal
68     ])
69     return intervals, X
70
71
72     def baseline_windows(trajecory_csv: str, fps: float, frame_width: float) ->
List[Interval]:
73         points = TrackNetRunner.parse_trajectory_csv(trajecory_csv)
74         vt, mg, mwd = BASELINE_LOCAL
75         wins = TrackNetRunner.trajectory_to_action_windows(
76             points, fps=fps, frame_width=frame_width,
77             velocity_thresh=vt, merge_gap=mg, min_window_duration=mwd)
78         return [(w["start"], w["end"]) for w in wins]
79
80
81     def load_video(spec: Dict, iou: float = 0.5) -> Dict:
82         fps, fw = float(spec["fps"]), float(spec["frame_width"])
83         intervals, X = build_candidates(spec["trajectory"], fps, fw)
84         gts = load_golden_gts(spec["labels"])
85         y = label_candidates(intervals, gts, iou)
86         return {"name": spec.get("name", spec["trajectory"]), "intervals": intervals, "X":
X,
87                 "y": y, "gts": gts, "baseline": baseline_windows(spec["trajectory"], fps,
fw)}
88
89
90     def predict_rallies(model: LogisticRegression, X: List[List[float]], intervals:
List[Interval],
91                       threshold: float = 0.5) -> List[Interval]:
92         if not intervals:
93             return []
94         proba = model.predict_proba(X)
95         pos = [iv for iv, p in zip(intervals, proba) if p >= threshold]
96         return merge_overlaps(pos, gap_tol=1.0)
97
98
99     def run(specs: List[Dict], iou: float = 0.5, threshold: float = 0.5, model_out: str =
None) -> dict:
100         videos = [load_video(s, iou) for s in specs]
101         print(f"=== Distilled local rally classifier vs Gemini silver-GT ")
102         f"({len(videos)} videos, IoU>={iou}, prob>={threshold}) ===")
103         for v in videos:
104             ceil = score(v["intervals"], v["gts"], iou).recall
105             print(f"[{v['name']}] {len(v['intervals'])} candidates ({sum(v['y'])} pos) | "
106                   f"proposal recall ceiling {ceil:.3f} | {len(v['gts'])} Gemini rallies")
107
108         result: dict = {"videos": [v["name"] for v in videos]}
109         if len(videos) >= 2:
110             print("\n--- HONEST leave-one-video-out: learned vs threshold baseline (held-
out) ---")
111             clf, base = [], []
112             for i, held in enumerate(videos):
113                 others = [v for j, v in enumerate(videos) if j != i]
114                 X = [row for v in others for row in v["X"]]
115                 y = [t for v in others for t in v["y"]]
116                 model = LogisticRegression().fit(X, y, FEATURE_NAMES)
117                 preds = predict_rallies(model, held["X"], held["intervals"], threshold)
118                 sc = score(preds, held["gts"], iou)
119                 bsc = score(held["baseline"], held["gts"], iou)

```

```

120         clf.append(sc)
121         base.append(bsc)
122         print(f" held-out {held['name']}: LEARNED P={sc.precision:.3f}
R={sc.recall:.3f} F1={sc.f1:.3f}"
123             f" | baseline P={bsc.precision:.3f} R={bsc.recall:.3f}
F1={bsc.f1:.3f}")
124         mean = lambda xs, a: sum(getattr(s, a) for s in xs) / len(xs)
125         print(f" MEAN held-out: LEARNED P={mean(clf, 'precision'):.3f}
R={mean(clf, 'recall'):.3f} "
126             f"F1={mean(clf, 'f1'):.3f} | baseline P={mean(base, 'precision'):.3f} "
127             f"R={mean(base, 'recall'):.3f} F1={mean(base, 'f1'):.3f}")
128         result["mean_learned_f1"] = mean(clf, "f1")
129         result["mean_baseline_f1"] = mean(base, "f1")
130
131         # Final model on ALL videos (for production / running on a fresh held-out video).
132         Xall = [row for v in videos for row in v["X"]]
133         yall = [t for v in videos for t in v["y"]]
134         final = LogisticRegression().fit(Xall, yall, FEATURE_NAMES)
135         weights = dict(zip(FEATURE_NAMES, [round(float(w), 3) for w in final.w]))
136         print(f"\nfinal model standardized weights (sign/size = influence): {weights}")
137         if model_out:
138             final.save(model_out)
139             print(f"final model (all videos) -> {model_out}")
140             result["model_out"] = model_out
141         return result
142
143
144     def main() -> None:
145         ap = argparse.ArgumentParser(description="Distill Gemini labels into a local rally
classifier.")
146         ap.add_argument("--manifest", required=True,
147                         help="JSON: [{name, trajectory, fps, frame_width, labels}, ...]
(labels = Gemini CSV)")
148         ap.add_argument("--iou", type=float, default=0.5)
149         ap.add_argument("--threshold", type=float, default=0.5, help="classifier probability
cutoff")
150         ap.add_argument("--model-out", default="output/local_rally_model.json")
151         args = ap.parse_args()
152         with open(args.manifest, encoding="utf-8") as f:
153             specs = json.load(f)
154         run(specs, iou=args.iou, threshold=args.threshold, model_out=args.model_out)
155
156
157     if __name__ == "__main__":
158         main()
159

```

20. user (2026-06-10T06:02:27.606Z)

No files found

21. user (2026-06-10T06:02:27.668Z)

```

backend\eval\__init__.py
backend\eval\annotations.py
backend\eval\classifier.py
backend\eval\metrics.py
backend\eval\training.py
backend\eval\calibration.py
backend\eval\gemini_refine.py
backend\eval\calibrate_local.py
backend\eval\calibrate_wasb.py
backend\eval\distill_local.py
backend\eval\export_wasb_segments.py
backend\eval\harness.py

```

backend\eval\regression.py
backend\eval\batch.py
backend\eval\eval_partial_labels.py
backend\eval\rally_seq_proto.py

22. user (2026-06-10T06:02:34.799Z)

docs\EVALUATION.md
docs\TRACKNET_WSL_SETUP.md
docs\GOLDEN_SET_IMPLEMENTATION.md
docs\GOLDEN_SET_PHASE1_VIDEOS.md
docs\HOW_RALLY_DETECTION_WORKS.md
docs\DEFERRED_HARDENING_PLAN.md
docs\PMF_MARKET_RESEARCH.md
docs\ROADMAP.md
docs\COMMERCIALIZATION.md
docs\PLATFORM_ARCHITECTURE.md
docs\BACKLOG.md
docs\GOLDEN_SET_VENDORING.md
docs\README.md
docs\VIDEO_RECORDING_GUIDELINES.md
docs\CODE_MAP.md
docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md
docs\NEXT_STEPS.md

23. user (2026-06-10T06:02:35.075Z)

docs\archives\COMMERCIAL_READINESS_REVIEW.md
docs\archives\MONETIZATION_AUDIT.md
docs\archives\README.md
docs\archives\checkpoints\2026-06-eval-baseline\README.md
docs\archives\checkpoints\2026-06-logging-reporting\README.md
docs\archives\checkpoints\2026-06-scaffolding-complete\README.md
docs\archives\decisions\DECISIONS.md
docs\archives\past_projects\low-regret-moves\LOW_REGRET_MOVES_PLAN.md
docs\archives\past_projects\README.md
docs\archives\research\AUDIO_E1_FINDINGS.md
docs\archives\research\AUDIO_RALLY_DETECTION_PLAN.md
docs\archives\research\RALLY_DETECTION_RESEARCH.md
docs\archives\checkpoints\2026-06-strategy-foundation\README.md
docs\archives\quality-iterations\2026-06-human-gt-tuning\README.md
docs\archives\quality-iterations\2026-06-initial\README.md
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md
docs\archives\quality-iterations\README.md
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md

24. user (2026-06-10T06:02:49.031Z)

run_all_tests.py
run_eval.py
run_phase3_eval.py
run_process_and_stitch.py
run_regression.py

25. user (2026-06-10T06:02:50.837Z)

Byte-compiled / optimized / DLL files

__pycache__/
*.py[cod]
*\$py.class
*.pt

Local environment secrets

.env
*.env
.env.*

C extensions

*.so

Distribution / packaging

.Python
build/
develop-eggs/
dist/
downloads/
eggs/
.eggs/
lib/
lib64/
parts/
sdist/
var/
wheels/
share/python-wheels/
*.egg-info/
.installed.cfg
*.egg
MANIFEST

SQLite databases & Local output files

*.db
*.db-journal
*.db-wal
output/
output_dir/
backend/static/highlights_*.mp4
backend/static/*_compiled.mp4

OS generated files

.DS_Store
Thumbs.db
desktop.ini

Coverage data

.coverage
.coverage.*
htmlcov/
.pytest_cache/

Local Claude Code config (machine-specific)

.claude/
.claude-verify-agent
■
■---OUTPUT DIR---
Compressed Videos
adarsh_avi_full_vs_human.csv
adarsh_avi_vs_human.csv
adarsh_baseline_gated.csv
adarsh_baseline_segments.csv
adarsh_gemini_fullvideo.csv
adarsh_gemini_refined.csv
adarsh_gemini_smoke.csv
adarsh_min5s_segments.csv

adarsh_tuned_segments.csv
badminton_indexer.db
gbaaddy_vs_human.csv
human_lovo_manifest.json
kushagra_vs_human.csv
largetestvideo_vs_human.csv
local_calib_manifest.json
local_rally_model.json
mahadevpura_vs_human.csv
phase3_probe.json
phase3_yolo.db
rally_clips
sports_indexer.db
test_indexer.db
testlarge_gemini_fullvideo.csv
testlarge_gemini_rallies.csv
testlarge_vs_human.csv
wasb_tuned_segments.csv
wasb_tuned_segments_baseline.csv
wasb_tuned_segments_vs_golden.csv

26. user (2026-06-10T06:03:03.085Z)

```
28:         CREATE TABLE IF NOT EXISTS videos (  
29:             id TEXT PRIMARY KEY,  
30:             filepath TEXT NOT NULL,  
31:             processed_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
32:             status TEXT NOT NULL,  
33:             validation_results TEXT  
34:         )  
35:     """)  
36:  
37:     # Play Segments Table (Extensible metadata JSON)  
38:     cursor.execute("""  
39:         CREATE TABLE IF NOT EXISTS play_segments (  
40:             id INTEGER PRIMARY KEY AUTOINCREMENT,  
41:             video_id TEXT NOT NULL,  
42:             start_time REAL NOT NULL,  
43:             end_time REAL NOT NULL,  
44:             duration REAL NOT NULL,  
45:             server TEXT,  
46:             receiver TEXT,  
47:             metadata TEXT,  
48:             FOREIGN KEY (video_id) REFERENCES videos(id) ON DELETE CASCADE  
49:         )  
50:     """)  
51:  
52:     # Events Table  
53:     cursor.execute("""  
54:         CREATE TABLE IF NOT EXISTS events (  
55:             id INTEGER PRIMARY KEY AUTOINCREMENT,  
56:             segment_id INTEGER NOT NULL,  
57:             timestamp REAL NOT NULL,  
58:             type TEXT NOT NULL,  
59:             player TEXT,  
60:             details TEXT,  
61:             FOREIGN KEY (segment_id) REFERENCES play_segments(id) ON DELETE CASCADE  
62:         )  
63:     """)  
64:  
65:     # Versioned migrations live here. PRAGMA user_version is the schema  
66:     # contract ? bump it for every change and add an ordered `if version < N`  
67:     # block below. Tests assert the expected version, so accidental drift fails CI.  
68:     version = cursor.execute("PRAGMA user_version").fetchone()[0]  
69:
```

```

70-         if version < 1:
71-             # v1: raw candidate detections (pre-confirmation), detector-agnostic.
72-             # Common fields are columns; detector-specific metrics go in `stats` JSON,
73-             # so a new segmenter reuses this table by setting its own `detector`/`stats`.
74-             cursor.execute("""
75-                 CREATE TABLE IF NOT EXISTS candidate_segments (
76-                     id INTEGER PRIMARY KEY AUTOINCREMENT,
77-                     video_id TEXT NOT NULL,
78-                     detector TEXT NOT NULL,
79-                     chunk_index INTEGER,
80-                     start_time REAL NOT NULL,
81-                     end_time REAL NOT NULL,
82-                     duration REAL NOT NULL,
83-                     confidence REAL,
84-                     stats TEXT,
85-                     detection_params TEXT,
86-                     confirmed_segment_id INTEGER,
87-                     human_label TEXT,
88-                     notes TEXT,
89-                     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
90-                     --
91-                 )
92-             """)
93-             cursor.execute("PRAGMA user_version = 1")
94-             # future: if version < 2: cursor.execute("ALTER TABLE ..."); PRAGMA user_version
95-             = 2
96-
97-         conn.commit()
98-
99-         def add_video(self, video_id: str, filepath: str, status: str, validation_results:
100-             List[Dict[str, Any]]) -> bool:
101-             try:
102-                 with self._get_connection() as conn:
103-                     conn.cursor().execute(
104-                         "INSERT OR REPLACE INTO videos (id, filepath, status,
105-                         validation_results) VALUES (?, ?, ?, ?)",
106-                         (video_id, filepath, status, json.dumps(validation_results))
107-                     )
108-                     conn.commit()
109-                     return True
110-             except Exception as e:
111-                 logger.error(f"Failed to add video: {e}")

```

27. user (2026-06-10T06:03:20.534Z)

docs\archives\decisions\DECISIONS.md:254:- The `yolo\_hybrid` registry key is now a slight misnomer (no YOLO); keep it for back-compat or rename in a follow-up.

docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md:73: feature / per-sport calibration). The collector + indexer are already sport-generic. ? **ACTION: rename the repo off

docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md:154:| 5 | Rename repo off "badminton" | **owner action** ? |

docs\DEFERRED_HARDENING_PLAN.md:38:- **Rename the diagnostic `setup.py`** (e.g. `scripts/doctor.py`) so it no longer

docs\DEFERRED_HARDENING_PLAN.md:44:- `setup.py` rename touches docs and any muscle-memory invocations ? grep first.

docs\NEXT_STEPS.md:39:3. **Housekeeping:** **rename repo off "badminton"** (multisport); confirm collector = corpus/event-store system-of-record.

docs\CODE_MAP.md:243:- **spec.yaml <-> harvest.py coupling:** rule `path` strings dotted-navigate harvest's stage/metric/detail names; rename a detail in harvest.py and the matching rule silently stops firing. Add rules in spec.yaml, not code.

docs\BACKLOG.md:25:- ? **`yolo\_hybrid` naming** ? the registry key, `yolo\_` config keys, and `yolo-hybrid-` DB detector tags keep the "yolo" name for back-compat despite no YOLO remaining. Optional rename. **Revisit:** if/when a clean break is worth the migration.

docs\archives\past_projects\low-regret-moves\LOW_REGRET_MOVES_PLAN.md:75:? ??? segmenters/ # (renamed/moved from current indexer/)

28. user (2026-06-10T06:03:22.350Z)

commit 26ee2de13ccfeda7c069a79b24e61edecd142b7f
Author: Avi Dullu <avi.dullu@gmail.com>
Date: Tue Jun 9 21:37:22 2026 +0530

docs: next-steps + multisport findings (WASB transfers to table tennis, not pickleball)

Live docs/NEXT_STEPS.md capturing the recommendation + prioritized next steps after the quality-iteration + multisport-recon session:

- Recommendation: quality-first TAL-head harness (decouples from multi-court); VLM (ms-swift + Qwen3-VL) later; on-device deferred; MIT/Apache/BSD only.
- Multisport finding: Roora pickleball/TT labels ingest cleanly (convert-cvat, 6 rallies/sport); WASB-transfer SURPRISE (n=6 directional) -- table tennis F1 0.909 out-of-the-box, pickleball 0.308 -> the per-sport-detector blocker is SPORT-DEPENDENT.
- Owner field notes: varied camera = generalization data; adjacent-court SOUND dominates -> reconfirms audio-foreground won't work + the robustness-first path.
- Prioritized: Phase-0 (M0.4 harness + ship-gate target, M0.1 corpus+event-index, M0.3 compliance) on greenlight; multisport (more pickleball/TT videos -> confirm transfer -> merged LOVO); repo rename.

docs/NEXT_STEPS.md | 46 +++++
=== NEXT_STEPS on master? ===
docs/NEXT_STEPS.md
13d4573 fix(stitcher): restore f-string quotes dropped in logging conversion (SyntaxError on master) (#63)
a5477c2 docs: next-steps + multisport findings (WASB transfers to table tennis, not pickleball) (#62)
4d04a73 Quality iteration: human-GT multi-video LOVO + learned prototype + multi-court finding + owned-model study/plan (#61)

29. user (2026-06-10T06:03:47.294Z)

```
60
61     ### 1. Launching the Local Web App Dashboard
62     To start the FastAPI server and open the web dashboard:
63     ```bash
64     python -m backend.main --server --port 8000
65     ```
66     Open your browser and navigate to **[http://localhost:8000](http://localhost:8000)**.
67     *Simply input your local raw video file path in the sidebar to process, validate, index,
68     and compile highlights instantly.*
69
70     ### 2. Running in CLI Processing Mode
71     To process and index a video completely via terminal commands:
72     ```bash
73     python -m backend.main --video-path "C:/path/to/my_video.mp4"
74     ```
75     This outputs a validation summary and indexes all rallies directly into the local SQLite
76     database, using whichever segmenter is set as `default_segmenter` in `config.json`
77     (`yolo_hybrid` by default).
78
79     To explicitly pick a segmenter for one run:
80     ```bash
81     python -m backend.main --video-path "C:/path/to/my_video.mp4" --segmenter yolo_hybrid
82     python -m backend.main --video-path "C:/path/to/my_video.mp4" --segmenter gemini
83     python -m backend.main --video-path "C:/path/to/my_video.mp4" --segmenter motion
84     ```
85
86     ### 3. Rich CLI Debugging (`--debug-clip`)
87     To diagnose exactly why a video segment was registered as active/dead play, or inspect
88     detailed focus/continuity metrics at a particular time in seconds:
89     ```bash
90     python -m backend.main --video-path "C:/path/to/my_video.mp4" --debug-clip 125.5
91     ```
92     *Outputs granular validation check logs and motion ratios within a +/- 3 second frame
93     window around the timestamp.*
```

30. user (2026-06-10T06:03:48.204Z)

```

        255
    ],
    "court_pixels_min_ratio": 0.05
}
},
"indexing": {
    "default_segmenter": "gemini",
    "use_gpu": true,
    "motion_threshold": 0.08,
    "min_segment_duration": 3.0,
    "dead_time_merge_gap": 2.0,
    "chunk_duration": 90,
    "chunk_overlap": 10,
    "detector_model": "fasterrcnn_resnet50_fpn",
    "detector_score_threshold": 0.5,
    "yolo_velocity_threshold": 0.15,

```

31. user (2026-06-10T06:04:06.728Z)

```

docs\NEXT_STEPS.md:4: docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md`. **Latest blow-by-blow:** memory
`current-state.md` / `session-handoff.md`.
docs\NEXT_STEPS.md:42:- Roadmap: `docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md` · Study:
`docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md`
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:12:## Update (2026-06-09, post-
PR-#61-review) ? authoritative roadmap is now `docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md`
docs\archives\quality-iterations\2026-06-multivideo-
lovo\README.md:116: `docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md` (Gen-0 first), gated on n?5 + owner
sign-off.
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:253: +
`docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md`, after scaffolding).

```

32. user (2026-06-10T06:04:07.956Z)

```

backend\eval\calibrate_local.py:9:Ground truth = Gemini full-context silver labels (one CSV per
video). With >=2
backend\eval\calibrate_local.py:21:(labels = a Gemini full-context CSV; trajectory = the WASB
CSV).
backend\eval\calibrate_local.py:80:     print(f"=== Local-only calibration vs Gemini silver-GT
({len(videos)} videos, "
backend\eval\calibrate_local.py:114:     ap = argparse.ArgumentParser(description="Calibrate the
local-only pipeline against Gemini silver labels.")
backend\eval\distill_local.py:4:further: learn "is this WASB candidate a real rally?" from
Gemini silver labels,
backend\eval\distill_local.py:101:     print(f"=== Distilled local rally classifier vs Gemini
silver-GT "
backend\eval\distill_local.py:147:                                     help="JSON: [{name, trajectory, fps,
frame_width, labels}, ...] (labels = Gemini CSV)")
backend\eval\audio\e1_probe.py:10:eval.metrics. Labels may be human-gold OR Gemini-silver ?
silver is fine for the
backend\eval\audio\e1_probe.py:101:     ap.add_argument("--labels", required=True, help="golden
or Gemini-silver rally CSV")
backend\eval\gemini_refine.py:18:     python -m backend.eval.gemini_refine --video Adarsh.MP4
--trajectory adarsh_traj.csv \
backend\eval\gemini_refine.py:19:                                     --fps 59.94 --frame-width 1920 --out
output\adarsh_gemini.csv [--limit 4]
backend\eval\gemini_refine.py:103:                                     max_workers: int = 4, out_csv: str =
"output/gemini_refined.csv",
backend\eval\gemini_refine.py:150:                                     out_csv: str =
"output/gemini_fullvideo.csv", clip_scale_width: Optional[int] = 1280,
backend\eval\gemini_refine.py:195:     print(f"[gemini_full] {len(rallies)} rallies, {play:.1f}s

```

```
play -> {out_csv}", flush=True)
backend\eval\gemini_refine.py:211:      ap.add_argument("--out",
default="output/gemini_refined.csv")
```

33. user (2026-06-10T06:04:21.860Z)

```
123:[Omitted long matching line]
176:[Omitted long matching line]
177:[Omitted long matching line]
284:| Resume a crashed WASB pass | nothing to do ?
`@backend/pipeline/detectors/wasb_infer.py::run` auto-resumes from
`det_raw.jsonl`+`manifest.json` next to `--out`; `--fresh` forces restart |
```

34. user (2026-06-10T06:04:22.783Z)

CODE_MAP.md

****Purpose:**** cold-start navigation map for the badminton-highlight-indexer ? a self-hosted video ? highlight-reel pipeline (CV/AI rally detection ? SQLite ? ffmpeg stitch) with a segmenter-agnostic eval/calibration stack, typed config, and per-video observability reporting.

****Regenerate:**** rerun the `build-code-map` workflow (parallel subsystem readers ? this doc). Re-run after any registry/contract/WSL-interface change, or after a directory restructure (this revision follows #40).

****LEGEND****

- `@path` = file (always repo-relative)
- `::sym` = symbol (class/func/const) in the nearest preceding `@path`
- `->` = dataflow / call / "produces"
- `\$` = data contract (canonical shape)
- `?` = gotcha / footgun
- `?` = registry / plugin extension point
- `WSL` = runs inside WSL2 conda env, NOT the Windows Python process

1. PIPELINE

1A. Runtime: video file -> highlight reel

...

typed config load (built-in defaults -> config.json overrides; absent/parse-fail -> all-defaults, never fatal)

@backend/config/loader.py::load_config ->

@backend/config/models.py::AppConfig

```
-> video_id = MD5(whole file)      @backend/utils/hashing.py::compute_video_id (single
shared helper, 64KB chunks; imported by main + all run_*.py ? no more per-entrypoint copies)
-> validate
```

@backend/pipeline/validators/base.py::registry.run_all_with_context(video_path, global_config)

FormatValidator -> ctx['ffprobe_meta'] -> ResolutionFPSValidator -> PTSContinuity ->

SceneCut -> Blur -> Relevance (? ACTUAL registration/exec order ? see

@backend/pipeline/validators/__init__.py)

```
[validation FAIL -> persist status="failed", return early; report STILL emitted in
finally]
```

```
-> db.add_video(video_id, filepath, status, [r.dict()]) @backend/infrastructure/database.py
```

```
-> seg_name = req.segmenter or global_config.indexing.default_segmenter (fallback 'motion')
```

```
-> seg = segmenter_registry.get(seg_name)(db, global_config) ?
```

@backend/pipeline/segmenters/base.py::segmenter_registry (400 + available list if missing)

35. user (2026-06-10T06:05:45.440Z)

Found 15 files

C:\Users\avidu\Projects\sports-data-collector\tests\test_normalize_annotations.py

C:\Users\avidu\Projects\sports-data-collector\src\cli\normalize_annotations.py

C:\Users\avidu\Projects\sports-data-collector\docs\MANUAL_INGESTION_POSITIONING.md
C:\Users\avidu\Projects\sports-data-collector\src\reporting\adapter.py
C:\Users\avidu\Projects\sports-data-collector\src\reporting__init__.py
C:\Users\avidu\Projects\sports-data-collector\src\cli\curate.py
C:\Users\avidu\Projects\sports-data-collector\docs\INGESTION_PIPELINE.md
C:\Users\avidu\Projects\sports-data-collector\src\parsers\cvat\rally_cvat.py
C:\Users\avidu\Projects\sports-data-collector\tests\test_cvat_parser.py
C:\Users\avidu\Projects\sports-data-collector\docs\annotation\R00RA_BRIEF.md
C:\Users\avidu\Projects\sports-data-collector\README.md
C:\Users\avidu\Projects\sports-data-collector\tests\test_cli.py
C:\Users\avidu\Projects\sports-data-collector\src\cli\ingest.py
C:\Users\avidu\Projects\sports-data-collector\docs\DECISIONS.md
C:\Users\avidu\Projects\sports-data-collector\tests\test_local_import.py

36. user (2026-06-10T06:05:48.468Z)

True
pb
tt
annotations.report.json
annotations.report.md
annotations.summary.md
pickleball_rallies.csv
tabletennis_rallies.csv
True
True

37. user (2026-06-10T06:09:01.122Z)

Structured output provided successfully

38. assistant (2026-06-10T06:09:07.888Z)

Audit complete. Structured findings returned.